SAMSUNG CONFIDENTIAL
THIS DOCUMENT CONTAINS CONFIDENTIAL
PROPRIETARY INFORMATION THAT IS
SAMSUNG ELECTRONICS CO'S PROPERTY.
DO NOT DISCLOSE TO OR DUPLICATE FOR OTHERS
EXCEPT AS AUTHORIZED BY SAMSUNG.

A35 5G NA/CAN REV05

2023.12

- Sheet01 : GSM / WCDMA / LTE / NR(N41) RF
- Sheet02 : TRANSCEIVER, SUB ANT
- Sheet03 : CONNECTIVITY(BT/WIFI/FM), NFC
- Sheet04 : AP(QUARTZ-INTERFACE,GPIO)
- Sheet05 : AP(POWER), MEM(uMCP)
- Sheet06 : PMIC/SUB PMIC, IF PMIC
- Sheet07 : AUDIO Codec, SENSOR, SIM, SD
- Sheet08 : CAMERA
- Sheet09 : DISPLAY PMIC, C2C, mmW

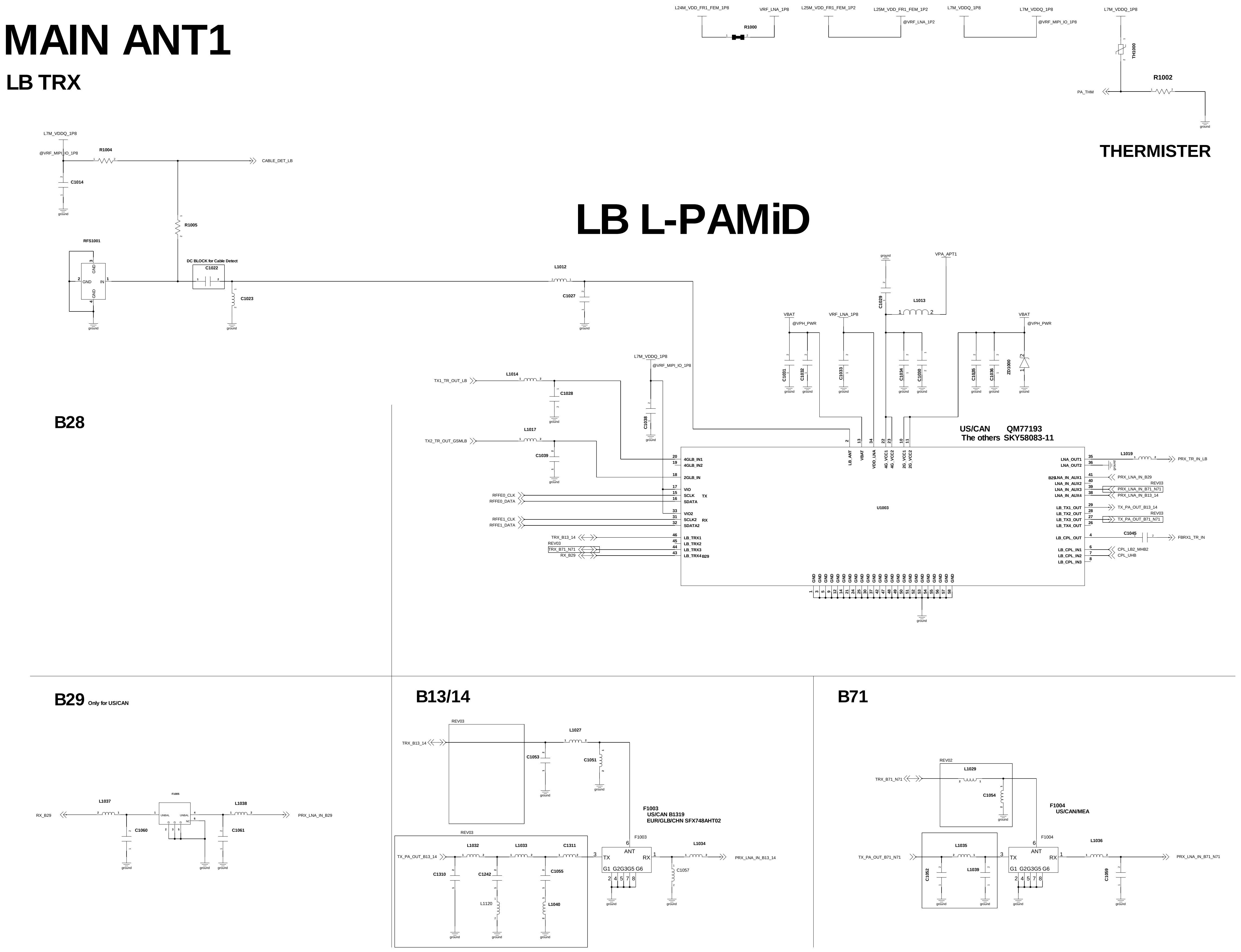
			Engineer:			
			Drawn by:			
			R&D CHK:	TITLE:		Size:
			DOC CTRL CHK:	A54		
			MFG ENGR CHK:			
Changed by:	Date Changed:	Time Changed:	QA CHK:	REV:	Drawing Number:	Page:

SAMSUNG CONFIDENTIAL
THIS DOCUMENT CONTAINS CONFIDENTIAL
PROPRIETARY INFORMATION THAT IS
SAMSUNG ELECTRONICS CO'S PROPERTY.
DO NOT DISCLOSE TO OR DUPLICATE FOR OTHERS
EXCEPT AS AUTHORIZED BY SAMSUNG.

RF PART 1

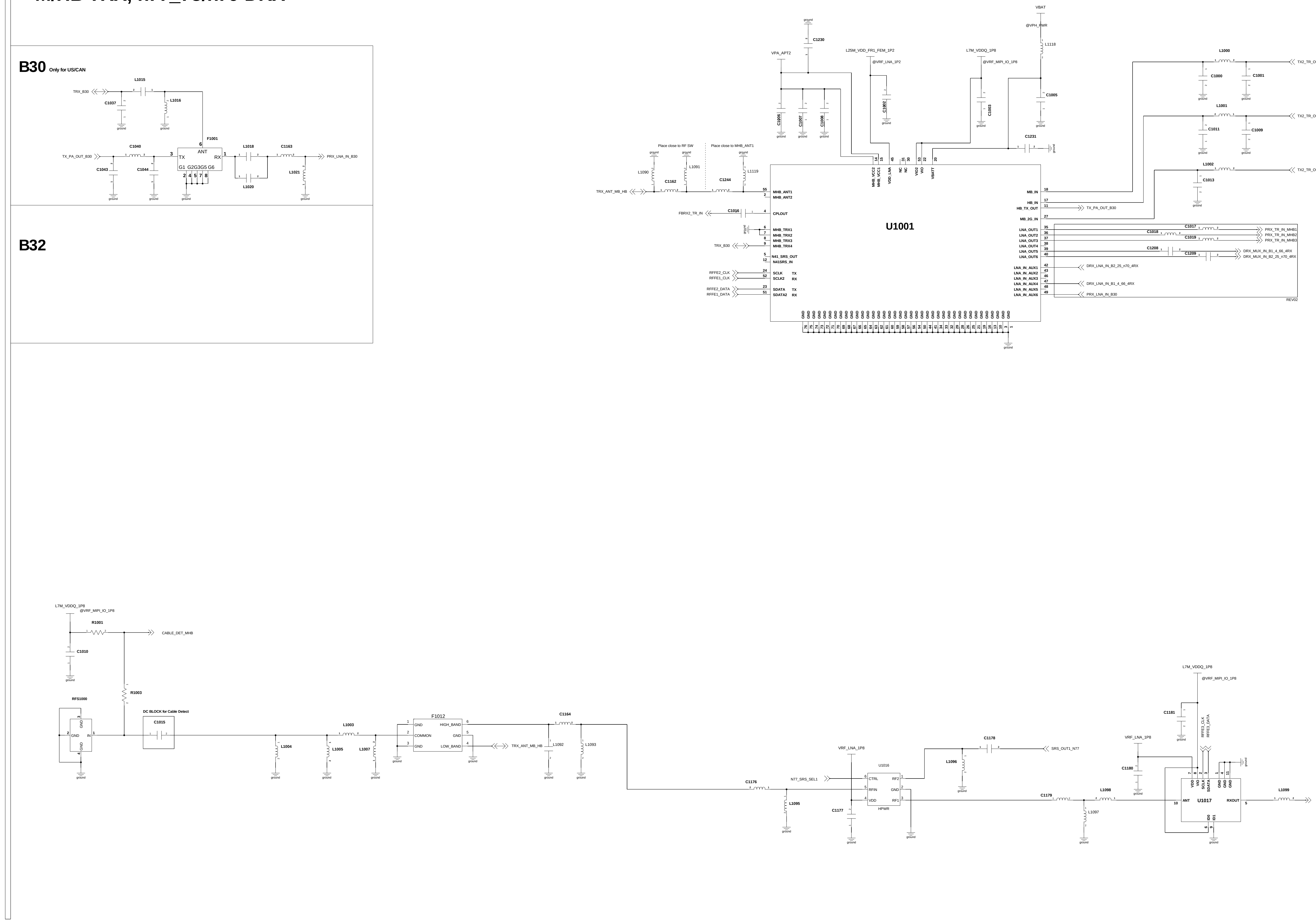
MAIN ANT1

LB TRX


L24M_VDD_FR1_FEM_1P8
VRF_LNA_1P8
R1000
L25M_VDD_FR1_FEM_1P2
L25M_VDD_FR1_FEM_1P2
@VRF_LNA_1P2
L7M_VDDQ_1P8
L7M_VDDQ_1P8
@VRF_MIPI_IO_1P8
L7M_VDDQ_1P8
TH1000
R1002
PA_THM
THERMISTER
L7M_VDDQ_1P8
@VRF_MIPI_IO_1P8
R1004
CABLE_DET_LB
C1014
R1005
RFS1001
DC BLOCK for Cable Detect
C1022
C1023
L1012
C1027
LB L-PAMiD
VPA_APT1
C1029
L1013
VBAT
@VPH_PWR
VRF_LNA_1P8
C1031
C1032
C1033
C1034
C1030
C1035
C1036
ZD1000
L7M_VDDQ_1P8
@VRF_MIPI_IO_1P8
C1038
TX1_TR_OUT_LB
L1014
C1028
TX2_TR_OUT_GSMLB
L1017
C1039
RFFE0_CLK
RFFE0_DATA
RFFE1_CLK
RFFE1_DATA
TRX_B13_14
REV03
TRX_B71_N71
RX_B29
US/CAN QM77193
The others SKY58083-11
U1003
4GLB_IN1
4GLB_IN2
2GLB_IN
VIO
SCLK
SDATA
TX
VIO2
SCLK2
SDATA2
RX
LB_TRX1
LB_TRX2
LB_TRX3
LB_TRX4
B29
LB_ANT
VBAT
VDD_LNA
4G_VCC1
4G_VCC2
2G_VCC1
2G_VCC2
LNA_OUT1
LNA_OUT2
LNA_IN_AUX1
LNA_IN_AUX2
LNA_IN_AUX3
LNA_IN_AUX4
LB_TX1_OUT
LB_TX2_OUT
LB_TX3_OUT
LB_TX4_OUT
LB_CPL_OUT
LB_CPL_IN1
LB_CPL_IN2
LB_CPL_IN3
GND
L1019
PRX_TR_IN_LB
PRX_LNA_IN_B29
PRX_LNA_IN_B71_N71
PRX_LNA_IN_B13_14
TX_PA_OUT_B13_14
TX_PA_OUT_B71_N71
C1045
FBRX1_TR_IN
CPL_LB2_MHB2
CPL_UHB
B28
B29 Only for US/CAN
RX_B29
L1037
C1060
F1005
UNBAL
NC
L1038
C1061
PRX_LNA_IN_B29
B13/14
REV03
TRX_B13_14
L1027
C1053
C1051
F1003
US/CAN B1319
EUR/GLB/CHN SFX748AHT02
TX_PA_OUT_B13_14
L1032
L1033
C1311
C1310
C1242
L1120
C1055
L1040
ANT
TX
RX
G1 G2G3G5 G6
L1034
C1057
PRX_LNA_IN_B13_14
B71
REV02
TRX_B71_N71
L1029
C1054
F1004
US/CAN/MEA
TX_PA_OUT_B71_N71
L1035
C1052
L1039
L1036
C1059
PRX_LNA_IN_B71_N71
ground

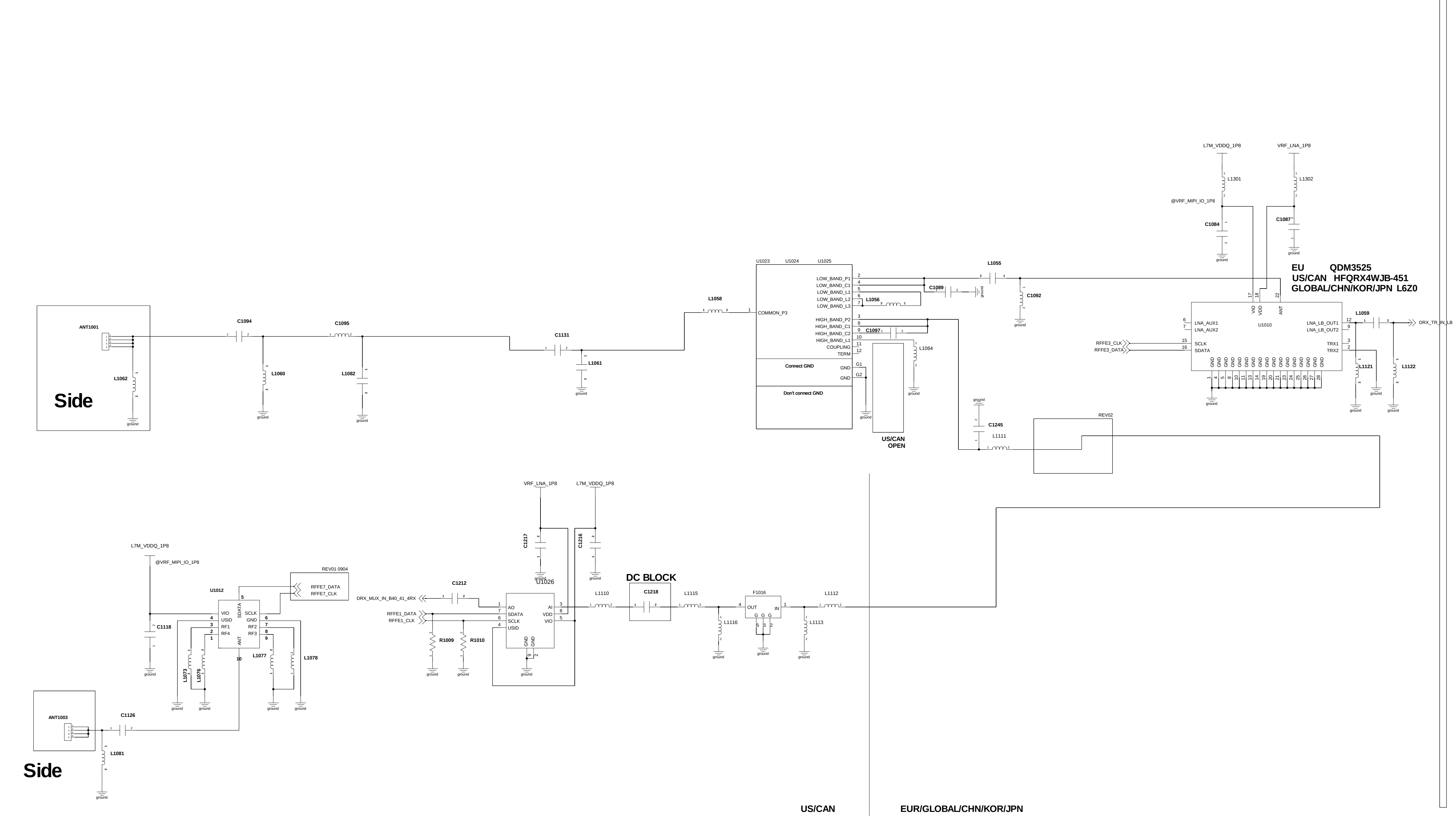
MAIN ANT2

M/HB TRX, n77_78/n79 DRX


B30 Only for US/CAN
TRX_B30
L1015
C1037
L1016
F1001
TX_PA_OUT_B30
C1040
C1043
C1044
ANT
TX
RX
G1 G2G3G5 G6
L1018
L1020
C1163
L1021
PRX_LNA_IN_B30
B32
VPA_APT2
C1230
C1006
C1007
C1008
L25M_VDD_FR1_FEM_1P2
@VRF_LNA_1P2
C1002
L7M_VDDQ_1P8
@VRF_MIPI_IO_1P8
C1003
VBAT
@VPH_PWR
L1118
C1005
C1231
Place close to RF SW
Place close to MHB_ANT1
L1090
L1091
L1119
C1162
C1244
TRX_ANT_MB_HB
FBRX2_TR_IN
C1016
TRX_B30
RFFE2_CLK
RFFE1_CLK
RFFE2_DATA
RFFE1_DATA
U1001
MHB_ANT1
MHB_ANT2
CPLOUT
MHB_TRX1
MHB_TRX2
MHB_TRX3
MHB_TRX4
N41_SRS_OUT
N41SRS_IN
SCLK
SCLK2
SDATA
SDATA2
TX
RX
MHB_VCC2
MHB_VCC1
VDD_LNA
NC
VIO2
VIO
VBATT
MB_IN
HB_IN
HB_TX_OUT
MB_2G_IN
LNA_OUT1
LNA_OUT2
LNA_OUT3
LNA_OUT4
LNA_OUT5
LNA_OUT6
LNA_IN_AUX1
LNA_IN_AUX2
LNA_IN_AUX3
LNA_IN_AUX4
LNA_IN_AUX5
LNA_IN_AUX6
GND
L1000
C1000
C1001
TX2_TR_OUT_MB
L1001
C1011
C1009
TX2_TR_OUT_HB
L1002
C1013
TX2_TR_OUT_GSMHB
TX_PA_OUT_B30
C1017
C1018
C1019
C1208
C1209
PRX_TR_IN_MHB1
PRX_TR_IN_MHB2
PRX_TR_IN_MHB3
DRX_MUX_IN_B1_4_66_4RX
DRX_MUX_IN_B2_25_n70_4RX
DRX_LNA_IN_B2_25_n70_4RX
DRX_LNA_IN_B1_4_66_4RX
PRX_LNA_IN_B30
REV02
L7M_VDDQ_1P8
@VRF_MIPI_IO_1P8
R1001
CABLE_DET_MHB
C1010
R1003
RFS1000
DC BLOCK for Cable Detect
C1015
L1004
L1005
L1003
L1007
F1012
HIGH_BAND
COMMON
LOW_BAND
TRX_ANT_MB_HB
C1164
L1092
L1093
C1176
L1095
VRF_LNA_1P8
N77_SRS_SEL1
U1016
CTRL
RFIN
VDD
RF2
RF1
HPWR
C1177
C1178
L1096
SRS_OUT1_N77
C1179
L1097
L1098
VRF_LNA_1P8
C1180
L7M_VDDQ_1P8
@VRF_MIPI_IO_1P8
C1181
RFFE3_CLK
RFFE3_DATA
U1017
ANT
RXOUT
ID0
ID1
L1099
DRX_TR_IN_N77
ground

SUB ANT1

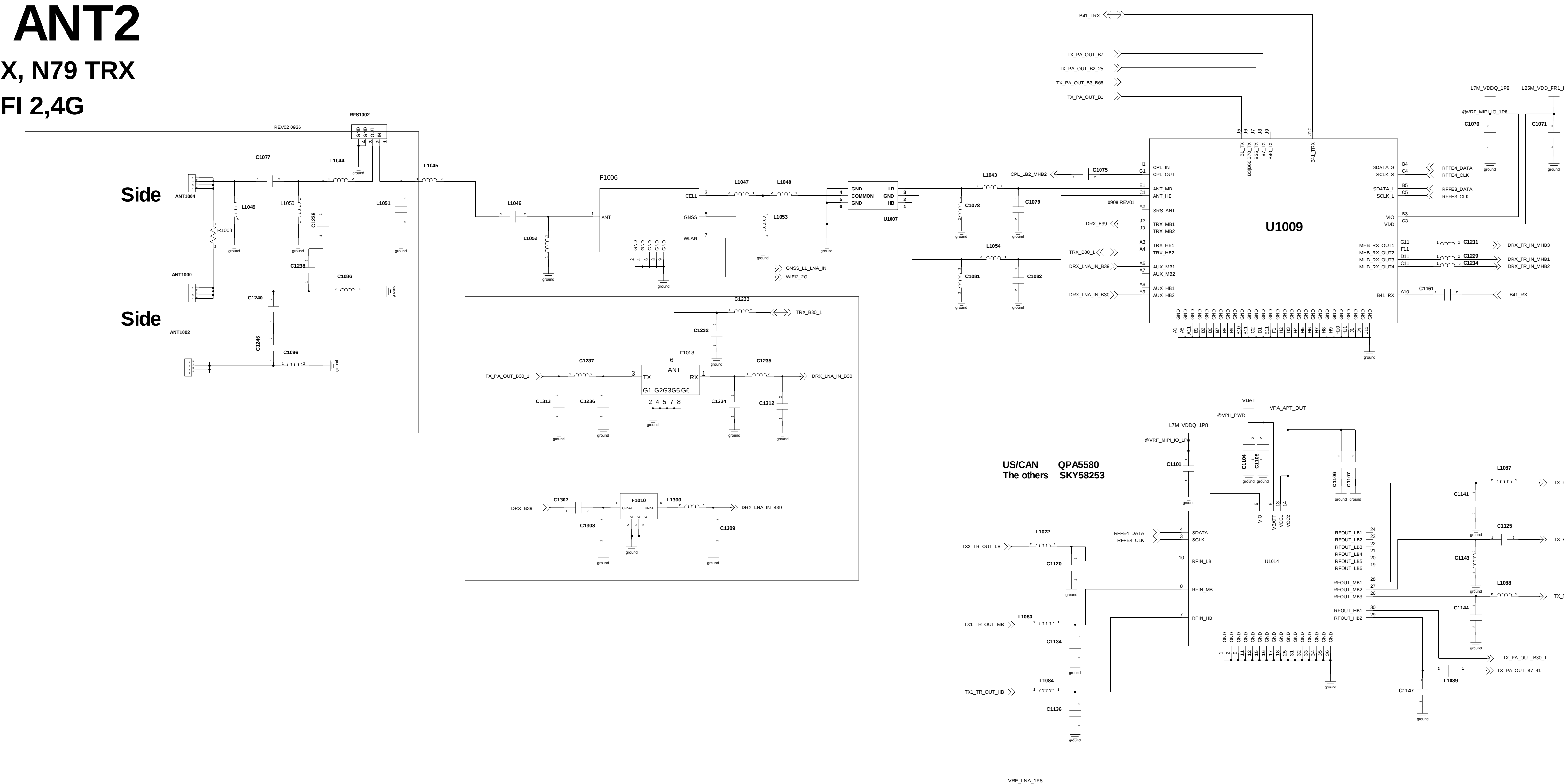
LB DRX, HB DRX MIMO, N79_PRX MIMO

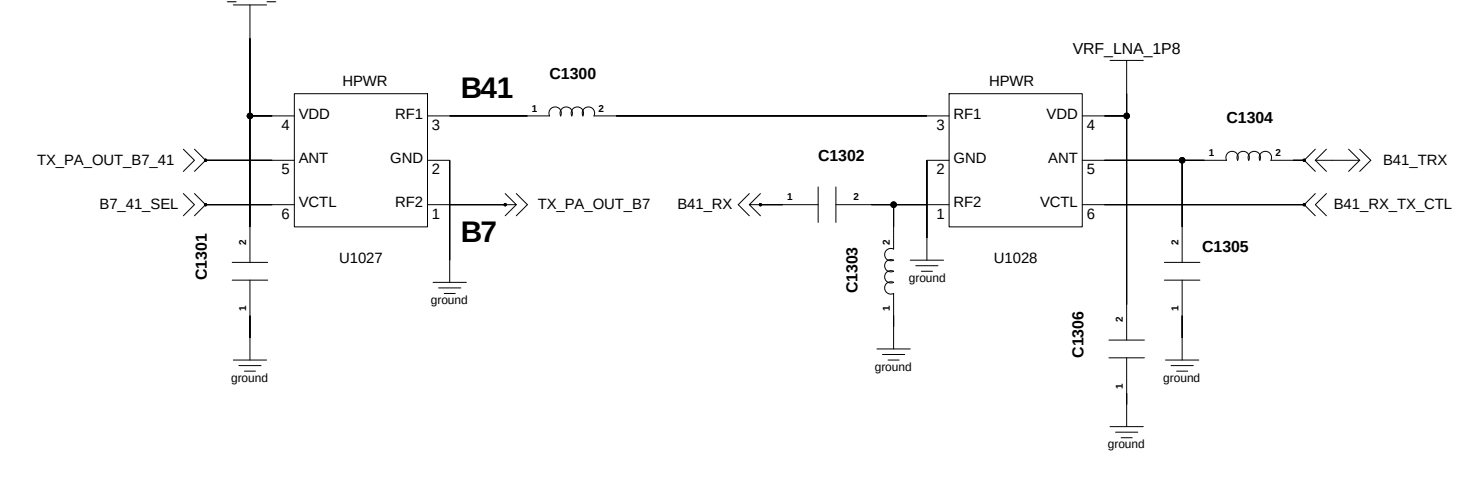

ANT1001
Side
L1062
C1094
L1060
C1095
L1082
C1131
L1061
L1058
U1023
U1024
U1025
COMMON_P3
LOW_BAND_P1
LOW_BAND_C1
LOW_BAND_L1
LOW_BAND_L2
LOW_BAND_L3
HIGH_BAND_P2
HIGH_BAND_C1
HIGH_BAND_C2
HIGH_BAND_L1
COUPLING
TERM
Connect GND
Don't connect GND
L1056
C1089
C1097
L1064
US/CAN
OPEN
L1055
C1092
C1245
L1111
REV02
L7M_VDDQ_1P8
L1301
@VRF_MIPI_IO_1P8
C1084
VRF_LNA_1P8
L1302
C1087
EU QDM3525
US/CAN HFQRX4WJB-451
GLOBAL/CHN/KOR/JPN L6Z0
U1010
LNA_AUX1
LNA_AUX2
SCLK
SDATA
VIO
VDD
ANT
LNA_LB_OUT1
LNA_LB_OUT2
TRX1
TRX2
RFFE3_CLK
RFFE3_DATA
L1059
DRX_TR_IN_LB
L1121
L1122
L7M_VDDQ_1P8
@VRF_MIPI_IO_1P8
C1118
U1012
REV01 0904
RFFE7_DATA
RFFE7_CLK
USID
RF1
RF2
RF3
RF4
L1073
L1076
L1077
L1078
ANT1003
Side
C1126
L1081
DRX_MUX_IN_B40_41_4RX
C1212
RFFE1_DATA
RFFE1_CLK
R1009
R1010
U1026
AO
AI
C1217
C1216
VRF_LNA_1P8
L7M_VDDQ_1P8
DC BLOCK
C1218
L1110
L1115
L1116
F1016
OUT
IN
L1112
L1113
US/CAN
EUR/GLOBAL/CHN/KOR/JPN
ground

SUB ANT2

MHB DRX, N79 TRX

GPS, WIFI 2,4G

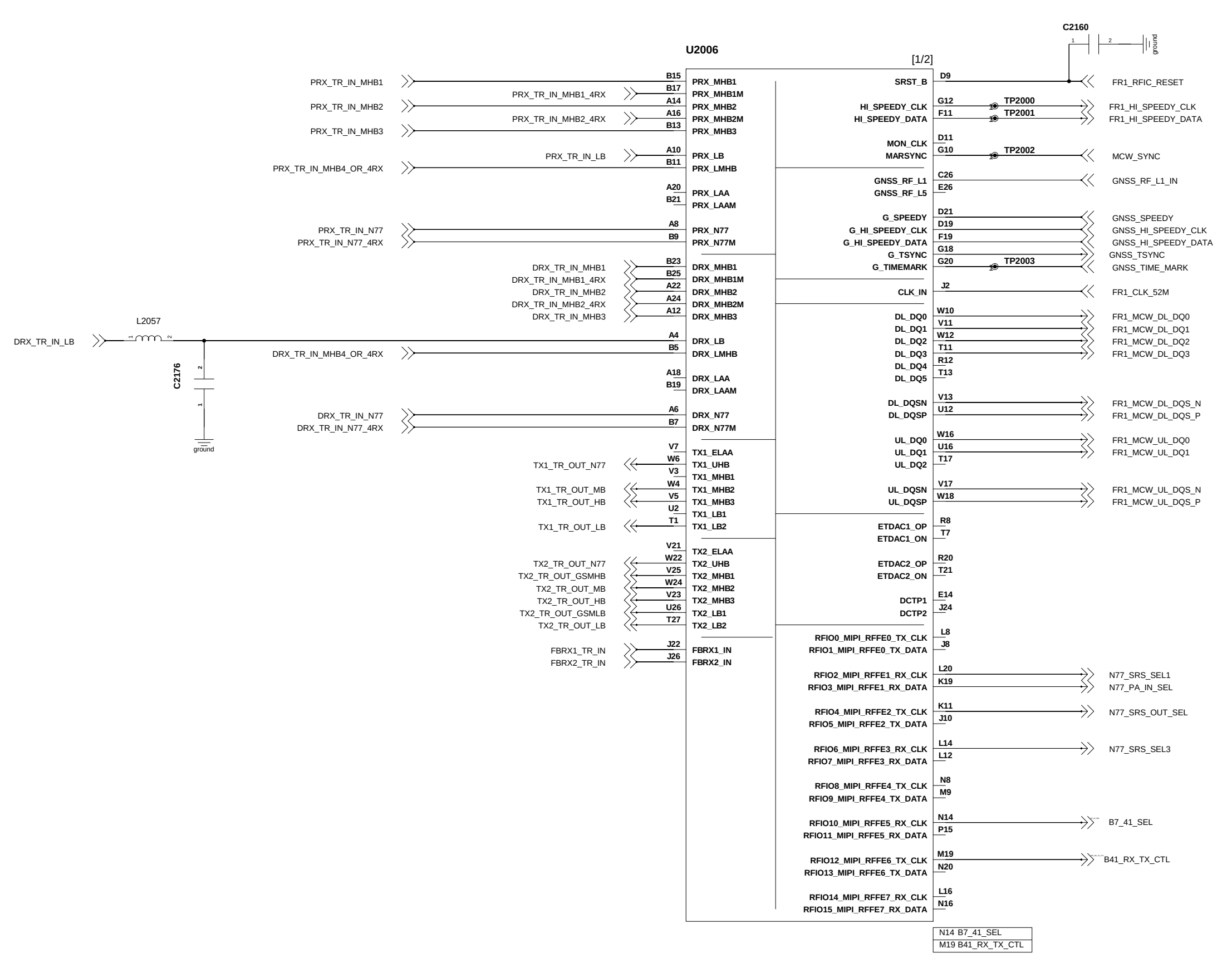

REV02 0926
RFS1002
Side
ANT1004
C1077
L1044
L1049
R1008
L1050
C1239
L1051
C1238
C1086
ANT1000
Side
ANT1002
C1240
C1246
C1096
L1045
L1046
L1052
F1006
ANT
CELL
GNSS
WLAN
L1047
L1048
L1053
GNSS_L1_LNA_IN
WIFI2_2G
U1007
COMMON
LB
HB
L1043
C1078
C1079
L1054
C1081
C1082
CPL_LB2_MHB2
C1075
B41_TRX
TX_PA_OUT_B7
TX_PA_OUT_B2_25
TX_PA_OUT_B3_B66
TX_PA_OUT_B1
U1009
CPL_IN
CPL_OUT
ANT_MB
ANT_HB
0908 REV01
SRS_ANT
TRX_MB1
TRX_MB2
TRX_HB1
TRX_HB2
AUX_MB1
AUX_MB2
AUX_HB1
AUX_HB2
DRX_B39
TRX_B30_1
DRX_LNA_IN_B39
DRX_LNA_IN_B30
SDATA_S
SCLK_S
SDATA_L
SCLK_L
VIO
VDD
RFFE4_DATA
RFFE4_CLK
RFFE3_DATA
RFFE3_CLK
L7M_VDDQ_1P8
@VRF_MIPI_IO_1P8
C1070
L25M_VDD_FR1_FEM_1P2
@VRF_LNA_1P2
C1071
MHB_RX_OUT1
MHB_RX_OUT2
MHB_RX_OUT3
MHB_RX_OUT4
C1211
C1229
C1214
DRX_TR_IN_MHB3
DRX_TR_IN_MHB1
DRX_TR_IN_MHB2
B41_RX
C1161
C1233
TRX_B30_1
C1232
F1018
TX_PA_OUT_B30_1
C1237
C1313
C1236
C1235
C1234
C1312
DRX_LNA_IN_B30
DRX_B39
C1307
C1308
F1010
L1300
C1309
DRX_LNA_IN_B39
US/CAN QPA5580
The others SKY58253
VBAT
@VPH_PWR
VPA_APT_OUT
L7M_VDDQ_1P8
@VRF_MIPI_IO_1P8
C1101
C1104
C1105
C1106
C1107
U1014
VBATT
VCC1
VCC2
RFIN_LB
RFIN_MB
RFIN_HB
RFOUT_LB1
RFOUT_LB2
RFOUT_LB3
RFOUT_LB4
RFOUT_LB5
RFOUT_LB6
RFOUT_MB1
RFOUT_MB2
RFOUT_MB3
RFOUT_HB1
RFOUT_HB2
TX2_TR_OUT_LB
L1072
C1120
TX1_TR_OUT_MB
L1083
C1134
TX1_TR_OUT_HB
L1084
C1136
L1087
C1141
TX_PA_OUT_B2_25
C1125
C1143
TX_PA_OUT_B3_B66
L1088
C1144
TX_PA_OUT_B1
TX_PA_OUT_B30_1
L1089
C1147
TX_PA_OUT_B7_41
ground

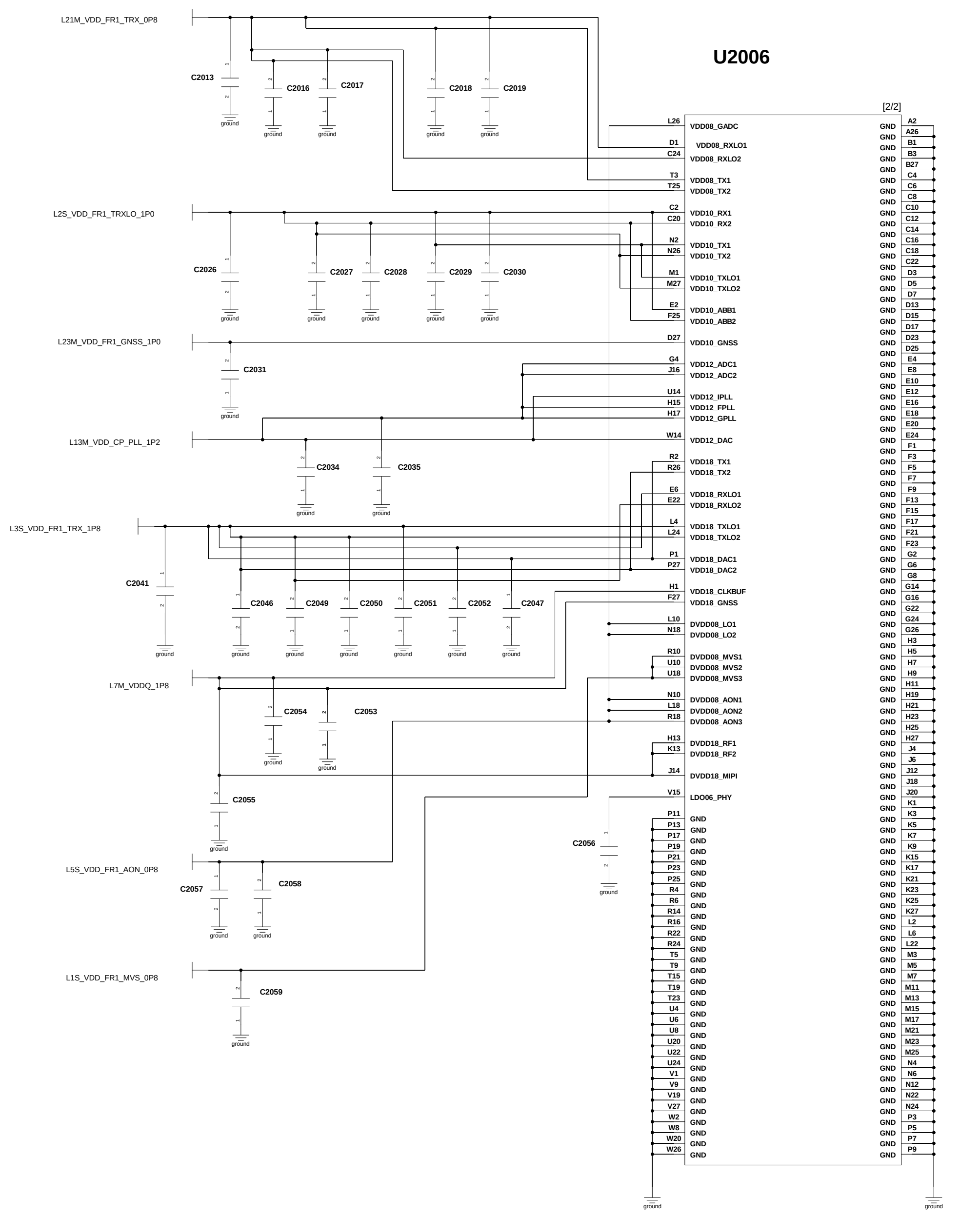

VRF_LNA_1P8
HPWR
B41
C1300
TX_PA_OUT_B7_41
B7_41_SEL
VDD
ANT
VCTL
RF1
RF2
GND
U1027
C1301
B7
TX_PA_OUT_B7
B41_RX
C1302
C1303
U1028
VRF_LNA_1P8
C1304
B41_TRX
B41_RX_TX_CTL
C1305
C1306
ground

Engineer:
Drawn by:
R&D CHK:
DOC CTRL CHK:
MFG ENGR CHK:
A35
TITLE:
RF PAM/FEM
Size:
Changed by:
Date Changed:
Time Changed:
QA CHK:
REV:
Drawing Number:

SAMSUNG CONFIDENTIAL
THIS DOCUMENT CONTAINS CONFIDENTIAL
PROPRIETARY INFORMATION THAT IS
SAMSUNG ELECTRONICS CO'S PROPERTY.
DO NOT DISCLOSE TO OR DUPLICATE FOR OTHERS
EXCEPT AS AUTHORIZED BY SAMSUNG.

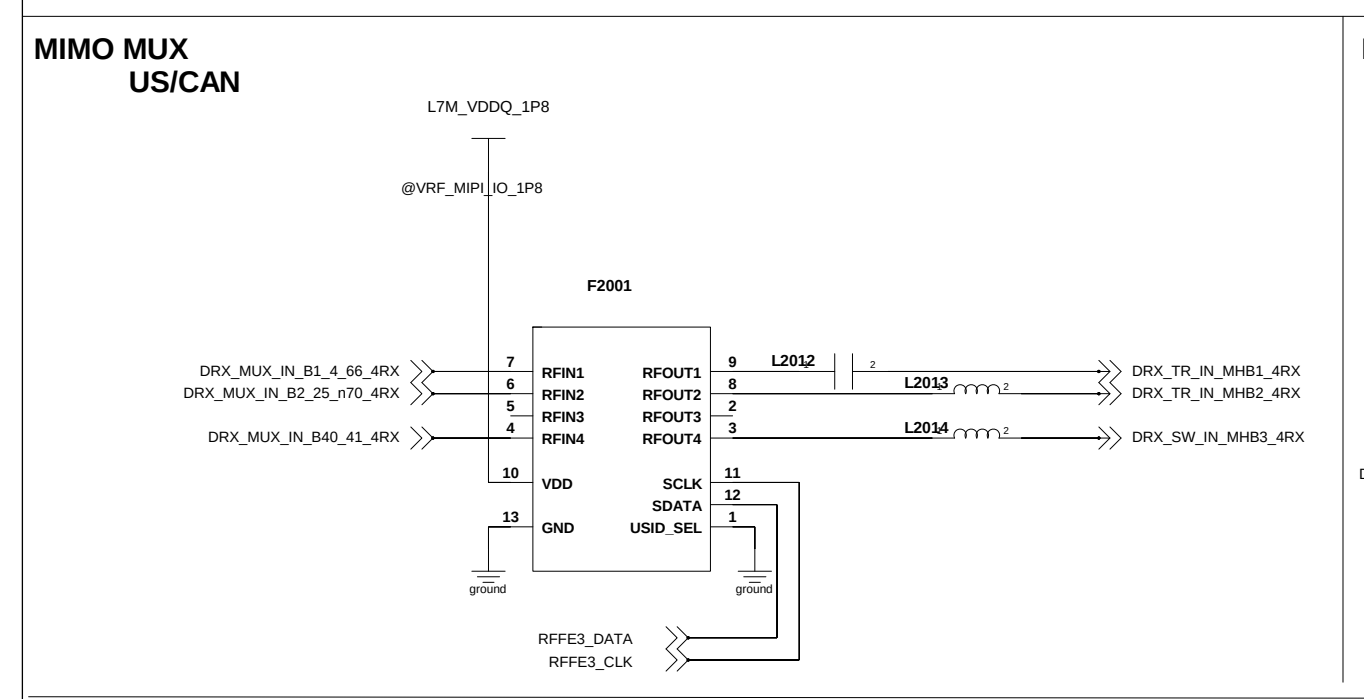
TRANSCEIVER (S5525)


U2006
[1/2]
C2160
PRX_TR_IN_MHB1
PRX_TR_IN_MHB1_4RX
PRX_TR_IN_MHB2
PRX_TR_IN_MHB2_4RX
PRX_TR_IN_MHB3
PRX_TR_IN_LB
PRX_TR_IN_MHB4_OR_4RX
PRX_TR_IN_N77
PRX_TR_IN_N77_4RX
DRX_TR_IN_MHB1
DRX_TR_IN_MHB1_4RX
DRX_TR_IN_MHB2
DRX_TR_IN_MHB2_4RX
DRX_TR_IN_MHB3
L2057
DRX_TR_IN_LB
C2176
DRX_TR_IN_MHB4_OR_4RX
DRX_TR_IN_N77
DRX_TR_IN_N77_4RX
TX1_TR_OUT_N77
TX1_TR_OUT_MB
TX1_TR_OUT_HB
TX1_TR_OUT_LB
TX2_TR_OUT_N77
TX2_TR_OUT_GSMHB
TX2_TR_OUT_MB
TX2_TR_OUT_HB
TX2_TR_OUT_GSMLB
TX2_TR_OUT_LB
FBRX1_TR_IN
FBRX2_TR_IN
PRX_MHB1
PRX_MHB1M
PRX_MHB2
PRX_MHB2M
PRX_MHB3
PRX_LB
PRX_LMHB
PRX_LAA
PRX_LAAM
PRX_N77
PRX_N77M
DRX_MHB1
DRX_MHB1M
DRX_MHB2
DRX_MHB2M
DRX_MHB3
DRX_LB
DRX_LMHB
DRX_LAA
DRX_LAAM
DRX_N77
DRX_N77M
TX1_ELAA
TX1_UHB
TX1_MHB1
TX1_MHB2
TX1_MHB3
TX1_LB1
TX1_LB2
TX2_ELAA
TX2_UHB
TX2_MHB1
TX2_MHB2
TX2_MHB3
TX2_LB1
TX2_LB2
FBRX1_IN
FBRX2_IN
SRST_B
HI_SPEEDY_CLK
HI_SPEEDY_DATA
MON_CLK
MARSYNC
GNSS_RF_L1
GNSS_RF_L5
G_SPEEDY
G_HI_SPEEDY_CLK
G_HI_SPEEDY_DATA
G_TSYNC
G_TIMEMARK
CLK_IN
DL_DQ0
DL_DQ1
DL_DQ2
DL_DQ3
DL_DQ4
DL_DQ5
DL_DQSN
DL_DQSP
UL_DQ0
UL_DQ1
UL_DQ2
UL_DQSN
UL_DQSP
ETDAC1_OP
ETDAC1_ON
ETDAC2_OP
ETDAC2_ON
DCTP1
DCTP2
RFIO0_MIPI_RFFE0_TX_CLK
RFIO1_MIPI_RFFE0_TX_DATA
RFIO2_MIPI_RFFE1_RX_CLK
RFIO3_MIPI_RFFE1_RX_DATA
RFIO4_MIPI_RFFE2_TX_CLK
RFIO5_MIPI_RFFE2_TX_DATA
RFIO6_MIPI_RFFE3_RX_CLK
RFIO7_MIPI_RFFE3_RX_DATA
RFIO8_MIPI_RFFE4_TX_CLK
RFIO9_MIPI_RFFE4_TX_DATA
RFIO10_MIPI_RFFE5_RX_CLK
RFIO11_MIPI_RFFE5_RX_DATA
RFIO12_MIPI_RFFE6_TX_CLK
RFIO13_MIPI_RFFE6_TX_DATA
RFIO14_MIPI_RFFE7_RX_CLK
RFIO15_MIPI_RFFE7_RX_DATA
TP2000
TP2001
TP2002
TP2003
FR1_RFIC_RESET
FR1_HI_SPEEDY_CLK
FR1_HI_SPEEDY_DATA
MCW_SYNC
GNSS_RF_L1_IN
GNSS_SPEEDY
GNSS_HI_SPEEDY_CLK
GNSS_HI_SPEEDY_DATA
GNSS_TSYNC
GNSS_TIME_MARK
FR1_CLK_52M
FR1_MCW_DL_DQ0
FR1_MCW_DL_DQ1
FR1_MCW_DL_DQ2
FR1_MCW_DL_DQ3
FR1_MCW_DL_DQS_N
FR1_MCW_DL_DQS_P
FR1_MCW_UL_DQ0
FR1_MCW_UL_DQ1
FR1_MCW_UL_DQS_N
FR1_MCW_UL_DQS_P
N77_SRS_SEL1
N77_PA_IN_SEL
N77_SRS_OUT_SEL
N77_SRS_SEL3
B7_41_SEL
B41_RX_TX_CTL
N14 B7_41_SEL
M19 B41_RX_TX_CTL


U2006
[2/2]
L21M_VDD_FR1_TRX_0P8
L2S_VDD_FR1_TRXLO_1P0
L23M_VDD_FR1_GNSS_1P0
L13M_VDD_CP_PLL_1P2
L3S_VDD_FR1_TRX_1P8
L7M_VDDQ_1P8
L5S_VDD_FR1_AON_0P8
L1S_VDD_FR1_MVS_0P8
C2013
C2016
C2017
C2018
C2019
C2026
C2027
C2028
C2029
C2030
C2031
C2034
C2035
C2041
C2046
C2049
C2050
C2051
C2052
C2047
C2054
C2053
C2055
C2056
C2057
C2058
C2059
VDD08_GADC
VDD08_RXLO1
VDD08_RXLO2
VDD08_TX1
VDD08_TX2
VDD10_RX1
VDD10_RX2
VDD10_TX1
VDD10_TX2
VDD10_TXLO1
VDD10_TXLO2
VDD10_ABB1
VDD10_ABB2
VDD10_GNSS
VDD12_ADC1
VDD12_ADC2
VDD12_IPLL
VDD12_FPLL
VDD12_GPLL
VDD12_DAC
VDD18_TX1
VDD18_TX2
VDD18_RXLO1
VDD18_RXLO2
VDD18_TXLO1
VDD18_TXLO2
VDD18_DAC1
VDD18_DAC2
VDD18_CLKBUF
VDD18_GNSS
DVDD08_LO1
DVDD08_LO2
DVDD08_MVS1
DVDD08_MVS2
DVDD08_MVS3
DVDD08_AON1
DVDD08_AON2
DVDD08_AON3
DVDD18_RF1
DVDD18_RF2
DVDD18_MIPI
LDO06_PHY
GND
ground

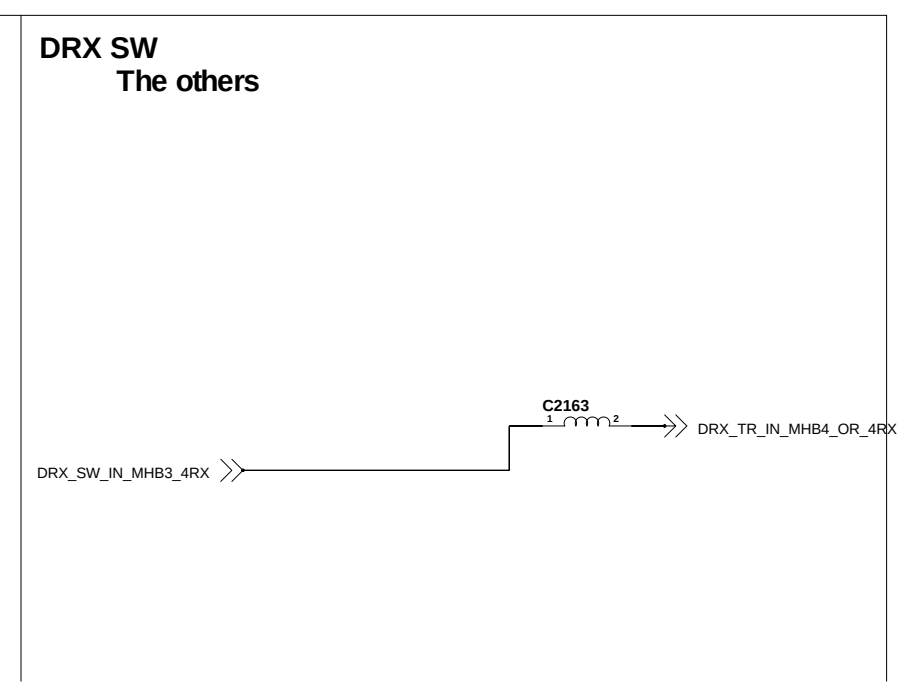
MIMO MUX

US/CAN

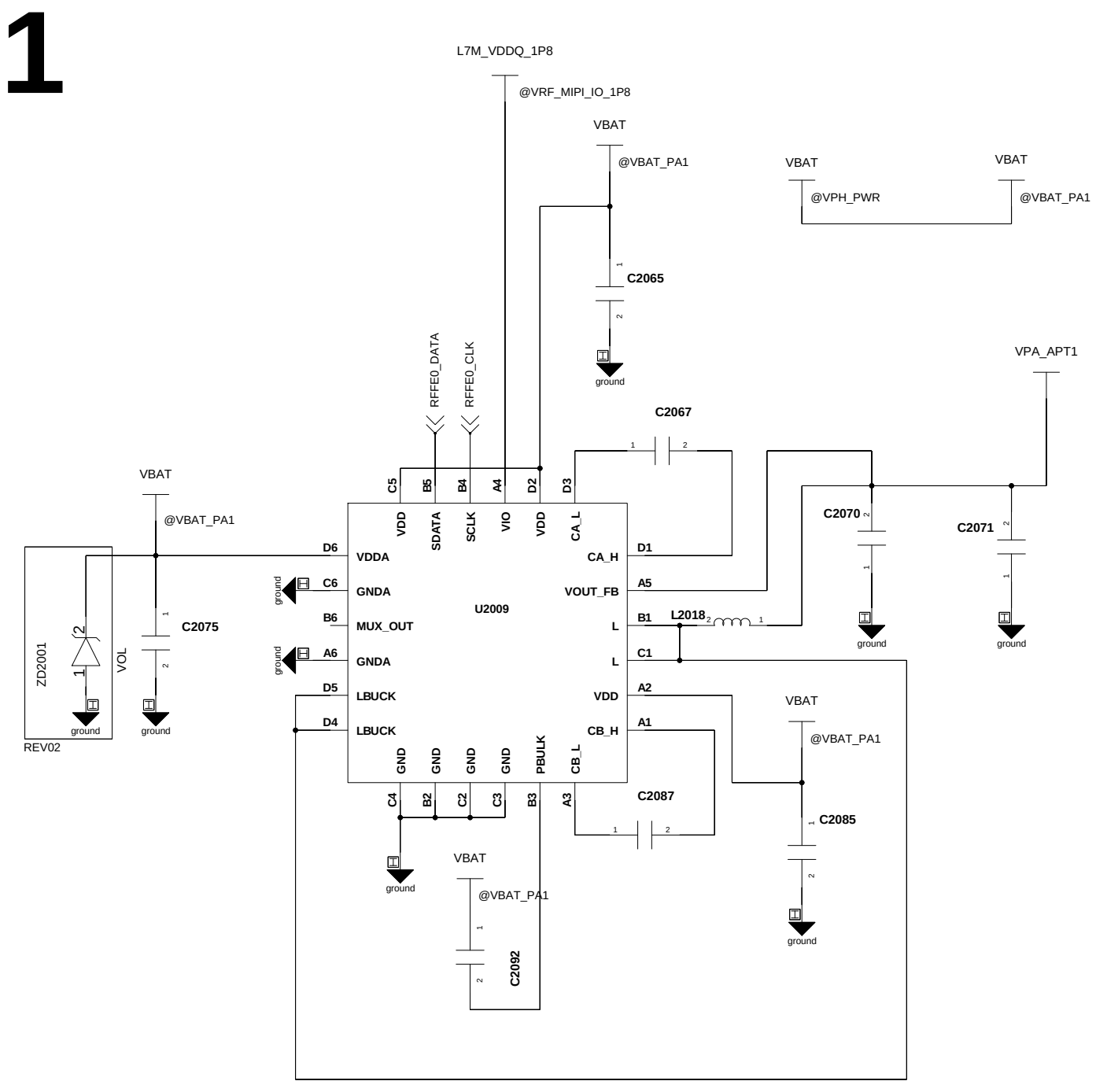

L7M_VDDQ_1P8
@VRF_MIPI_IO_1P8
F2001
DRX_MUX_IN_B1_4_66_4RX
DRX_MUX_IN_B2_25_n70_4RX
DRX_MUX_IN_B40_41_4RX
RFIN1
RFIN2
RFIN3
RFIN4
VDD
GND
RFOUT1
RFOUT2
RFOUT3
RFOUT4
SCLK
SDATA
USID_SEL
L2012
L2013
L2014
DRX_TR_IN_MHB1_4RX
DRX_TR_IN_MHB2_4RX
DRX_SW_IN_MHB3_4RX
RFFE3_DATA
RFFE3_CLK
ground

DRX SW

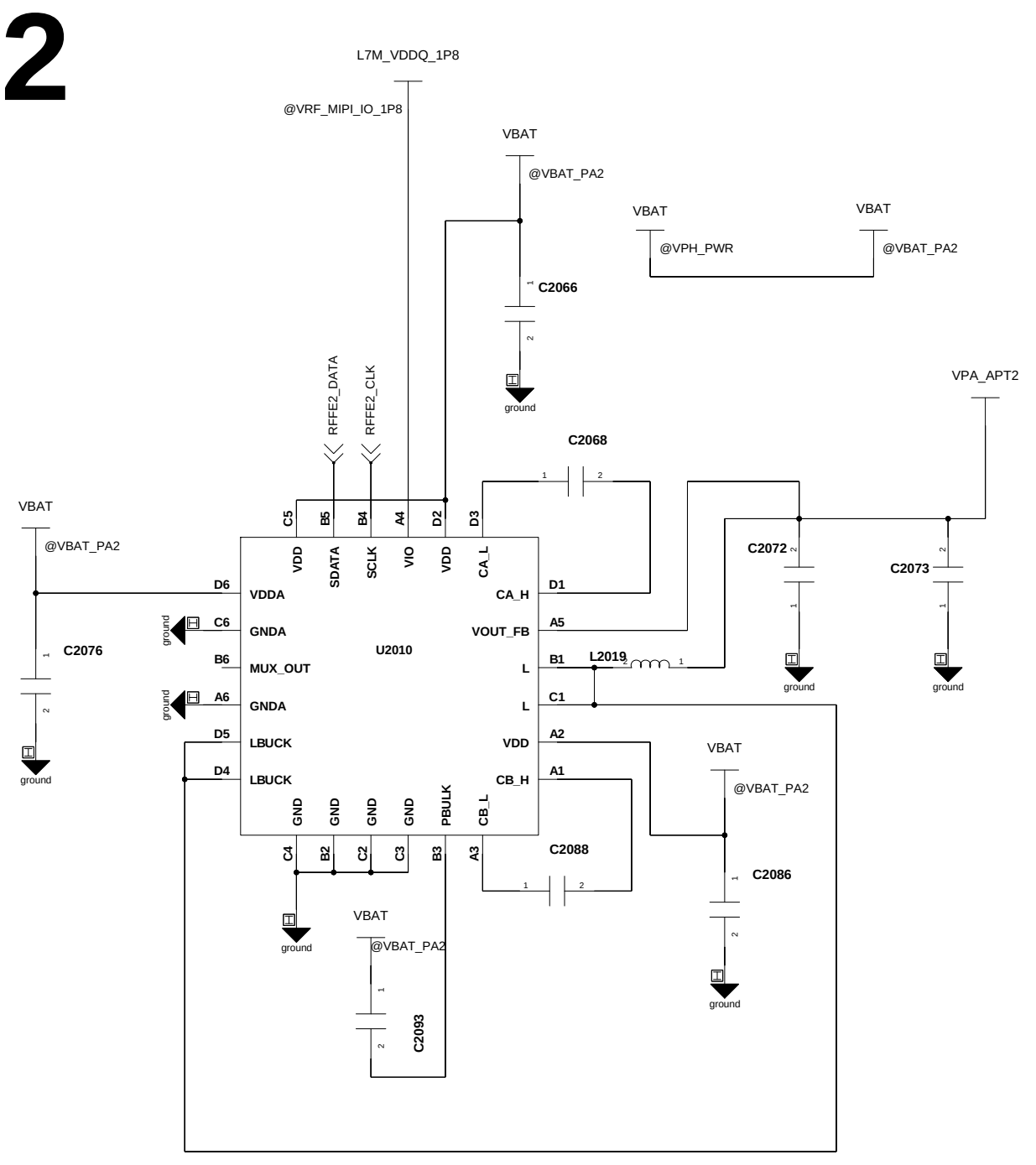
The others


C2163
DRX_SW_IN_MHB3_4RX
DRX_TR_IN_MHB4_OR_4RX

APT1

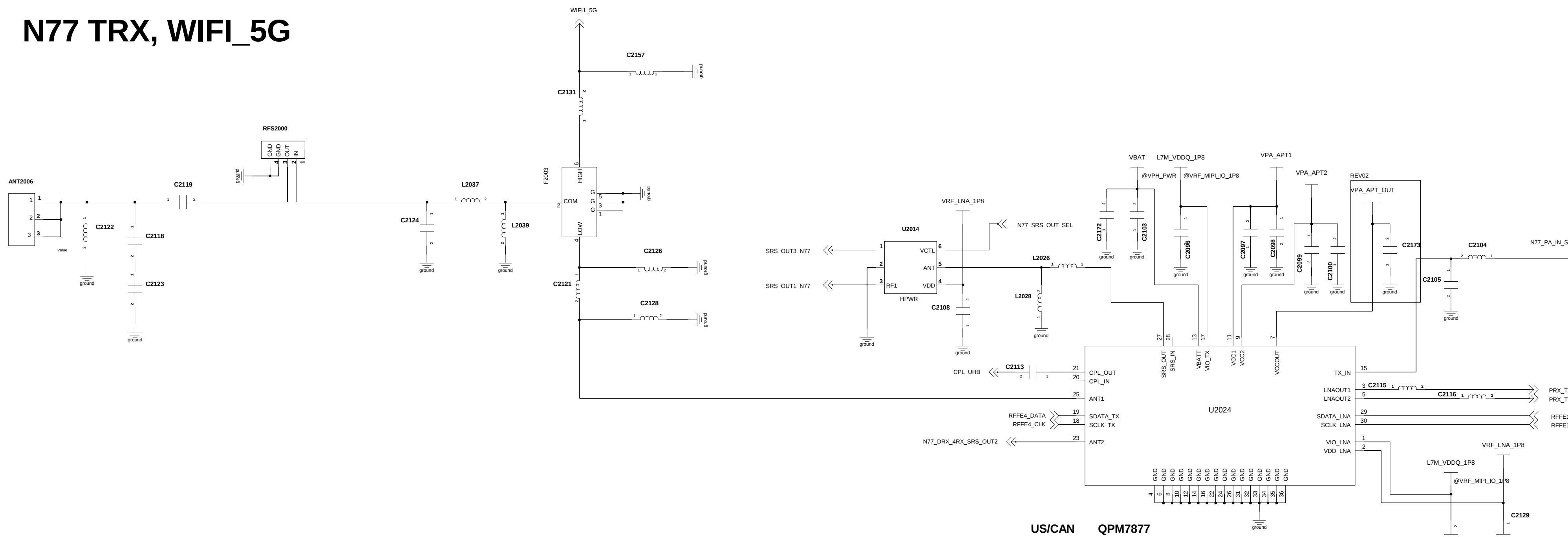

L7M_VDDQ_1P8
@VRF_MIPI_IO_1P8
VBAT
@VBAT_PA1
@VPH_PWR
C2065
RFFE0_DATA
RFFE0_CLK
C2067
VPA_APT1
C2070
C2071
C2075
ZD2001
VOL
REV02
U2009
VDDA
GNDA
MUX_OUT
LBUCK
VDD
SDATA
SCLK
VIO
CA_L
CA_H
VOUT_FB
L
CB_H
CB_L
PBULK
L2018
C2087
C2085
C2092
ground

APT2


L7M_VDDQ_1P8
@VRF_MIPI_IO_1P8
VBAT
@VBAT_PA2
@VPH_PWR
C2066
RFFE2_DATA
RFFE2_CLK
C2068
VPA_APT2
C2072
C2073
C2076
U2010
VDDA
GNDA
MUX_OUT
LBUCK
VDD
SDATA
SCLK
VIO
CA_L
CA_H
VOUT_FB
L
CB_H
CB_L
PBULK
L2019
C2088
C2086
C2093
ground

SUB ANT3

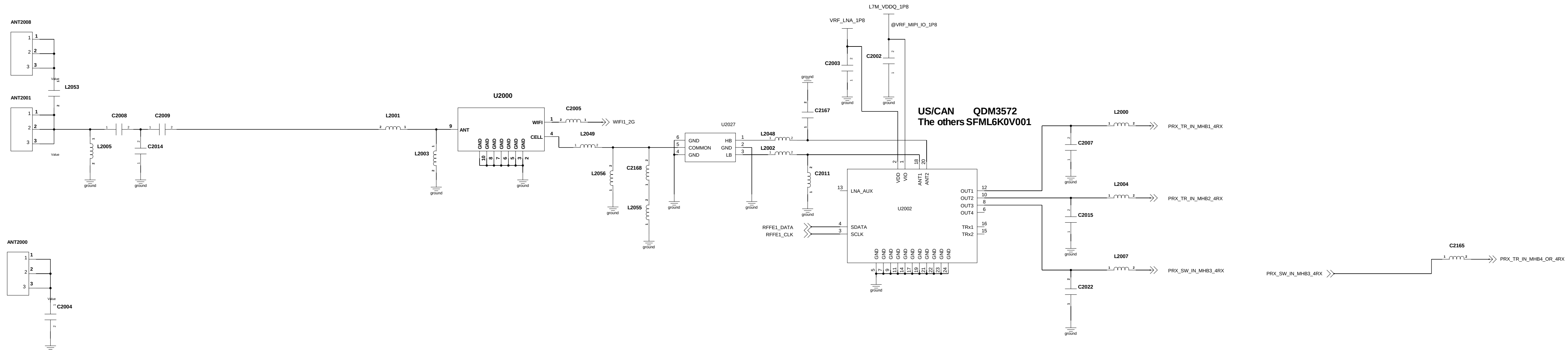
N77 TRX, WIFI_5G


WIFI1_5G
C2157
C2131
RFS2000
ANT2006
C2119
C2122
C2118
C2123
C2124
L2037
L2039
F2003
HIGH
COM
LOW
C2121
C2126
C2128
U2014
VCTL
ANT
RF1
VDD
HPWR
VRF_LNA_1P8
N77_SRS_OUT_SEL
SRS_OUT3_N77
SRS_OUT1_N77
L2026
L2028
C2108
VBAT
@VPH_PWR
L7M_VDDQ_1P8
@VRF_MIPI_IO_1P8
VPA_APT1
VPA_APT2
REV02
VPA_APT_OUT
C2172
C2103
C2096
C2097
C2098
C2099
C2100
C2173
C2104
C2105
U2024
CPL_OUT
CPL_IN
ANT1
ANT2
SDATA_TX
SCLK_TX
SRS_OUT
SRS_IN
VBATT
VIO_TX
VCC1
VCC2
VCCOUT
TX_IN
LNAOUT1
LNAOUT2
SDATA_LNA
SCLK_LNA
VIO_LNA
VDD_LNA
CPL_UHB
C2113
RFFE4_DATA
RFFE4_CLK
N77_DRX_4RX_SRS_OUT2
C2115
C2116
PRX_TR_IN_N77
PRX_TR_IN_N77_4RX
RFFE1_DATA
RFFE1_CLK
C2132
C2129
U2013
N77_PA_IN_SEL
RF2
RF1
GND
LPWR
L2025
L2027
C2161
C2162
C2117
TX2_TR_OUT_N77
TX1_TR_OUT_N77
ground

US/CAN QPM7877
The others SKY58261

SUB ANT5

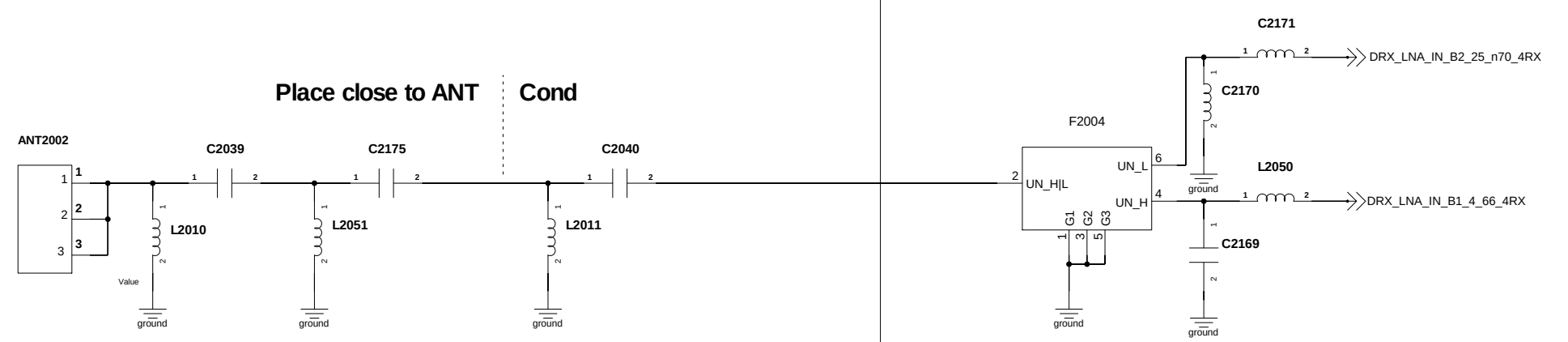
MHB PRX MIMO, WIFI1_2.4G


ANT2008
ANT2001
ANT2000
L2053
C2008
C2009
L2005
C2014
C2004
L2001
L2003
U2000
ANT
WIFI
CELL
C2005
WIFI1_2G
L2049
L2056
C2168
L2055
U2027
GND
COMMON
HB
LB
L2048
L2002
C2167
C2011
VRF_LNA_1P8
L7M_VDDQ_1P8
@VRF_MIPI_IO_1P8
C2003
C2002
U2002
VDD
VIO
ANT1
ANT2
LNA_AUX
SDATA
SCLK
OUT1
OUT2
OUT3
OUT4
TRx1
TRx2
RFFE1_DATA
RFFE1_CLK
L2000
L2004
L2007
C2007
C2015
C2022
PRX_TR_IN_MHB1_4RX
PRX_TR_IN_MHB2_4RX
PRX_SW_IN_MHB3_4RX
PRX_SW_IN_MHB3_4RX
C2165
PRX_TR_IN_MHB4_OR_4RX
ground

US/CAN QDM3572
The others SFML6K0V001

SUB ANT6

MB DRX MIMO

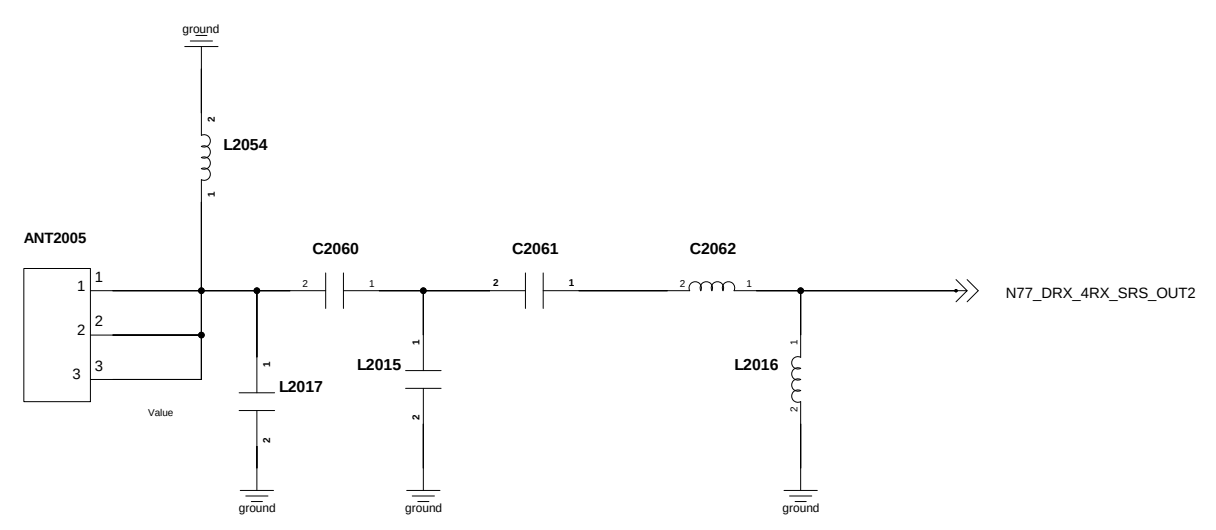

ANT2002
C2039
C2175
C2040
L2010
L2051
L2011
F2004
UN_H|L
UN_L
UN_H
C2171
C2170
L2050
C2169
DRX_LNA_IN_B2_25_n70_4RX
DRX_LNA_IN_B1_4_66_4RX
ground

Place close to ANT Cond

The others US/CAN

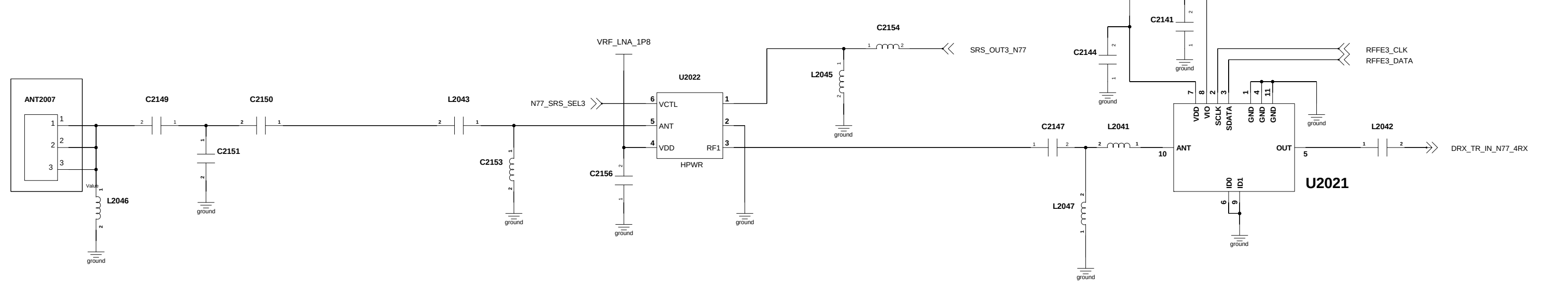
SUB ANT7

N77 PRX MIMO


ANT2005
L2054
C2060
C2061
C2062
L2017
L2015
L2016
N77_DRX_4RX_SRS_OUT2
ground

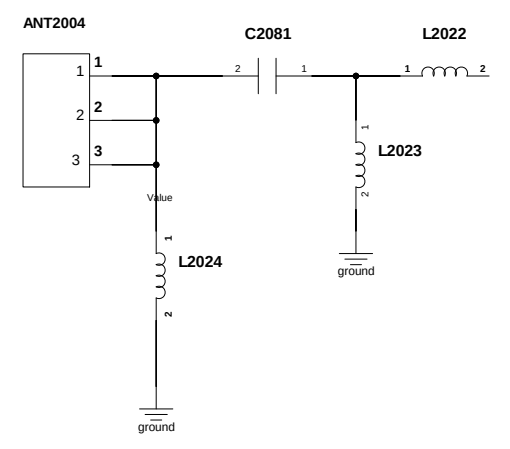
SUB ANT8

N77_DRX_MIMO


ANT2007
C2149
C2150
C2151
L2046
L2043
C2153
VRF_LNA_1P8
N77_SRS_SEL3
U2022
VCTL
ANT
VDD
RF1
HPWR
C2156
C2154
L2045
SRS_OUT3_N77
C2147
L2041
L2047
L7M_VDDQ_1P8
@VRF_MIPI_IO_1P8
C2144
C2141
RFFE3_CLK
RFFE3_DATA
U2021
ANT
OUT
VDD
VIO
SCLK
SDATA
L2042
DRX_TR_IN_N77_4RX
ground

SUB ANT10

N79_DRX_MIMO(CHN/JPN ONLY)


ANT2004
C2081
L2022
L2023
L2024
ground

Engineer:
Drawn by:
A35
R&D CHK:
TITLE: RF TRCV/PAM/APT
Size:
DOC CTRL CHK:
MFG ENGR CHK:
Changed by:
Date Changed:
Time Changed:
QA CHK:
REV:
Drawing Number:

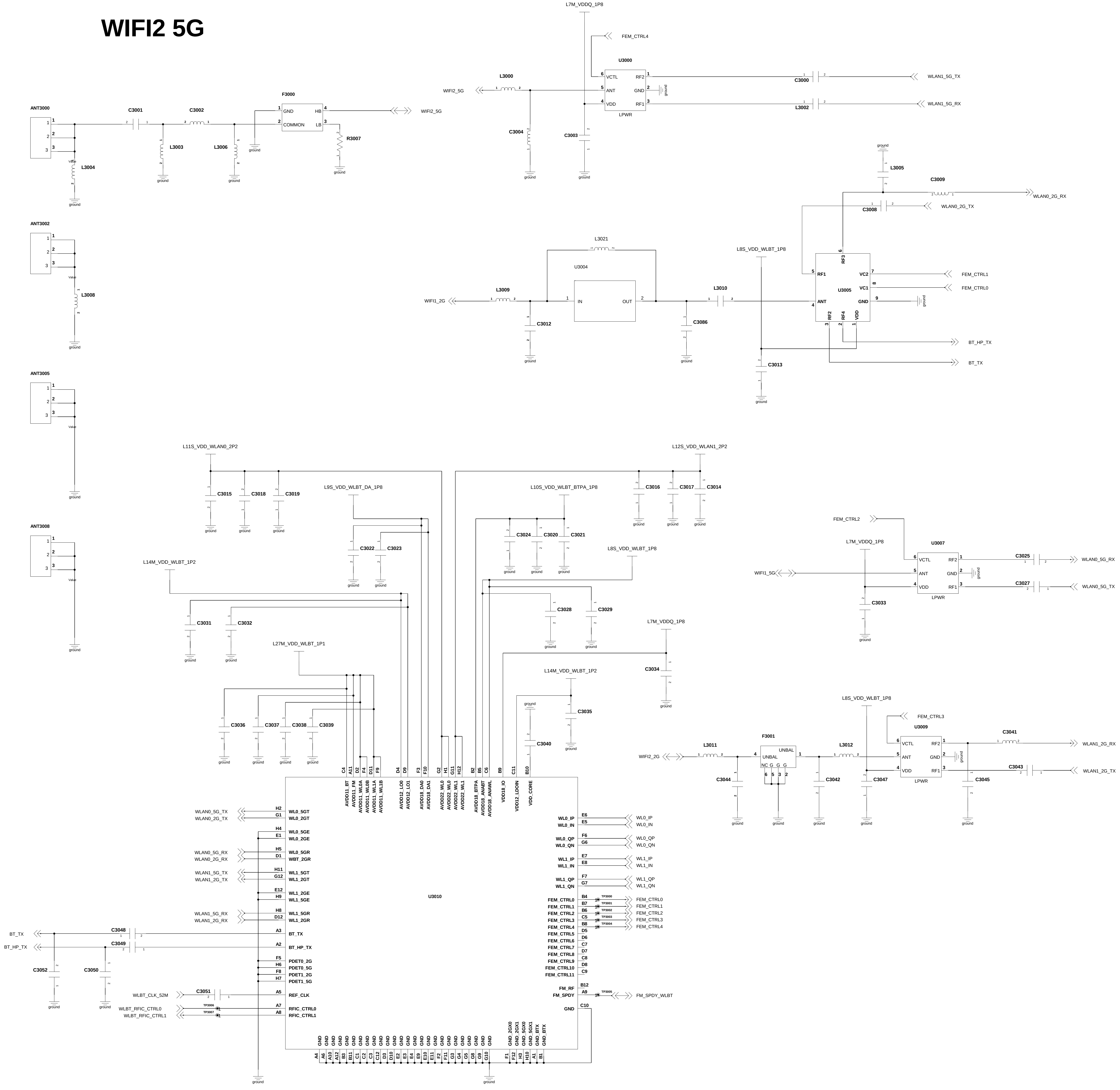
SAMSUNG CONFIDENTIAL
THIS DOCUMENT CONTAINS CONFIDENTIAL
PROPRIETARY INFORMATION THAT IS
SAMSUNG ELECTRONICS CO'S PROPERTY.
DO NOT DISCLOSE TO OR DUPLICATE FOR OTHERS
EXCEPT AS AUTHORIZED BY SAMSUNG.

SUB ANT9

WIFI2 5G

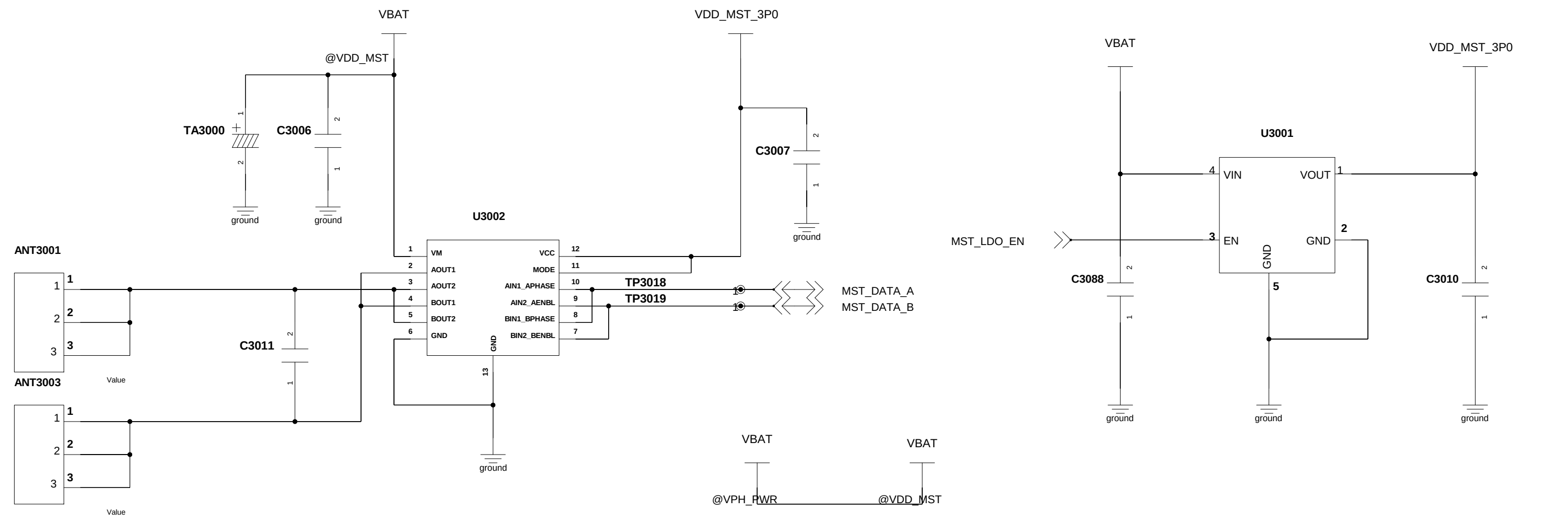
WIFI/BT/FM

WIFI 2.4G - sub5/sub2
WIFI 5G - sub3/sub9

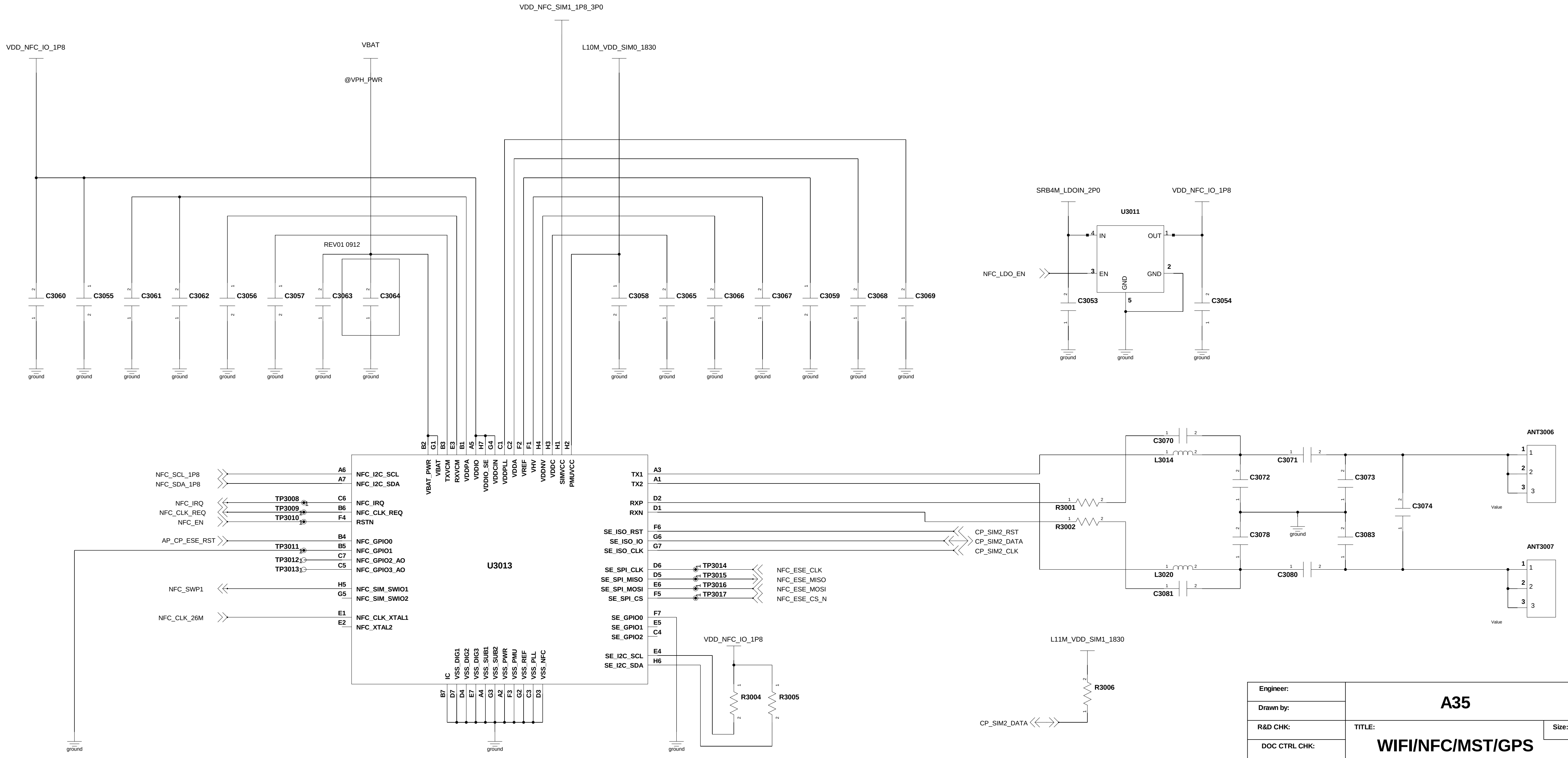

ANT3000
C3001
C3002
F3000
GND
COMMON
HB
LB
WIFI2_5G
R3007
L3003
L3006
L3004
ANT3002
L3008
ANT3005
ANT3008
Value
ground
L7M_VDDQ_1P8
FEM_CTRL4
U3000
VCTL
ANT
VDD
RF2
GND
RF1
LPWR
L3000
C3004
C3003
C3000
L3002
WLAN1_5G_TX
WLAN1_5G_RX
L3021
U3004
IN
OUT
L3009
WIFI1_2G
C3012
C3086
L3010
L8S_VDD_WLBT_1P8
C3013
L3005
C3009
C3008
WLAN0_2G_RX
WLAN0_2G_TX
U3005
RF3
VC2
VC1
RF4
FEM_CTRL1
FEM_CTRL0
BT_HP_TX
BT_TX
L11S_VDD_WLAN0_2P2
C3015
C3018
C3019
L9S_VDD_WLBT_DA_1P8
C3022
C3023
L10S_VDD_WLBT_BTPA_1P8
C3024
C3020
C3021
L12S_VDD_WLAN1_2P2
C3016
C3017
C3014
L8S_VDD_WLBT_1P8
C3028
C3029
L7M_VDDQ_1P8
C3034
L14M_VDD_WLBT_1P2
C3031
C3032
L27M_VDD_WLBT_1P1
C3036
C3037
C3038
C3039
L14M_VDD_WLBT_1P2
C3035
C3040
FEM_CTRL2
U3007
WIFI1_5G
C3025
C3027
WLAN0_5G_RX
WLAN0_5G_TX
C3033
FEM_CTRL3
U3009
F3001
UNBAL
NC G G G
L3011
L3012
WIFI2_2G
C3044
C3042
C3047
C3041
C3043
C3045
WLAN1_2G_RX
WLAN1_2G_TX
U3010
AVDD11_BT
AVDD11_FM
AVDD11_WL0A
AVDD11_WL0B
AVDD11_WL1A
AVDD11_WL1B
AVDD12_LO0
AVDD12_LO1
AVDD18_DA0
AVDD18_DA1
AVDD22_WL0
AVDD22_WL0
AVDD22_WL1
AVDD22_WL1
AVDD18_BTPA
AVDD18_ANABT
AVDD18_ANAWL
VDD18_IO
VDD12_LDOIN
VDD_CORE
WLAN0_5G_TX
WLAN0_2G_TX
WL0_5GT
WL0_2GT
WL0_5GE
WL0_2GE
WLAN0_5G_RX
WLAN0_2G_RX
WL0_5GR
WBT_2GR
WLAN1_5G_TX
WLAN1_2G_TX
WL1_5GT
WL1_2GT
WL1_2GE
WL1_5GE
WLAN1_5G_RX
WLAN1_2G_RX
WL1_5GR
WL1_2GR
BT_TX
BT_HP_TX
C3048
C3049
C3052
C3050
PDET0_2G
PDET0_5G
PDET1_2G
PDET1_5G
WLBT_CLK_52M
C3051
REF_CLK
WLBT_RFIC_CTRL0
WLBT_RFIC_CTRL1
TP3006
TP3007
RFIC_CTRL0
RFIC_CTRL1
WL0_IP
WL0_IN
WL0_QP
WL0_QN
WL1_IP
WL1_IN
WL1_QP
WL1_QN
FEM_CTRL0
FEM_CTRL1
FEM_CTRL2
FEM_CTRL3
FEM_CTRL4
FEM_CTRL5
FEM_CTRL6
FEM_CTRL7
FEM_CTRL8
FEM_CTRL9
FEM_CTRL10
FEM_CTRL11
TP3000
TP3001
TP3002
TP3003
TP3004
FM_RF
FM_SPDY
TP3005
FM_SPDY_WLBT
GND
GND_2GX0
GND_2GX1
GND_5GX0
GND_5GX1
GND_BTX
GND_BTX

MST

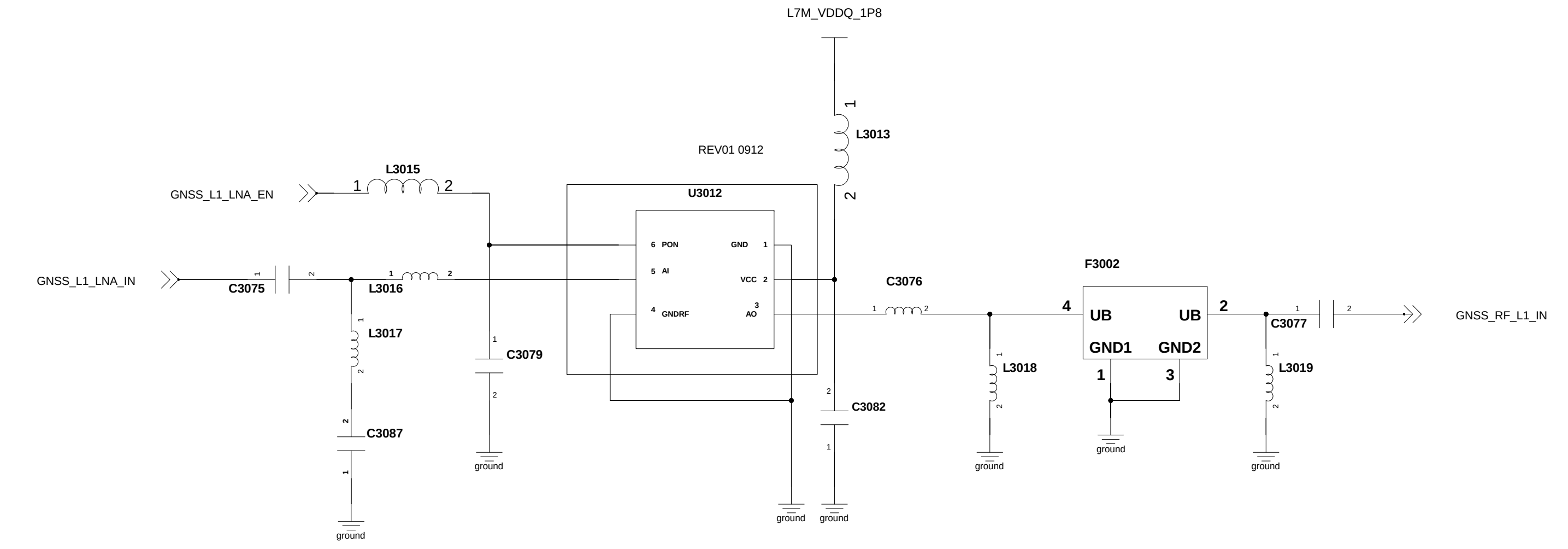
(KOR only)


VBAT
@VDD_MST
TA3000
C3006
ANT3001
ANT3003
C3011
U3002
VM
AOUT1
AOUT2
BOUT1
BOUT2
GND
VCC
MODE
AIN1_APHASE
AIN2_AENBL
BIN1_BPHASE
BIN2_BENBL
VDD_MST_3P0
C3007
TP3018
TP3019
MST_DATA_A
MST_DATA_B
VBAT
@VPH_PWR
VBAT
@VDD_MST
VBAT
U3001
VIN
VOUT
EN
GND
MST_LDO_EN
C3088
VDD_MST_3P0
C3010
ground

NFC


VDD_NFC_IO_1P8
VBAT
@VPH_PWR
VDD_NFC_SIM1_1P8_3P0
L10M_VDD_SIM0_1830
REV01 0912
C3060
C3055
C3061
C3062
C3056
C3057
C3063
C3064
C3058
C3065
C3066
C3067
C3059
C3068
C3069
U3013
VBAT_PWR
VBAT
TXVCM
RXVCM
VDDPA
VDDIO
VDDIO_SE
VDDCIN
VDDPLL
VDDA
VREF
VHV
VDDNV
VDDC
SIMVCC
PMUVCC
NFC_SCL_1P8
NFC_SDA_1P8
NFC_I2C_SCL
NFC_I2C_SDA
NFC_IRQ
NFC_CLK_REQ
NFC_EN
TP3008
TP3009
TP3010
RSTN
AP_CP_ESE_RST
TP3011
TP3012
TP3013
NFC_GPIO0
NFC_GPIO1
NFC_GPIO2_AO
NFC_GPIO3_AO
NFC_SWP1
NFC_SIM_SWIO1
NFC_SIM_SWIO2
NFC_CLK_26M
NFC_CLK_XTAL1
NFC_XTAL2
IC
VSS_DIG1
VSS_DIG2
VSS_DIG3
VSS_SUB1
VSS_SUB2
VSS_PWR
VSS_PMU
VSS_REF
VSS_PLL
VSS_NFC
TX1
TX2
RXP
RXN
SE_ISO_RST
SE_ISO_IO
SE_ISO_CLK
CP_SIM2_RST
CP_SIM2_DATA
CP_SIM2_CLK
SE_SPI_CLK
SE_SPI_MISO
SE_SPI_MOSI
SE_SPI_CS
TP3014
TP3015
TP3016
TP3017
NFC_ESE_CLK
NFC_ESE_MISO
NFC_ESE_MOSI
NFC_ESE_CS_N
SE_GPIO0
SE_GPIO1
SE_GPIO2
SE_I2C_SCL
SE_I2C_SDA
VDD_NFC_IO_1P8
R3004
R3005
L11M_VDD_SIM1_1830
R3006
CP_SIM2_DATA
SRB4M_LDOIN_2P0
VDD_NFC_IO_1P8
U3011
IN
OUT
EN
GND
NFC_LDO_EN
C3053
C3054
C3070
L3014
C3071
C3072
C3073
C3074
R3001
R3002
C3078
C3083
L3020
C3080
C3081
ANT3006
ANT3007
Value
ground

GPS L1 / GNSS/BEIDOU


L7M_VDDQ_1P8
L3013
REV01 0912
U3012
PON
AI
GNDRF
GND
VCC
AO
L3015
GNSS_L1_LNA_EN
GNSS_L1_LNA_IN
C3075
L3016
L3017
C3087
C3079
C3082
C3076
L3018
F3002
UB
UB
GND1
GND2
C3077
L3019
GNSS_RF_L1_IN
ground

Engineer:
Drawn by:
R&D CHK:
DOC CTRL CHK:
MFG ENGR CHK:
QA CHK:

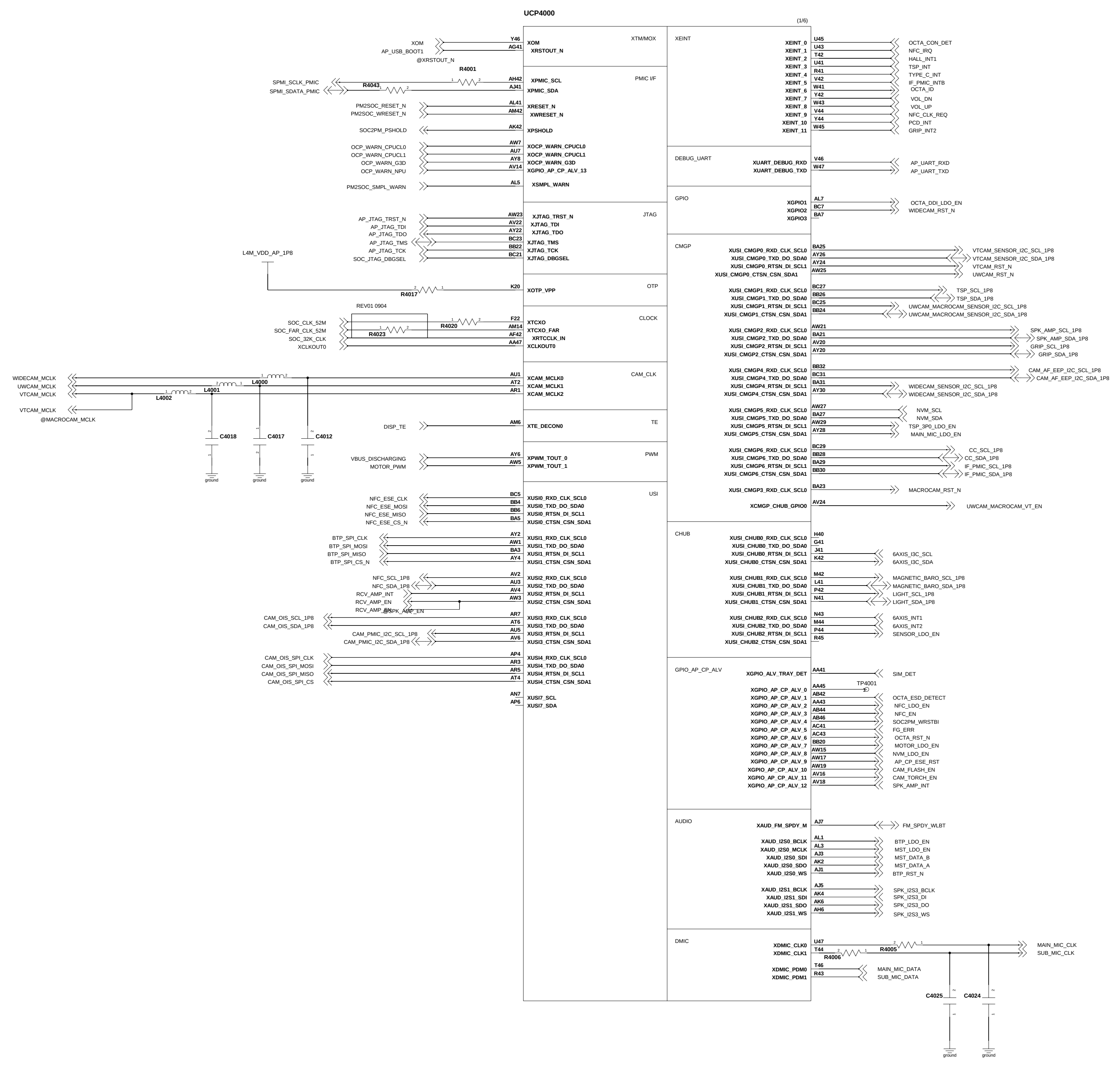
A35

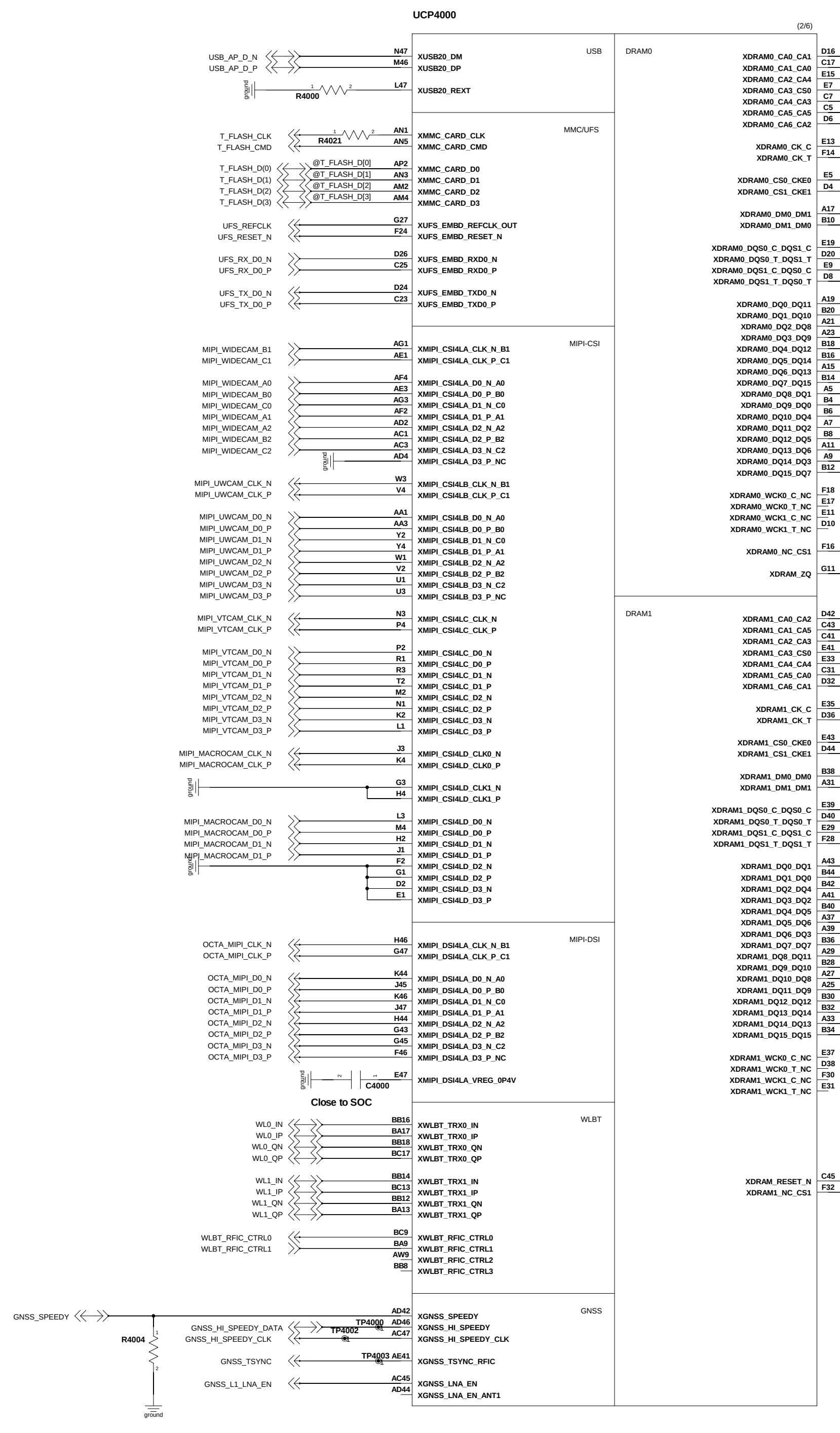
TITLE:
WIFI/NFC/MST/GPS
Size:

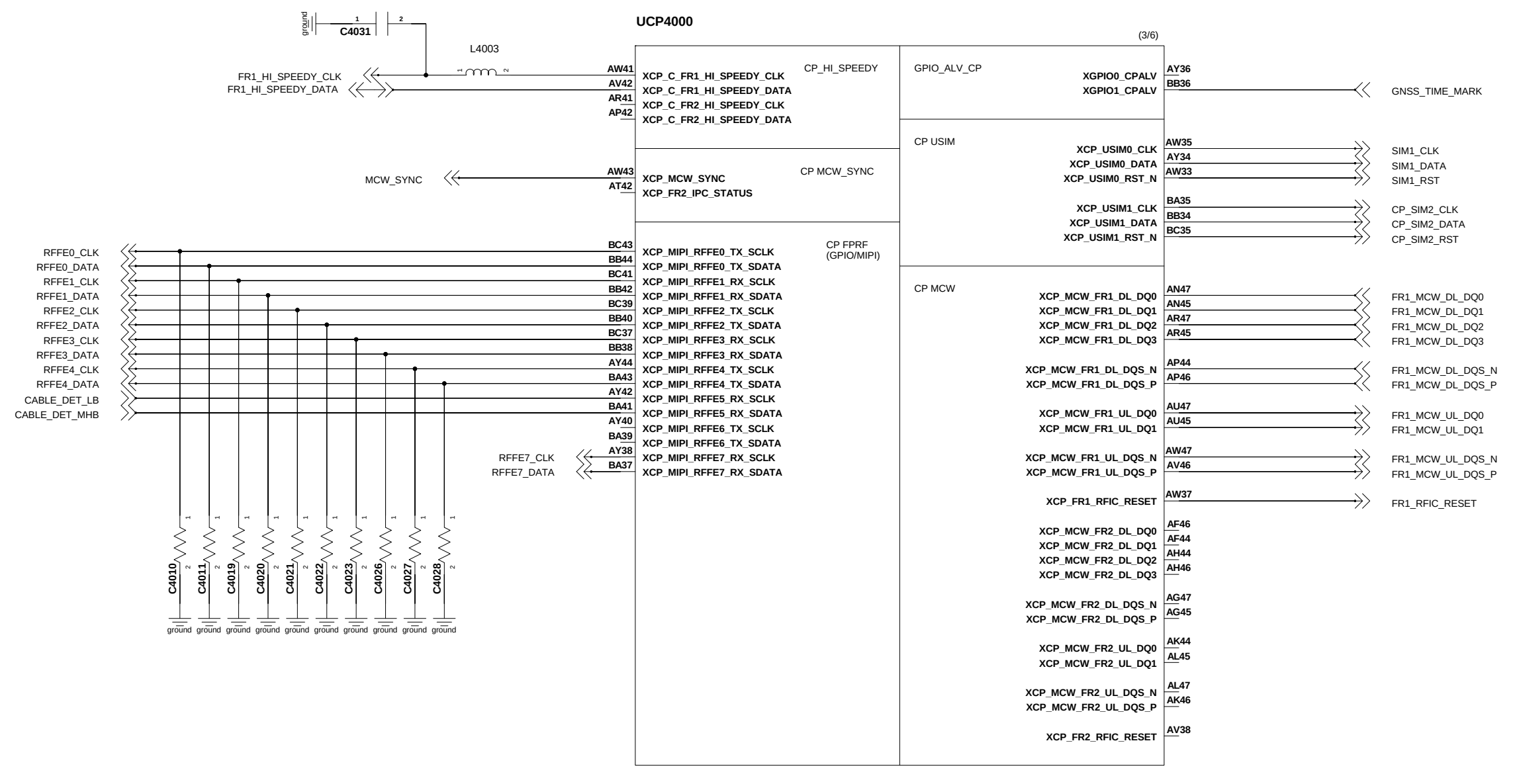
Changed by: Date Changed: Time Changed:
REV: Drawing Number: Page:

SAMSUNG CONFIDENTIAL
THIS DOCUMENT CONTAINS CONFIDENTIAL
PROPRIETARY INFORMATION THAT IS
SAMSUNG ELECTRONICS CO'S PROPERTY.
DO NOT DISCLOSE TO OR DUPLICATE FOR OTHERS
EXCEPT AS AUTHORIZED BY SAMSUNG.

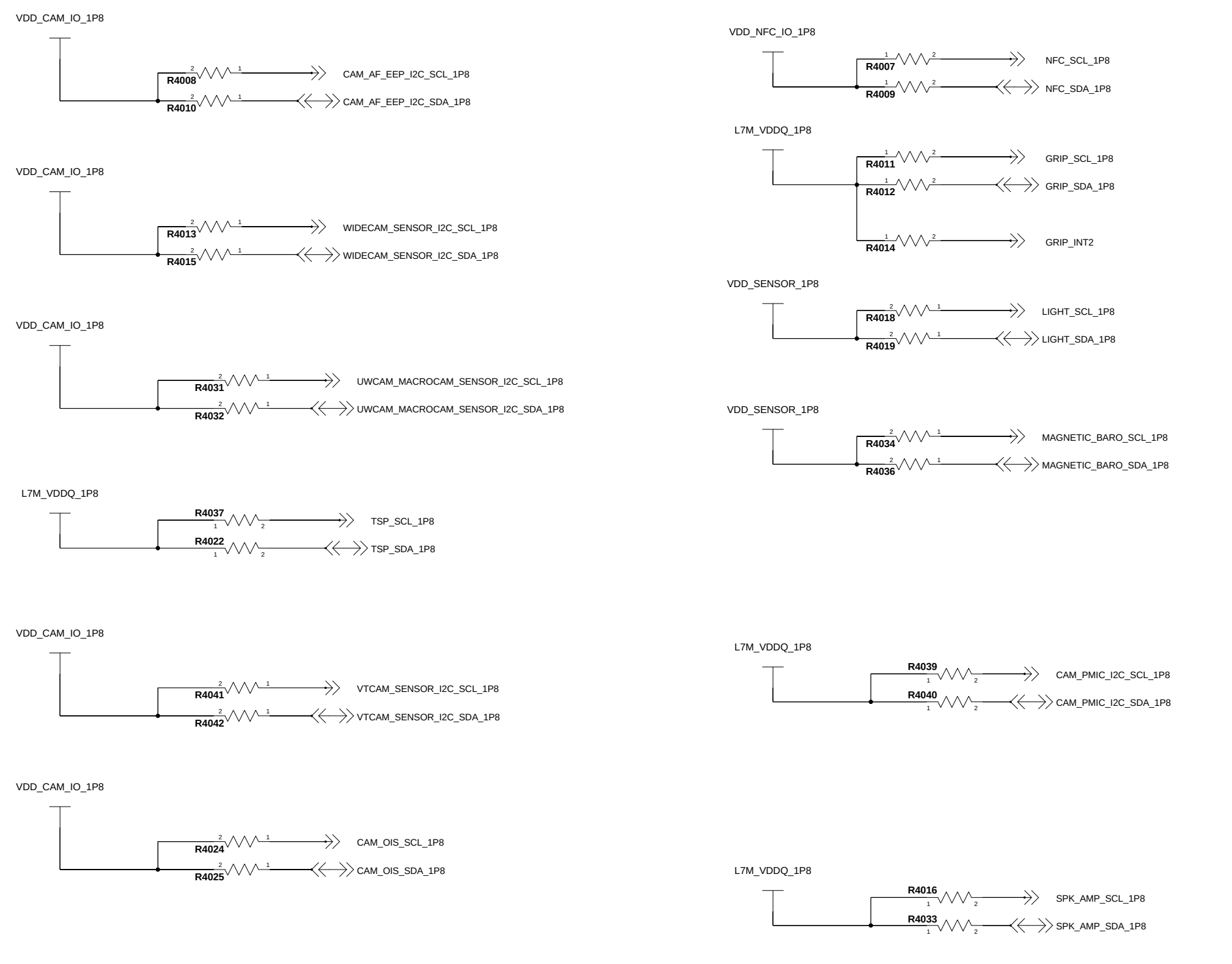
Quartz

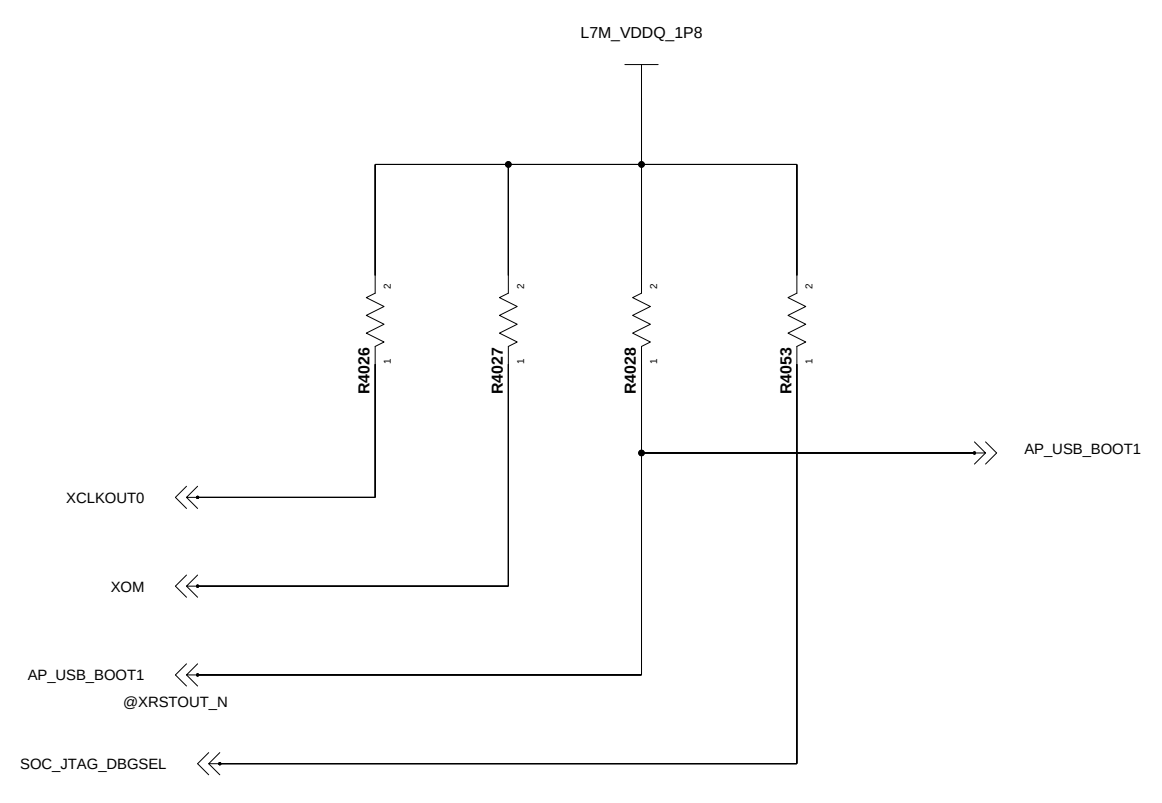

UCP4000
(1/6)
XTM/MOX
XOM
AP_USB_BOOT1
@XRSTOUT_N
XRSTOUT_N
PMIC I/F
R4001
R4043
SPMI_SCLK_PMIC
SPMI_SDATA_PMIC
XPMIC_SCL
XPMIC_SDA
PM2SOC_RESET_N
PM2SOC_WRESET_N
XRESET_N
XWRESET_N
SOC2PM_PSHOLD
XPSHOLD
OCP_WARN_CPUCL0
OCP_WARN_CPUCL1
OCP_WARN_G3D
OCP_WARN_NPU
XOCP_WARN_CPUCL0
XOCP_WARN_CPUCL1
XOCP_WARN_G3D
XGPIO_AP_CP_ALV_13
PM2SOC_SMPL_WARN
XSMPL_WARN
JTAG
AP_JTAG_TRST_N
AP_JTAG_TDI
AP_JTAG_TDO
AP_JTAG_TMS
AP_JTAG_TCK
SOC_JTAG_DBGSEL
XJTAG_TRST_N
XJTAG_TDI
XJTAG_TDO
XJTAG_TMS
XJTAG_TCK
XJTAG_DBGSEL
L4M_VDD_AP_1P8
R4017
OTP
XOTP_VPP
REV01 0904
CLOCK
SOC_CLK_52M
SOC_FAR_CLK_52M
SOC_32K_CLK
XCLKOUT0
R4020
R4023
XTCXO
XTCXO_FAR
XRTCCLK_IN
XCLKOUT0
CAM_CLK
WIDECAM_MCLK
UWCAM_MCLK
VTCAM_MCLK
@MACROCAM_MCLK
L4000
L4001
L4002
C4018
C4017
C4012
XCAM_MCLK0
XCAM_MCLK1
XCAM_MCLK2
TE
DISP_TE
XTE_DECON0
PWM
VBUS_DISCHARGING
MOTOR_PWM
XPWM_TOUT_0
XPWM_TOUT_1
USI
NFC_ESE_CLK
NFC_ESE_MOSI
NFC_ESE_MISO
NFC_ESE_CS_N
BTP_SPI_CLK
BTP_SPI_MOSI
BTP_SPI_MISO
BTP_SPI_CS_N
NFC_SCL_1P8
NFC_SDA_1P8
RCV_AMP_INT
RCV_AMP_EN
RCV_AMP_EN
@SPK_AMP_EN
CAM_OIS_SCL_1P8
CAM_OIS_SDA_1P8
CAM_PMIC_I2C_SCL_1P8
CAM_PMIC_I2C_SDA_1P8
CAM_OIS_SPI_CLK
CAM_OIS_SPI_MOSI
CAM_OIS_SPI_MISO
CAM_OIS_SPI_CS
XUSI0_RXD_CLK_SCL0
XUSI0_TXD_DO_SDA0
XUSI0_RTSN_DI_SCL1
XUSI0_CTSN_CSN_SDA1
XUSI7_SCL
XUSI7_SDA
XEINT
XEINT_0 … XEINT_11
OCTA_CON_DET
NFC_IRQ
HALL_INT1
TSP_INT
TYPE_C_INT
IF_PMIC_INTB
OCTA_ID
VOL_DN
VOL_UP
NFC_CLK_REQ
PCD_INT
GRIP_INT2
DEBUG_UART
XUART_DEBUG_RXD
XUART_DEBUG_TXD
AP_UART_RXD
AP_UART_TXD
GPIO
XGPIO1
XGPIO2
XGPIO3
OCTA_DDI_LDO_EN
WIDECAM_RST_N
CMGP
VTCAM_SENSOR_I2C_SCL_1P8
VTCAM_SENSOR_I2C_SDA_1P8
VTCAM_RST_N
UWCAM_RST_N
TSP_SCL_1P8
TSP_SDA_1P8
UWCAM_MACROCAM_SENSOR_I2C_SCL_1P8
UWCAM_MACROCAM_SENSOR_I2C_SDA_1P8
SPK_AMP_SCL_1P8
SPK_AMP_SDA_1P8
GRIP_SCL_1P8
GRIP_SDA_1P8
CAM_AF_EEP_I2C_SCL_1P8
CAM_AF_EEP_I2C_SDA_1P8
WIDECAM_SENSOR_I2C_SCL_1P8
WIDECAM_SENSOR_I2C_SDA_1P8
NVM_SCL
NVM_SDA
TSP_3P0_LDO_EN
MAIN_MIC_LDO_EN
CC_SCL_1P8
CC_SDA_1P8
IF_PMIC_SCL_1P8
IF_PMIC_SDA_1P8
MACROCAM_RST_N
XCMGP_CHUB_GPIO0
UWCAM_MACROCAM_VT_EN
CHUB
6AXIS_I3C_SCL
6AXIS_I3C_SDA
MAGNETIC_BARO_SCL_1P8
MAGNETIC_BARO_SDA_1P8
LIGHT_SCL_1P8
LIGHT_SDA_1P8
6AXIS_INT1
6AXIS_INT2
SENSOR_LDO_EN
GPIO_AP_CP_ALV
XGPIO_ALV_TRAY_DET
SIM_DET
TP4001
XGPIO_AP_CP_ALV_0 … XGPIO_AP_CP_ALV_12
OCTA_ESD_DETECT
NFC_LDO_EN
NFC_EN
SOC2PM_WRSTBI
FG_ERR
OCTA_RST_N
MOTOR_LDO_EN
NVM_LDO_EN
AP_CP_ESE_RST
CAM_FLASH_EN
CAM_TORCH_EN
SPK_AMP_INT
AUDIO
XAUD_FM_SPDY_M
FM_SPDY_WLBT
XAUD_I2S0_BCLK
XAUD_I2S0_MCLK
XAUD_I2S0_SDI
XAUD_I2S0_SDO
XAUD_I2S0_WS
BTP_LDO_EN
MST_LDO_EN
MST_DATA_B
MST_DATA_A
BTP_RST_N
XAUD_I2S1_BCLK
XAUD_I2S1_SDI
XAUD_I2S1_SDO
XAUD_I2S1_WS
SPK_I2S3_BCLK
SPK_I2S3_DI
SPK_I2S3_DO
SPK_I2S3_WS
DMIC
XDMIC_CLK0
XDMIC_CLK1
XDMIC_PDM0
XDMIC_PDM1
R4005
R4006
MAIN_MIC_CLK
SUB_MIC_CLK
MAIN_MIC_DATA
SUB_MIC_DATA
C4025
C4024
ground


UCP4000
(2/6)
USB
USB_AP_D_N
USB_AP_D_P
XUSB20_DM
XUSB20_DP
XUSB20_REXT
R4000
MMC/UFS
T_FLASH_CLK
T_FLASH_CMD
R4021
T_FLASH_D(0)
T_FLASH_D(1)
T_FLASH_D(2)
T_FLASH_D(3)
XMMC_CARD_CLK
XMMC_CARD_CMD
XMMC_CARD_D0 … XMMC_CARD_D3
UFS_REFCLK
UFS_RESET_N
UFS_RX_D0_N
UFS_RX_D0_P
UFS_TX_D0_N
UFS_TX_D0_P
XUFS_EMBD_REFCLK_OUT
XUFS_EMBD_RESET_N
XUFS_EMBD_RXD0_N
XUFS_EMBD_RXD0_P
XUFS_EMBD_TXD0_N
XUFS_EMBD_TXD0_P
MIPI-CSI
MIPI_WIDECAM_B1
MIPI_WIDECAM_C1
MIPI_WIDECAM_A0
MIPI_WIDECAM_B0
MIPI_WIDECAM_C0
MIPI_WIDECAM_A1
MIPI_WIDECAM_A2
MIPI_WIDECAM_B2
MIPI_WIDECAM_C2
MIPI_UWCAM_CLK_N
MIPI_UWCAM_CLK_P
MIPI_UWCAM_D0_N … MIPI_UWCAM_D3_P
MIPI_VTCAM_CLK_N
MIPI_VTCAM_CLK_P
MIPI_VTCAM_D0_N … MIPI_VTCAM_D3_P
MIPI_MACROCAM_CLK_N
MIPI_MACROCAM_CLK_P
MIPI_MACROCAM_D0_N
MIPI_MACROCAM_D0_P
MIPI_MACROCAM_D1_N
MIPI_MACROCAM_D1_P
MIPI-DSI
OCTA_MIPI_CLK_N
OCTA_MIPI_CLK_P
OCTA_MIPI_D0_N … OCTA_MIPI_D3_P
XMIPI_DSI4LA_VREG_0P4V
C4000
Close to SOC
WLBT
WL0_IN
WL0_IP
WL0_QN
WL0_QP
WL1_IN
WL1_IP
WL1_QN
WL1_QP
WLBT_RFIC_CTRL0
WLBT_RFIC_CTRL1
XWLBT_RFIC_CTRL0 … XWLBT_RFIC_CTRL3
GNSS
GNSS_SPEEDY
R4004
GNSS_HI_SPEEDY_DATA
GNSS_HI_SPEEDY_CLK
GNSS_TSYNC
GNSS_L1_LNA_EN
TP4000
TP4002
TP4003
XGNSS_SPEEDY
XGNSS_HI_SPEEDY
XGNSS_HI_SPEEDY_CLK
XGNSS_TSYNC_RFIC
XGNSS_LNA_EN
XGNSS_LNA_EN_ANT1
DRAM0
XDRAM0_CA0_CA1 … XDRAM0_CA6_CA2
DRAM0_ADCT_1
DRAM0_ADCT_0
DRAM0_ADCT_4
DRAM0_CS_0
DRAM0_ADCT_3
DRAM0_ADCT_5
DRAM0_ADCT_2
DRAM0_CLK_M
DRAM0_CLK_P
DRAM0_CKE_0
DRAM0_CKE_1
DRAM0_DMI_1
DRAM0_DMI_0
DRAM0_DQS_1_M
DRAM0_DQS_1_P
DRAM0_DQS_0_M
DRAM0_DQS_0_P
DRAM0_DQ(0) … DRAM0_DQ(15)
XDRAM0_WCK0_C_NC
XDRAM0_WCK0_T_NC
XDRAM0_WCK1_C_NC
XDRAM0_WCK1_T_NC
XDRAM0_NC_CS1
DRAM0_CS_1
XDRAM_ZQ
R4003
B1M_VDDQ_MEM_0P6
DRAM1
XDRAM1_CA0_CA2 … XDRAM1_CA6_CA1
DRAM1_ADCT_2
DRAM1_ADCT_5
DRAM1_ADCT_3
DRAM1_CS_0
DRAM1_ADCT_4
DRAM1_ADCT_0
DRAM1_ADCT_1
DRAM1_CLK_M
DRAM1_CLK_P
DRAM1_CKE_0
DRAM1_CKE_1
DRAM1_DMI_0
DRAM1_DMI_1
DRAM1_DQS_0_M
DRAM1_DQS_0_P
DRAM1_DQS_1_M
DRAM1_DQS_1_P
DRAM1_DQ(0) … DRAM1_DQ(15)
XDRAM_RESET_N
XDRAM1_NC_CS1
DRAM_RESET_N
DRAM1_CS_1


UCP4000
(3/6)
C4031
L4003
FR1_HI_SPEEDY_CLK
FR1_HI_SPEEDY_DATA
CP_HI_SPEEDY
XCP_C_FR1_HI_SPEEDY_CLK
XCP_C_FR1_HI_SPEEDY_DATA
XCP_C_FR2_HI_SPEEDY_CLK
XCP_C_FR2_HI_SPEEDY_DATA
MCW_SYNC
CP MCW_SYNC
XCP_MCW_SYNC
XCP_FR2_IPC_STATUS
CP FPRF (GPIO/MIPI)
RFFE0_CLK
RFFE0_DATA
RFFE1_CLK
RFFE1_DATA
RFFE2_CLK
RFFE2_DATA
RFFE3_CLK
RFFE3_DATA
RFFE4_CLK
RFFE4_DATA
CABLE_DET_LB
CABLE_DET_MHB
RFFE7_CLK
RFFE7_DATA
C4010
C4011
C4019
C4020
C4021
C4022
C4023
C4026
C4027
C4028
GPIO_ALV_CP
XGPIO0_CPALV
XGPIO1_CPALV
GNSS_TIME_MARK
CP USIM
XCP_USIM0_CLK
XCP_USIM0_DATA
XCP_USIM0_RST_N
XCP_USIM1_CLK
XCP_USIM1_DATA
XCP_USIM1_RST_N
SIM1_CLK
SIM1_DATA
SIM1_RST
CP_SIM2_CLK
CP_SIM2_DATA
CP_SIM2_RST
CP MCW
FR1_MCW_DL_DQ0
FR1_MCW_DL_DQ1
FR1_MCW_DL_DQ2
FR1_MCW_DL_DQ3
FR1_MCW_DL_DQS_N
FR1_MCW_DL_DQS_P
FR1_MCW_UL_DQ0
FR1_MCW_UL_DQ1
FR1_MCW_UL_DQS_N
FR1_MCW_UL_DQS_P
FR1_RFIC_RESET
XCP_FR2_RFIC_RESET

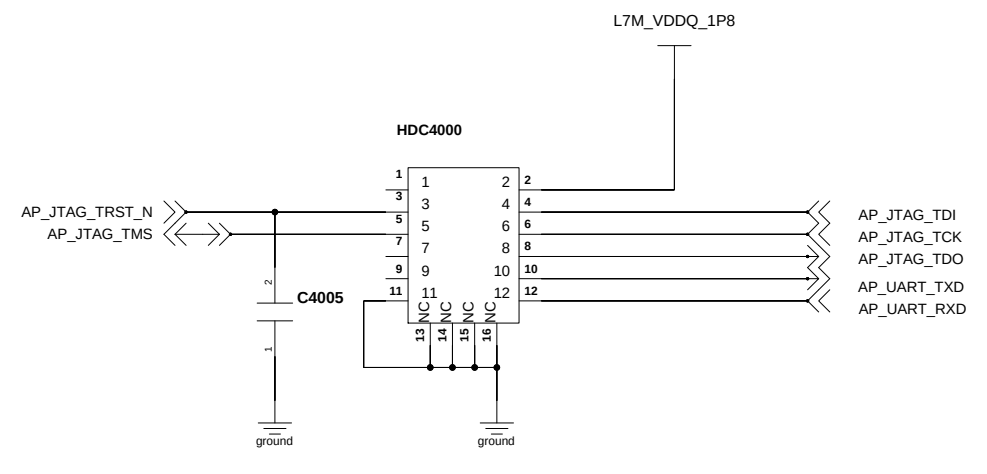
I2C PU RESISTOR


VDD_CAM_IO_1P8
R4008
R4010
CAM_AF_EEP_I2C_SCL_1P8
CAM_AF_EEP_I2C_SDA_1P8
R4013
R4015
WIDECAM_SENSOR_I2C_SCL_1P8
WIDECAM_SENSOR_I2C_SDA_1P8
R4031
R4032
UWCAM_MACROCAM_SENSOR_I2C_SCL_1P8
UWCAM_MACROCAM_SENSOR_I2C_SDA_1P8
L7M_VDDQ_1P8
R4037
R4022
TSP_SCL_1P8
TSP_SDA_1P8
R4041
R4042
VTCAM_SENSOR_I2C_SCL_1P8
VTCAM_SENSOR_I2C_SDA_1P8
R4024
R4025
CAM_OIS_SCL_1P8
CAM_OIS_SDA_1P8
VDD_NFC_IO_1P8
R4007
R4009
NFC_SCL_1P8
NFC_SDA_1P8
R4011
R4012
R4014
GRIP_SCL_1P8
GRIP_SDA_1P8
GRIP_INT2
VDD_SENSOR_1P8
R4018
R4019
LIGHT_SCL_1P8
LIGHT_SDA_1P8
R4034
R4036
MAGNETIC_BARO_SCL_1P8
MAGNETIC_BARO_SDA_1P8
R4039
R4040
CAM_PMIC_I2C_SCL_1P8
CAM_PMIC_I2C_SDA_1P8
R4016
R4033
SPK_AMP_SCL_1P8
SPK_AMP_SDA_1P8

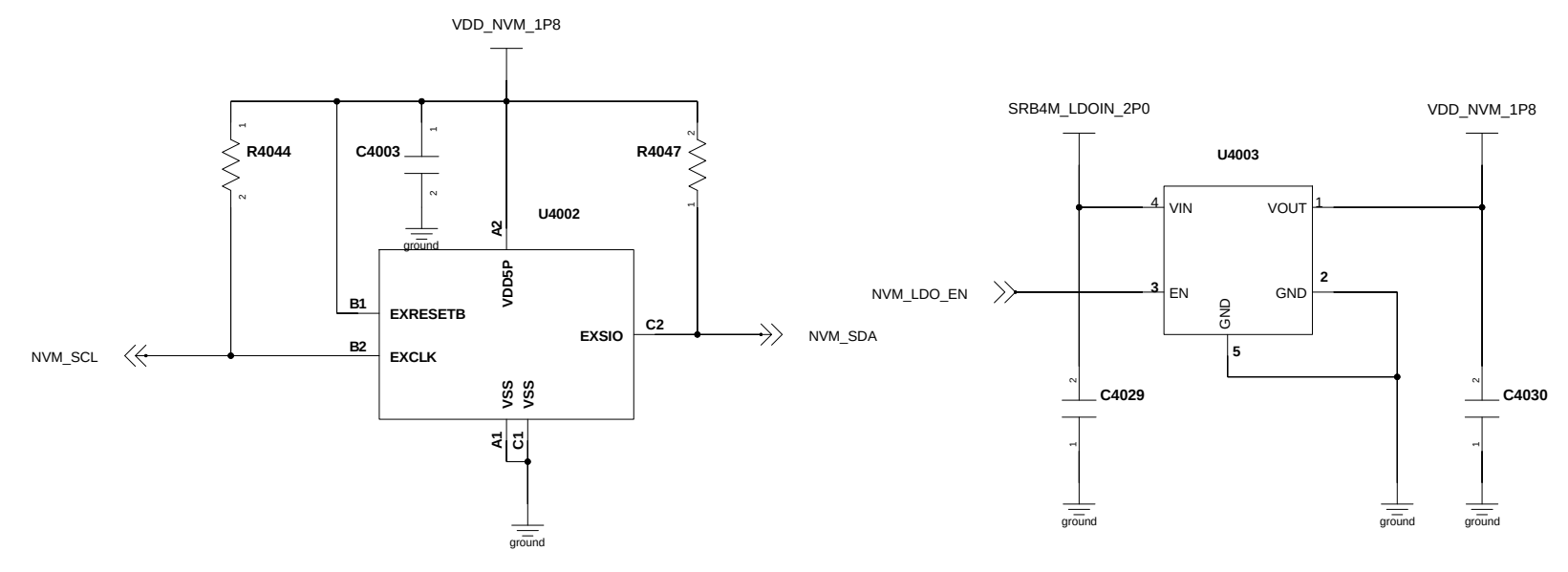

L7M_VDDQ_1P8
R4026
R4027
R4028
R4053
AP_USB_BOOT1
XCLKOUT0
XOM
AP_USB_BOOT1
@XRSTOUT_N
SOC_JTAG_DBGSEL

PIN			Function	
XCLKOUT0	XRSTOUT_N	XOM	1st Boot	2nd Boot
0	0	0	UFS	USB
0	1	X	USB	----
1	0	0	SD	USB

BOOT CONFIGURATIONS


L7M_VDDQ_1P8
HDC4000
AP_JTAG_TRST_N
AP_JTAG_TMS
C4005
AP_JTAG_TDI
AP_JTAG_TCK
AP_JTAG_TDO
AP_UART_TXD
AP_UART_RXD
NC
ground

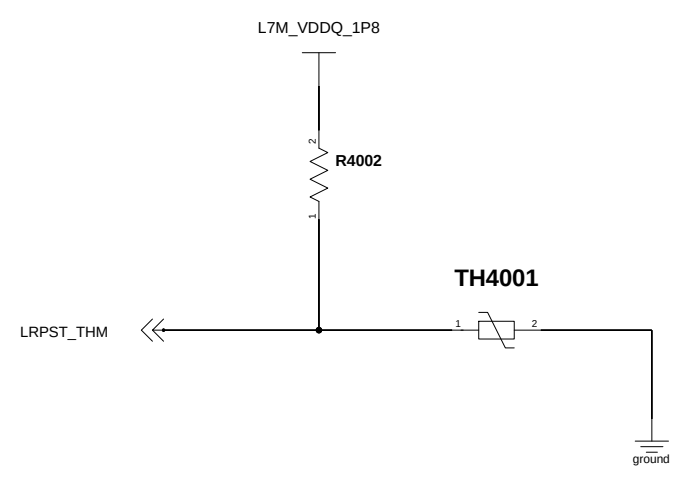
AP JTAG

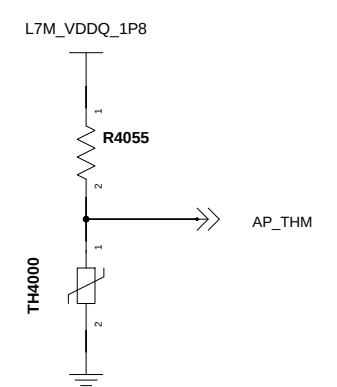

VDD_NVM_1P8
R4044
C4003
U4002
R4047
EXRESETB
EXCLK
VDD5P
EXSIO
VSS
NVM_SCL
NVM_SDA
SRB4M_LDOIN_2P0
U4003
VIN
VOUT
EN
GND
NVM_LDO_EN
VDD_NVM_1P8
C4029
C4030
ground

NVM

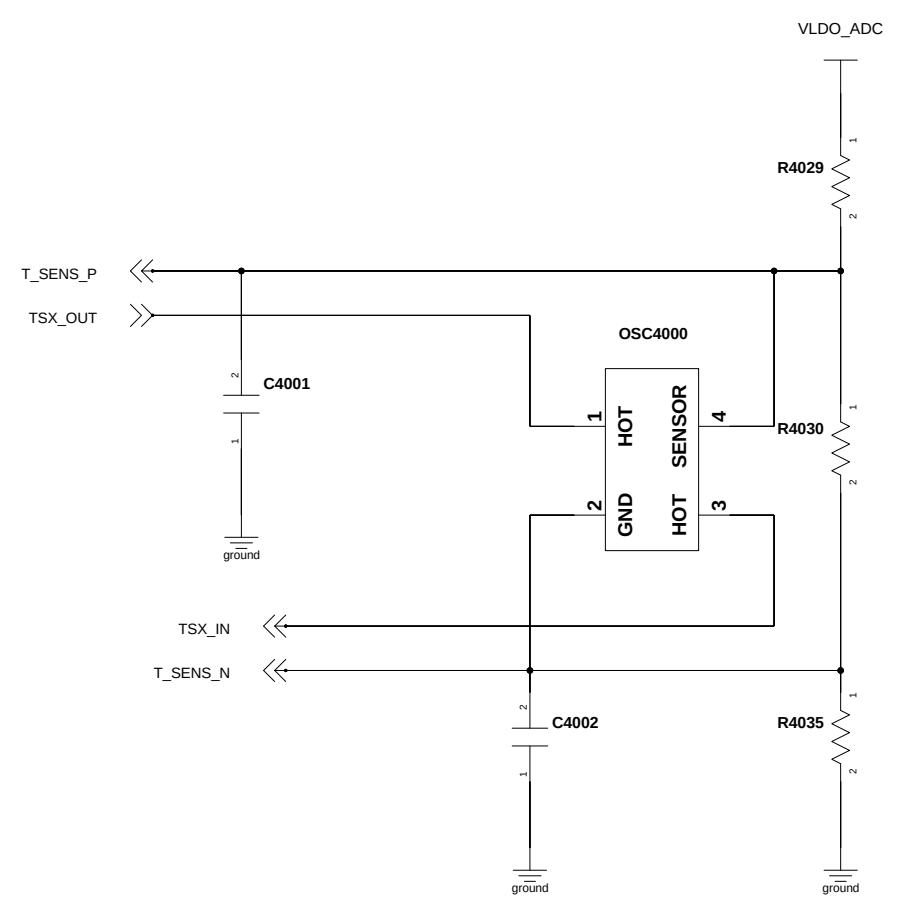
SYSTEM CLOCK

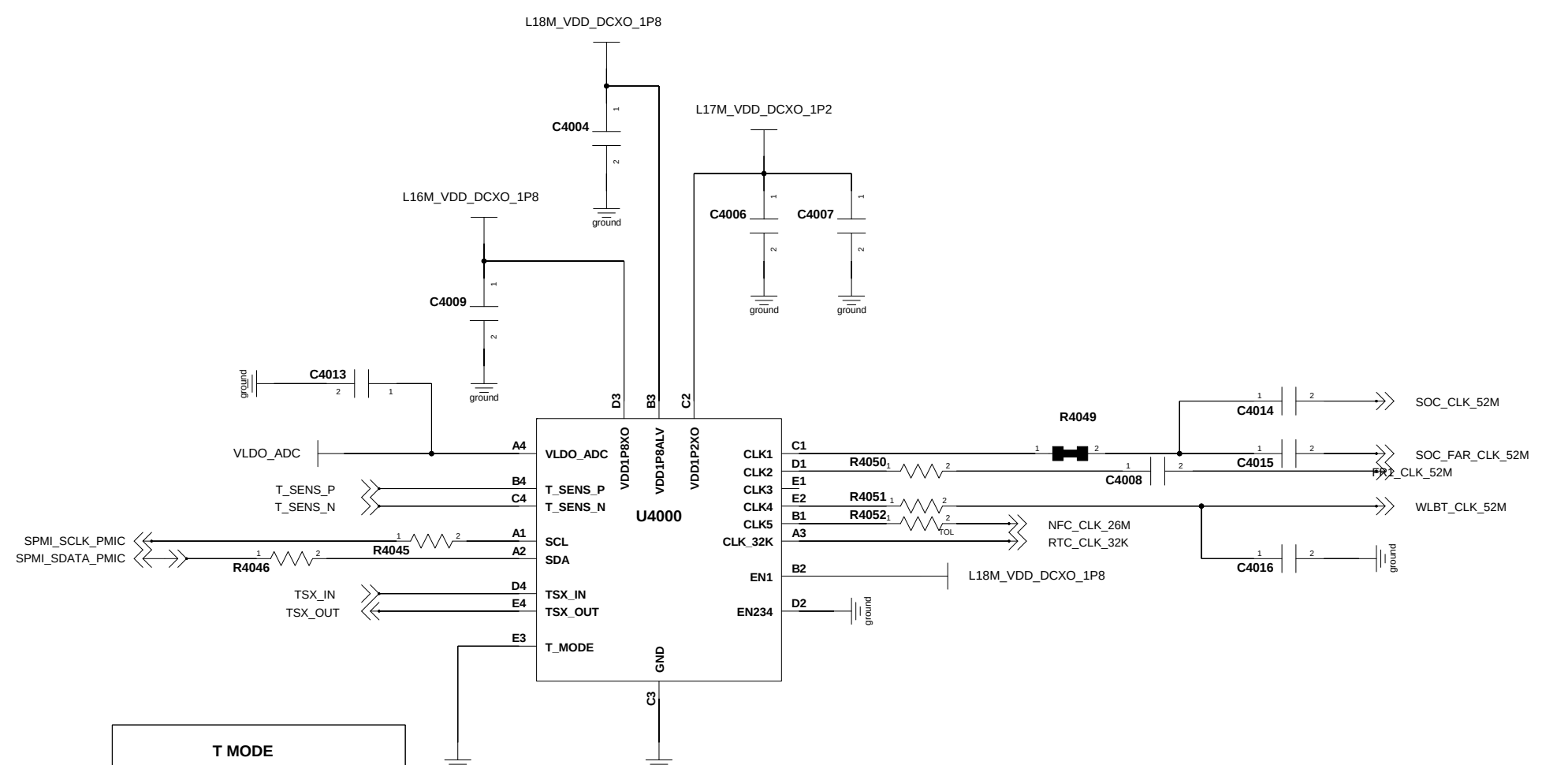
SOC THERMISTOR


L7M_VDDQ_1P8
R4002
TH4001
LRPST_THM
ground


L7M_VDDQ_1P8
R4055
AP_THM
TH4000
ground

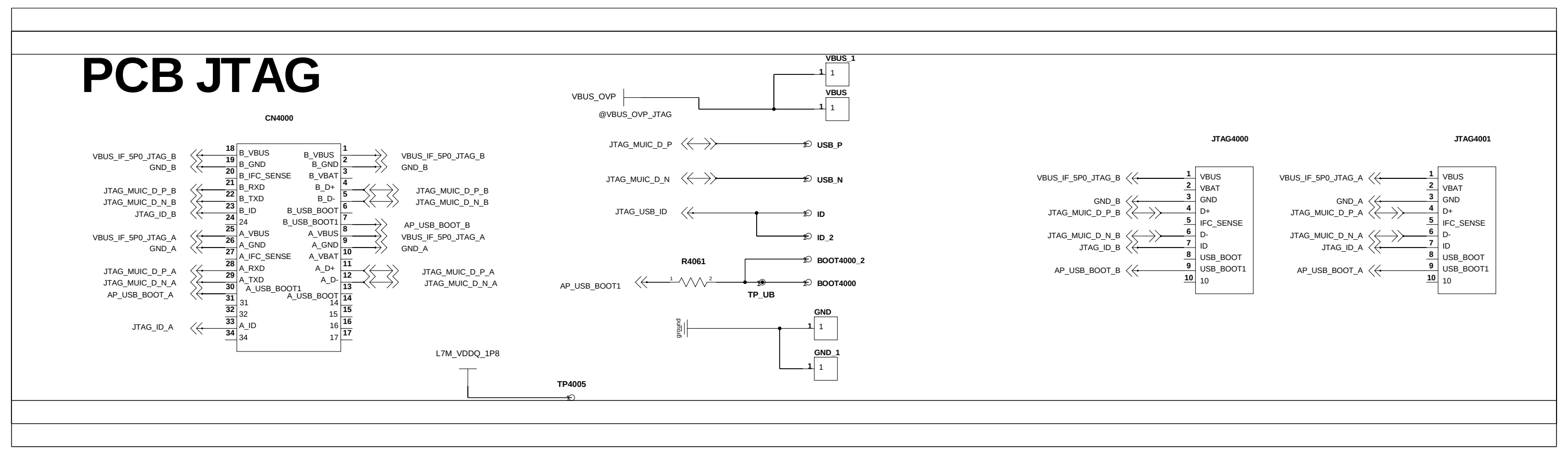
AP_THM


VLDO_ADC
R4029
T_SENS_P
TSX_OUT
C4001
OSC4000
HOT
SENSOR
GND
HOT
R4030
TSX_IN
T_SENS_N
C4002
R4035
ground


L18M_VDD_DCXO_1P8
C4004
L17M_VDD_DCXO_1P2
C4006
C4007
L16M_VDD_DCXO_1P8
C4009
C4013
VLDO_ADC
T_SENS_P
T_SENS_N
SPMI_SCLK_PMIC
SPMI_SDATA_PMIC
R4045
R4046
TSX_IN
TSX_OUT
U4000
VDD1P8XO
VDD1P8ALV
VDD1P2XO
VLDO_ADC
T_SENS_P
T_SENS_N
SCL
SDA
TSX_IN
TSX_OUT
T_MODE
GND
CLK1
CLK2
CLK3
CLK4
CLK5
CLK_32K
EN1
EN234
R4049
R4050
R4051
R4052
C4008
C4014
C4015
C4016
SOC_CLK_52M
SOC_FAR_CLK_52M
FR1_CLK_52M
WLBT_CLK_52M
NFC_CLK_26M
RTC_CLK_32K
L18M_VDD_DCXO_1P8
ground

T MODE	
GND	DCXO(Default)
PWR	TCXO

PCB JTAG


CN4000
VBUS_IF_5P0_JTAG_B
GND_B
JTAG_MUIC_D_P_B
JTAG_MUIC_D_N_B
JTAG_ID_B
VBUS_IF_5P0_JTAG_A
GND_A
JTAG_MUIC_D_P_A
JTAG_MUIC_D_N_A
AP_USB_BOOT_A
JTAG_ID_A
AP_USB_BOOT_B
B_VBUS
B_GND
B_IFC_SENSE
B_RXD
B_TXD
B_ID
A_VBUS
A_GND
A_IFC_SENSE
A_RXD
A_TXD
A_ID
B_VBAT
B_D+
B_D-
B_USB_BOOT
B_USB_BOOT1
A_VBAT
A_D+
A_D-
A_USB_BOOT1
A_USB_BOOT
VBUS_1
VBUS
VBUS_OVP
@VBUS_OVP_JTAG
JTAG_MUIC_D_P
JTAG_MUIC_D_N
JTAG_USB_ID
USB_P
USB_N
ID
ID_2
R4061
AP_USB_BOOT1
TP_UB
BOOT4000_2
BOOT4000
GND
GND_1
L7M_VDDQ_1P8
TP4005
JTAG4000
JTAG4001
VBUS
VBAT
GND
D+
IFC_SENSE
D-
ID
USB_BOOT
USB_BOOT1

Engineer:	A35
Drawn by:	
R&D CHK:	TITLE: Size:
DOC CTRL CHK:	
MFG ENGR CHK:	
QA CHK:	REV: Drawing Number: Page:

Changed by: Date Changed: Time Changed:

SAMSUNG CONFIDENTIAL

THIS DOCUMENT CONTAINS CONFIDENTIAL
PROPRIETARY INFORMATION THAT IS
SAMSUNG ELECTRONICS CO'S PROPERTY.
DO NOT DISCLOSE TO OR DUPLICATE FOR OTHERS
EXCEPT AS AUTHORIZED BY SAMSUNG.

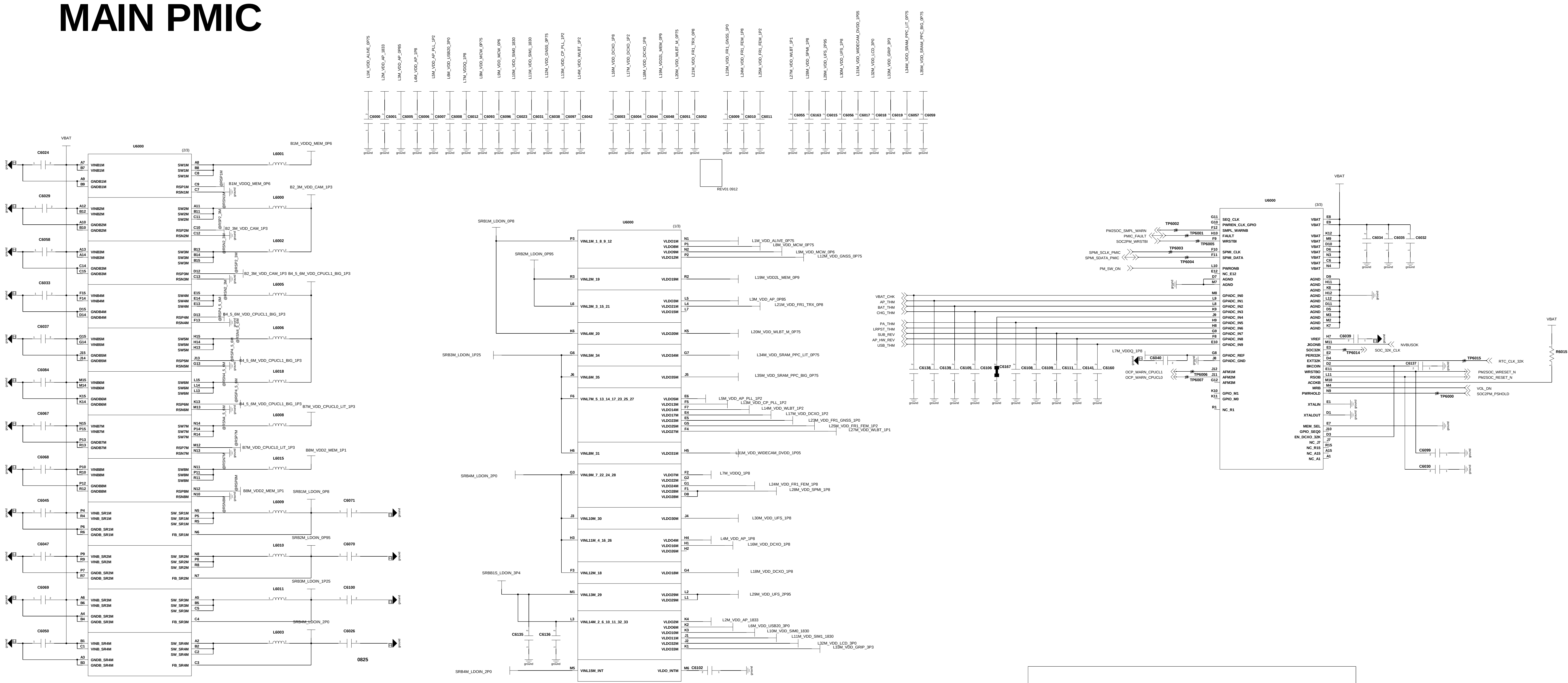
UCP4000
(4/6)
(5/6)
(6/6)
UME5000
B1M_VDDQ_MEM_0P6
@RSP1M
@RSN1M
230526
VDDQ06_DRAM0
VDDQ06_DRAM1
B2_3M_VDD_CAM_1P3
@RSP2_3M
@RSN2_3M
VDD_CAM
B4_5_6M_VDD_CPUCL1_BIG_1P3
@RSP4_5_6M
@RSN4_5_6M
VDD_CPUCL1_BIG
B7M_VDD_CPUCL0_LIT_1P3
@RSP7M
@RSN7M
VDD_CPUCL0_LIT
B8M_VDD2_MEM_1P1
VDDQ11_DRAM0_CKE
VDDQ11_DRAM1_CKE
L1M_VDD_ALIVE_0P75
VDD_ALIVE
VDD_ALIVE_JSC
VDD_CHUBVTS
AVDD075_USB20
VDD075_DLL_GNSS
VDD_GNSS_ALV
VDD_ALV_CP
L2M_VDD_AP_1833
VDDQ1833_MMC_CARD
AVDD085_UFS_EMBD
AVDD085_MIPI_DSI
AVDD085_MIPI_CSI
VDD_ICPU_CSILINK
L3M_VDD_AP_0P85
AVDD18_UFS_EMBD
AVDD18_MIPI_DSI
AVDD18_MIPI_CSI
AVDD18_OTP
AVDD18_TS
L4M_VDD_AP_1P8
AVDD12_PLL_AUD
AVDD12_PLL_CPU
AVDD12_PLL_SHARED
AVDD12_PLL_MIF
AVDD12_MIPI_DSI
VDDQ12_UFS_EMBD
L5M_VDD_AP_PLL_1P2
AVDD33_USB20
L6M_VDD_USB20_3P0
AVDD18_TCXO_FAR
AVDD18_TCXO
AVDD18_USB20
VDDQ18_GPIO_ALV_CP
VDDQ18_MCW_CP
VDDQ1218_GNSS
VDDQ18_WLBT
VDDQ18_GNSS
VDDQ18_GPIO_B
VDDQ18_GPIO_T
VDDQ18_CMGP
L7M_VDDQ_1P8
VDD_MCW_CP
L8M_VDD_MCW_0P75
VDD06_MCW_FR12_CP
L9M_VDD_MCW_0P6
VDDQ1830_USIM0_CP
L10M_VDD_SIM0_1830
VDDQ1830_USIM1_CP
L11M_VDD_SIM1_1830
VDD_GNSS
L12M_VDD_GNSS_0P75
AVDD12_CDPLL_CP
AVDD12_IMPLL_CP
L13M_VDD_CP_PLL_1P2
AVDD12_WLBT
L14M_VDD_WLBT_1P2
VDD_WLBT_M
L20M_VDD_WLBT_M_0P75
VDD_SRAM_PPC_LIT
L34M_VDD_SRAM_PPC_LIT_0P75
VDD_SRAM_PPC_BIG
L35M_VDD_SRAM_PPC_BIG_0P75
VSS
ground
B1S_VDD_MIF_0P75
@RSP1S
@RSN1S
VDD_MIF
B2S_VDD_WLBT_0P75
@RSP2S
@RSN2S
VDD_WLBT
B3S_VDD_NPU_0P75
@RSP3S
@RSN3S
VDD_NPU
B4_5S_VDD_G3D_0P75
@RSP4_5S
@RSN4_5S
VDD_G3D
B6S_VDD_CPU_CP_0P75
@RSP6S
@RSN6S
VDD_CPU_CP
VDD_CP
B7_8S_VDD_CP_0P75
@RSP7_8S
@RSN7_8S
VDD_INT
B9S_VDD_INT_0P75
@RSP9S
@RSN9S
VDD_SRAM
B10S_VDD_SRAM_0P75
@RSP10S
@RSN10S
AVDD18_WLBT
L8S_VDD_WLBT_1P8
VDD_SRAM_PPC_NPU
L15S_VDD_SRAM_PPC_NPU_0P75
L29M_VDD_UFS_2P95
L30M_VDD_UFS_1P8
L7M_VDDQ_1P8
B8M_VDD2_MEM_1P1
@RSP8M
@RSN8M
B1M_VDDQ_MEM_0P6
R5000
R5001
VCC
VCCQ
VCCQ2
VDD1
VDD2
VDDI
VDDIQ
VDDIQ2
VDDQ
ZQ0_A
ZQ1_A
DRAM0_ADCT_0
DRAM0_ADCT_1
DRAM0_ADCT_2
DRAM0_ADCT_3
DRAM0_ADCT_4
DRAM0_ADCT_5
DRAM0_CLK_M
DRAM0_CLK_P
DRAM0_CKE_0
DRAM0_CKE_1
DRAM0_CS_0
DRAM0_CS_1
DRAM0_DMI_0
DRAM0_DMI_1
DRAM0_DQ(0)–DRAM0_DQ(15)
DRAM0_DQS_0_M
DRAM0_DQS_0_P
DRAM0_DQS_1_M
DRAM0_DQS_1_P
ODT_A
UFS_TX_D0_N
UFS_TX_D0_P
UFS_RX_D0_N
UFS_RX_D0_P
UFS_REFCLK
TP5001–TP5009
DIN0_C
DIN0_T
DIN1_C
DIN1_T
DOUT0_C
DOUT0_T
DOUT1_C
DOUT1_T
REF_CLK
RFU
NC
VSS_G
DRAM1_ADCT_0
DRAM1_ADCT_1
DRAM1_ADCT_2
DRAM1_ADCT_3
DRAM1_ADCT_4
DRAM1_ADCT_5
DRAM1_CLK_M
DRAM1_CLK_P
DRAM1_CKE_0
DRAM1_CKE_1
DRAM1_CS_0
DRAM1_CS_1
DRAM1_DMI_0
DRAM1_DMI_1
DRAM1_DQ(0)–DRAM1_DQ(15)
DRAM1_DQS_0_M
DRAM1_DQS_0_P
DRAM1_DQS_1_M
DRAM1_DQS_1_P
ODT_B
RESET_N
RST_N
DRAM_RESET_N
UFS_RESET_N
DNU
VSF1
VSF2
VSF3
VSF4
VSF5
VSF6
A B C D E F G H I J K L
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

Engineer:	A35
Drawn by:	
R&D CHK:	TITLE: Size:
DOC CTRL CHK:	AP/CP/uMCP
MFG ENGR CHK:	
QA CHK:	REV: Drawing Number: Page:

Changed by: Date Changed: Time Changed:

SAMSUNG CONFIDENTIAL
THIS DOCUMENT CONTAINS CONFIDENTIAL
PROPRIETARY INFORMATION THAT IS
SAMSUNG ELECTRONICS CO'S PROPERTY.
DO NOT DISCLOSE TO OR DUPLICATE FOR OTHERS
EXCEPT AS AUTHORIZED BY SAMSUNG.

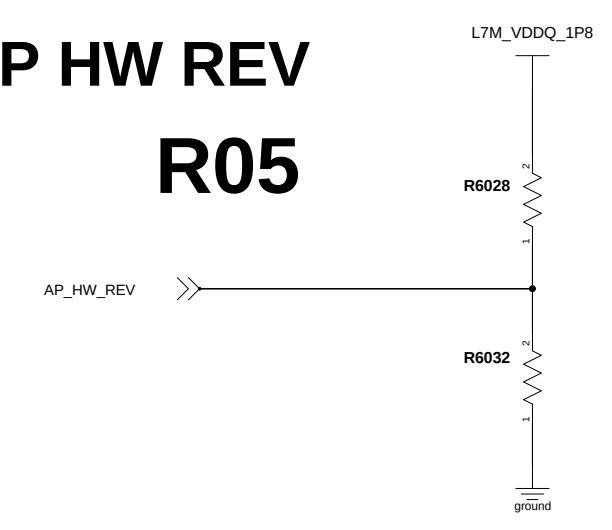
MAIN PMIC


U6000 (2/3)
VBAT
C6024 C6029 C6058 C6033 C6037 C6084 C6067 C6068 C6045 C6047 C6069 C6050
VINB1M GNDB1M SW1M RSP1M RSN1M ... VINB8M GNDB8M SW8M RSP8M RSN8M
VINB_SR1M ... VINB_SR4M SW_SR1M ... SW_SR4M FB_SR1M ... FB_SR4M
L6001 L6000 L6002 L6005 L6006 L6018 L6008 L6015 L6009 L6010 L6011 L6003
B1M_VDDQ_MEM_0P6
B2_3M_VDD_CAM_1P3
B4_5_6M_VDD_CPUCL1_BIG_1P3
B7M_VDD_CPUCL0_LIT_1P3
B8M_VDD2_MEM_1P1
SRB1M_LDOIN_0P8
SRB2M_LDOIN_0P95
SRB3M_LDOIN_1P25
SRB4M_LDOIN_2P0
C6071 C6070 C6100 C6026
0825
L1M_VDD_ALIVE_0P75 L2M_VDD_AP_1833 L3M_VDD_AP_0P85 L4M_VDD_AP_1P8 L5M_VDD_AP_PLL_1P2 L6M_VDD_USB20_3P0 L7M_VDDQ_1P8 L8M_VDD_MCW_0P75 L9M_VDD_MCW_0P6 L10M_VDD_SIM0_1830 L11M_VDD_SIM1_1830 L12M_VDD_GNSS_0P75 L13M_VDD_CP_PLL_1P2 L14M_VDD_WLBT_1P2 L16M_VDD_DCXO_1P8 L17M_VDD_DCXO_1P2 L18M_VDD_DCXO_1P8 L19M_VDD2L_MEM_0P9 L20M_VDD_WLBT_M_0P75 L21M_VDD_FR1_TRX_0P8 L23M_VDD_FR1_GNSS_1P0 L24M_VDD_FR1_FEM_1P8 L25M_VDD_FR1_FEM_1P2 L27M_VDD_WLBT_1P1 L28M_VDD_SPMI_1P8 L29M_VDD_UFS_2P95 L30M_VDD_UFS_1P8 L31M_VDD_WIDECAM_DVDD_1P05 L32M_VDD_LCD_3P0 L33M_VDD_GRIP_3P3 L34M_VDD_SRAM_PPC_LIT_0P75 L35M_VDD_SRAM_PPC_BIG_0P75
C6000 C6001 C6005 C6006 C6007 C6008 C6012 C6093 C6096 C6023 C6031 C6038 C6097 C6042 C6003 C6004 C6044 C6048 C6051 C6052 C6009 C6010 C6011 C6055 C6163 C6015 C6056 C6017 C6018 C6019 C6057 C6059
REV01 0912
U6000 (1/3)
VINL1M_1_8_9_12 VINL2M_19 VINL3M_3_15_21 VINL4M_20 VINL5M_34 VINL6M_35 VINL7M_5_13_14_17_23_25_27 VINL8M_31 VINL9M_7_22_24_28 VINL10M_30 VINL11M_4_16_26 VINL12M_18 VINL13M_29 VINL14M_2_6_10_11_32_33 VINL15M_INT
SRBB1S_LDOIN_3P4
C6135 C6136 C6102
VLDO_INTM
U6000 (3/3)
SEQ_CLK PWREN_CLK_GPIO SMPL_WARNB FAULT WRSTBI SPMI_CLK SPMI_DATA PWRONB NC_E12 AGND GPADC_IN0 ... GPADC_IN9 GPADC_REF GPADC_GND AFM1M AFM2M AFM3M GPIO_M1 GPIO_M0 NC_R1
VBAT VREF JIGONB SOC32K PERI32K EXT32K BKCOIN WRSTBO RSOB ACOKB MRB PWRHOLD XTALIN XTALOUT MEM_SEL GPIO_SEQ0 EN_DCXO_32K NC_J7 NC_R15 NC_A15 NC_A1
TP6002 TP6001 TP6005 TP6003 TP6004 TP6006 TP6007 TP6014 TP6015 TP6000
PM2SOC_SMPL_WARN PMIC_FAULT SOC2PM_WRSTBI SPMI_SCLK_PMIC SPMI_SDATA_PMIC PM_SW_ON
VBAT_CHK AP_THM BAT_THM CHG_THM PA_THM LRPST_THM SUB_REV AP_HW_REV USB_THM
C6138 C6139 C6105 C6106 C6167 C6108 C6109 C6111 C6141 C6160
L7M_VDDQ_1P8 C6040 OCP_WARN_CPUCL1 OCP_WARN_CPUCL0
C6034 C6035 C6032 C6039 C6137 C6099 C6030 R6015
NVBUSOK SOC_32K_CLK RTC_CLK_32K PM2SOC_WRESET_N PM2SOC_RESET_N VOL_DN SOC2PM_PSHOLD

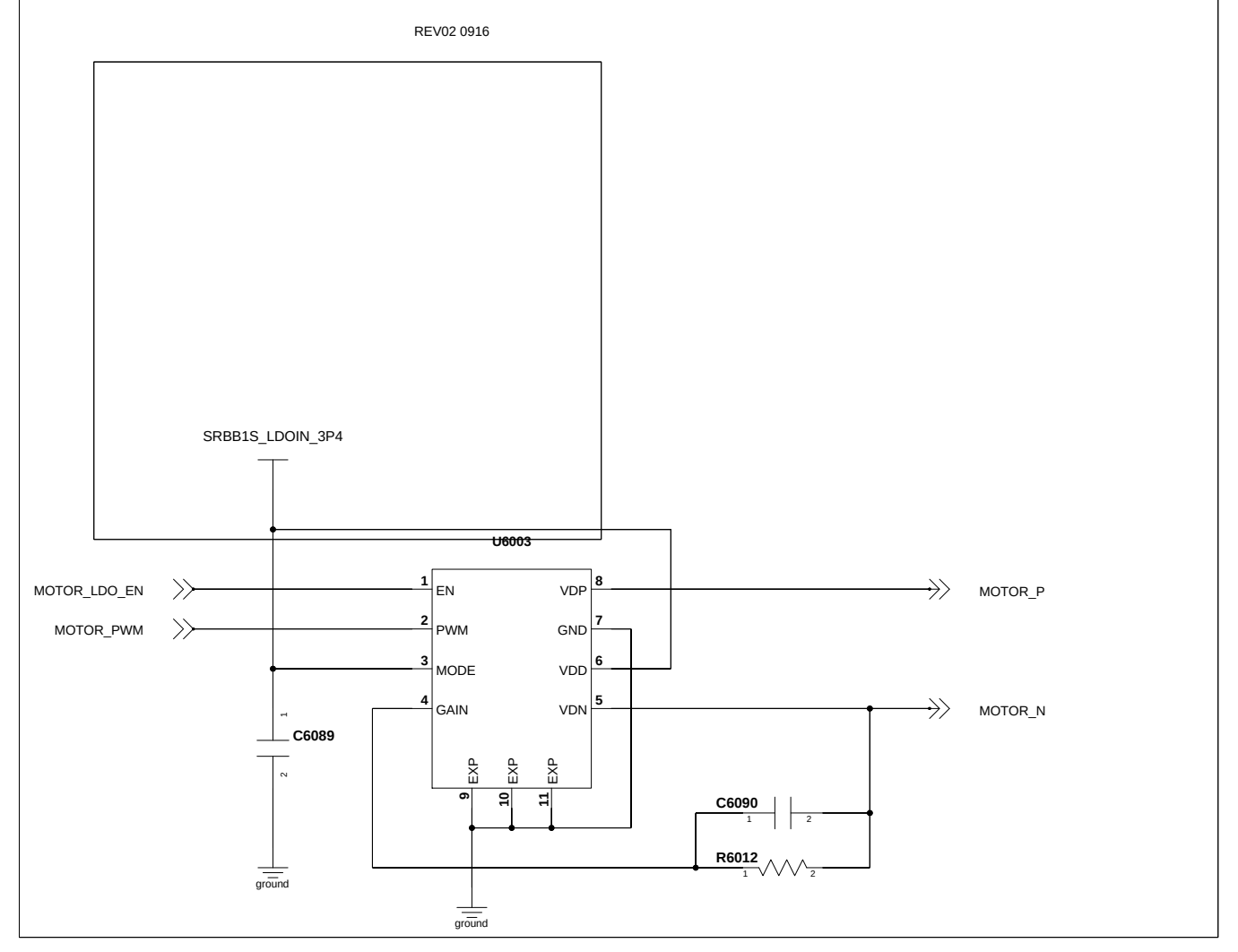
REV	R1(Kohm)	R2(Kohm)	AP	HW ID
0	NC	0		
1	470	30	R00	1
2	470	82	R01	2
3	360	91	R02	3
4	330	120	R03	4
5	360	180	R04	5
6	270	180	R05	6
7	240	220		7
8	200	240		8
9	180	270		9
10	180	360		10

AP HW REV

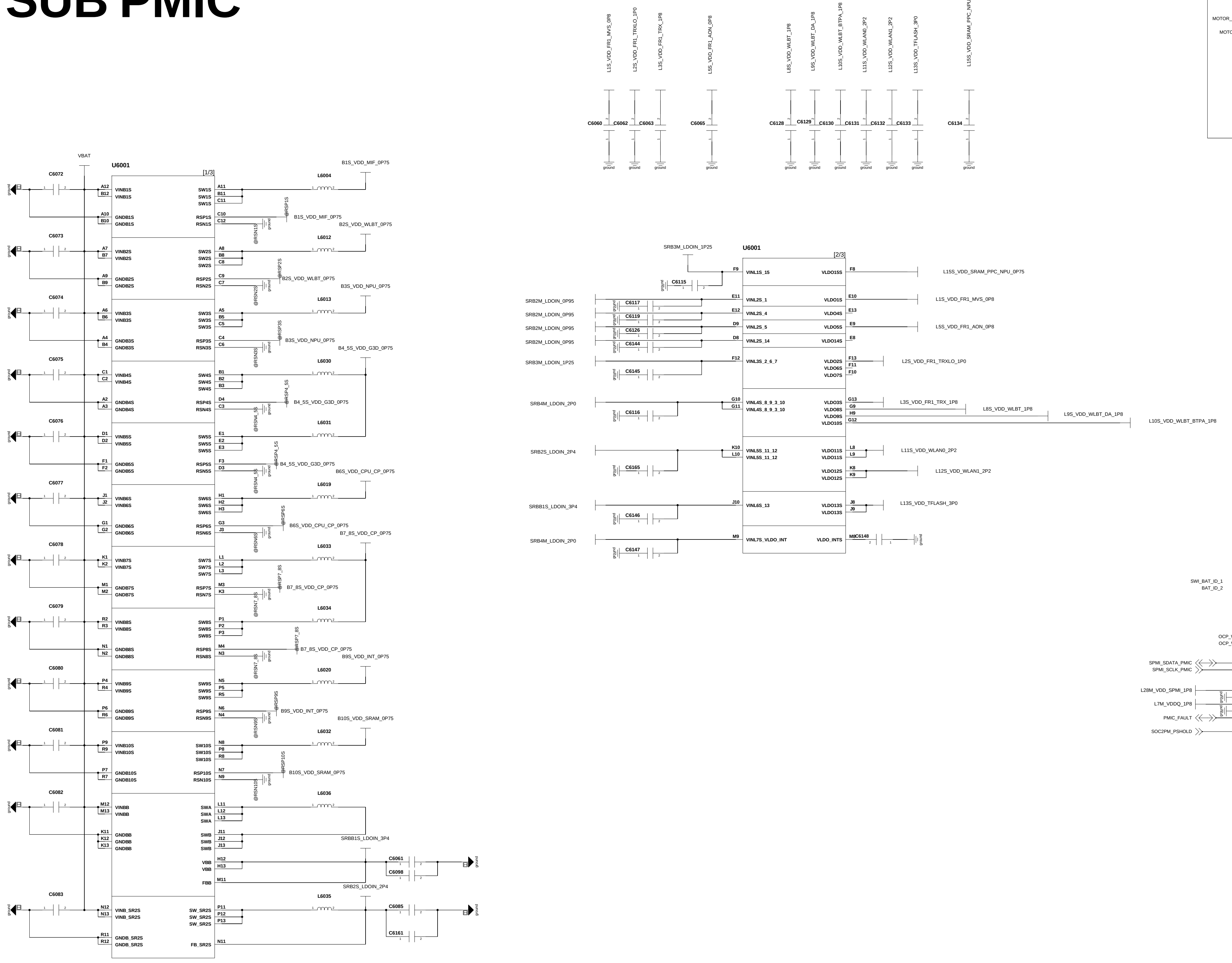
R05


L7M_VDDQ_1P8
R6028
R1
AP_HW_REV
R6032
R2
ground

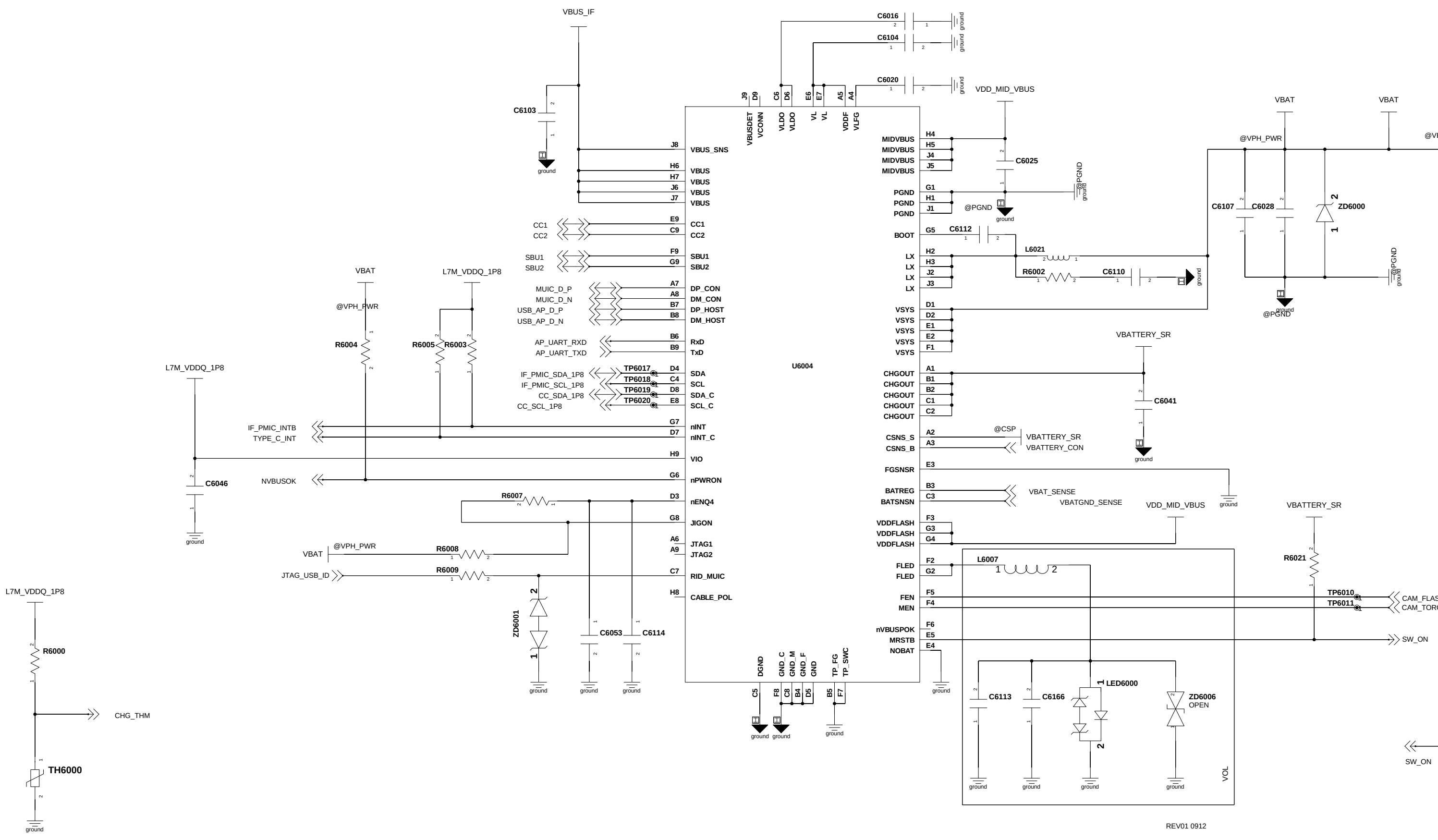
MOT DRV


REV02 0916
SRBB1S_LDOIN_3P4
U6003
MOTOR_LDO_EN
MOTOR_PWM
EN PWM MODE GAIN VDP GND VDD VDN EXP
MOTOR_P
MOTOR_N
C6089 C6090 R6012
ground

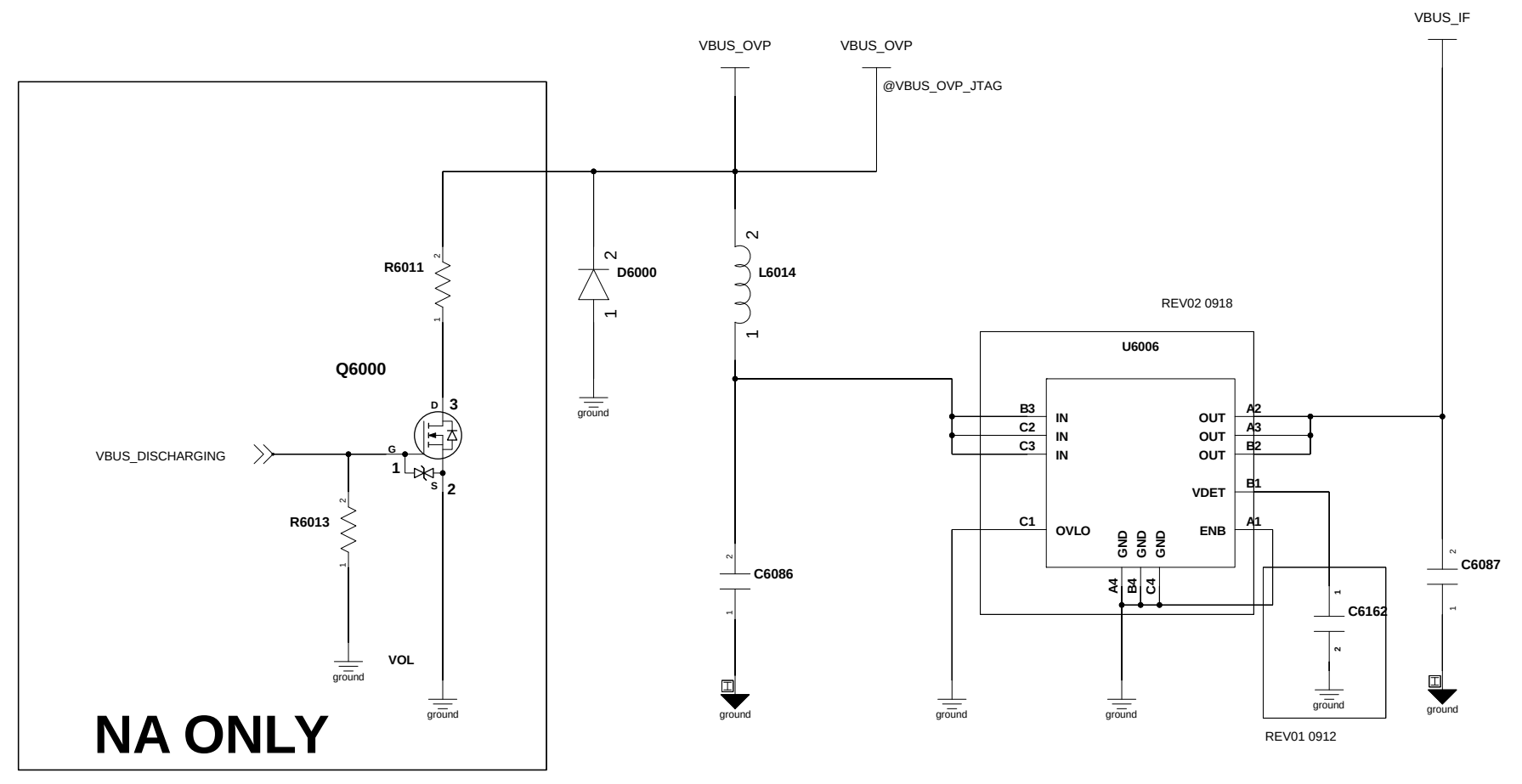
SUB PMIC


L1S_VDD_FR1_MVS_0P8 L2S_VDD_FR1_TRXLO_1P0 L3S_VDD_FR1_TRX_1P8 L5S_VDD_FR1_AON_0P8 L8S_VDD_WLBT_1P8 L9S_VDD_WLBT_DA_1P8 L10S_VDD_WLBT_BTPA_1P8 L11S_VDD_WLAN0_2P2 L12S_VDD_WLAN1_2P2 L13S_VDD_TFLASH_3P0 L15S_VDD_SRAM_PPC_NPU_0P75
C6060 C6062 C6063 C6065 C6128 C6129 C6130 C6131 C6132 C6133 C6134
VBAT
U6001 [1/3]
C6072 C6073 C6074 C6075 C6076 C6077 C6078 C6079 C6080 C6081 C6082 C6083
VINB1S ... VINB10S VINBB VINB_SR2S GNDB1S ... GNDB10S GNDBB GNDB_SR2S
SW1S ... SW10S SWA SWB VBB FBB SW_SR2S FB_SR2S
L6004 L6012 L6013 L6030 L6031 L6019 L6033 L6034 L6020 L6032 L6036 L6035
B1S_VDD_MIF_0P75
B2S_VDD_WLBT_0P75
B3S_VDD_NPU_0P75
B4_5S_VDD_G3D_0P75
B6S_VDD_CPU_CP_0P75
B7_8S_VDD_CP_0P75
B9S_VDD_INT_0P75
B10S_VDD_SRAM_0P75
SRBB1S_LDOIN_3P4
SRB2S_LDOIN_2P4
C6061 C6098 C6085 C6161
U6001 [2/3]
SRB3M_LDOIN_1P25 SRB2M_LDOIN_0P95 SRB4M_LDOIN_2P0 SRB2S_LDOIN_2P4 SRBB1S_LDOIN_3P4
VINL1S_15 VINL2S_1 VINL2S_4 VINL2S_5 VINL2S_14 VINL3S_2_6_7 VINL4S_8_9_3_10 VINL5S_11_12 VINL6S_13 VINL7S_VLDO_INT
C6115 C6117 C6119 C6126 C6144 C6145 C6116 C6165 C6146 C6147 C6148
VLDO_INTS
U6001 [3/3]
L7M_VDDQ_1P8 R6043 VBAT
GPIO_HV GPIO_S0 ... GPIO_S9 AFM1S AFM2S AFM3S SPMI_DATA SPMI_CLK VLDO_SPMI VLDO_SYS FAULT PWRHOLD SEQ_CLK
SWI_BAT_ID_1 BAT_ID_2 TP6012 TP9001
OCP_WARN_NPU OCP_WARN_G3D TP6008 TP6009
SPMI_SDATA_PMIC SPMI_SCLK_PMIC L28M_VDD_SPMI_1P8 L7M_VDDQ_1P8 PMIC_FAULT SOC2PM_PSHOLD
C6153 C6154 C6149 C6150 C6151 C6152
MEM_SEL VREF WRSTBI SOC2PM_WRSTBI

IF PMIC(SM5714)

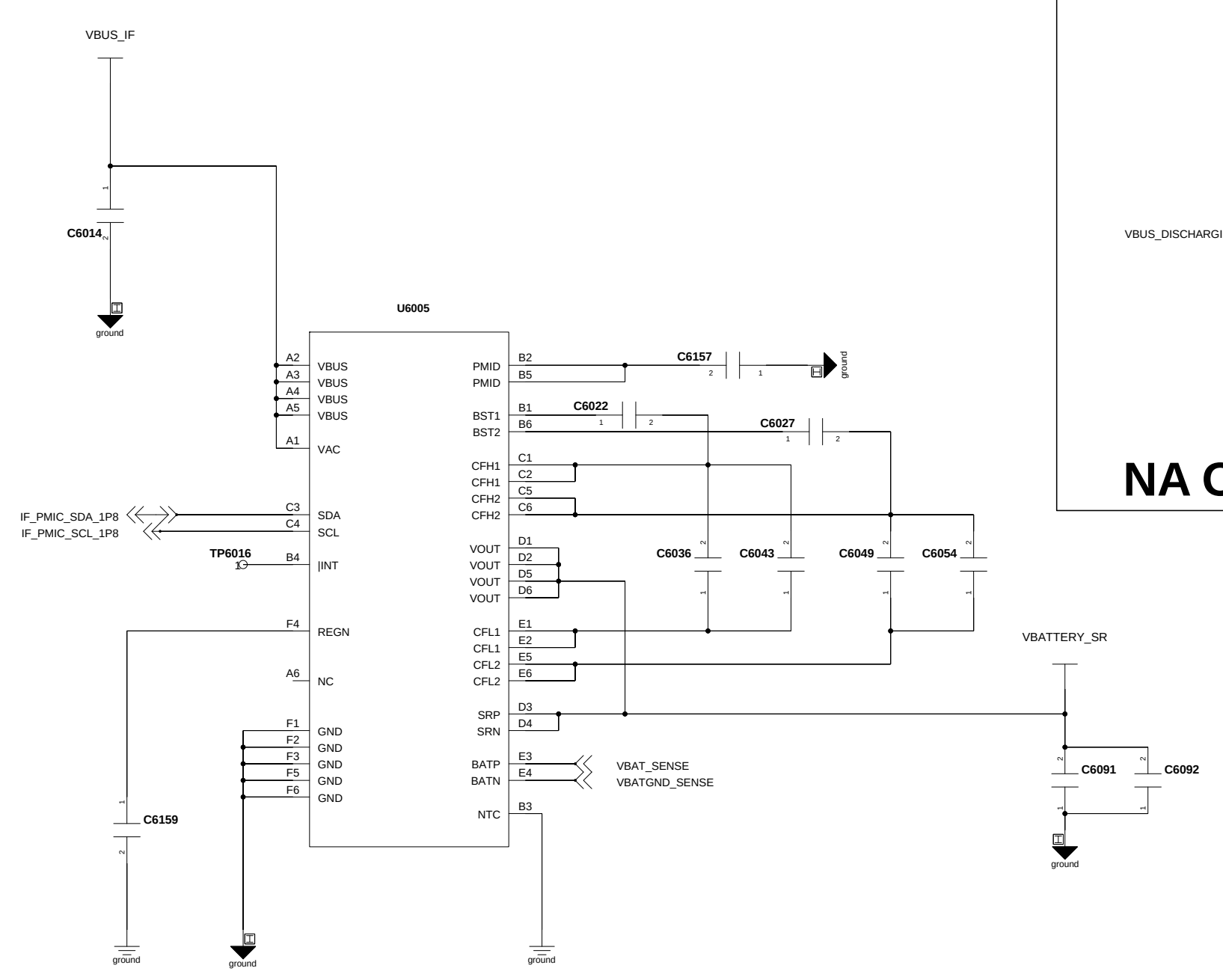

VBUS_IF
C6103 C6016 C6104 C6020
VDD_MID_VBUS
U6004
VBUS_SNS VBUS CC1 CC2 SBU1 SBU2 DP_CON DM_CON DP_HOST DM_HOST RxD TxD SDA SCL SDA_C SCL_C nINT nINT_C VIO nPWRON nENQ4 JIGON JTAG1 JTAG2 RID_MUIC CABLE_POL
MIDVBUS PGND BOOT LX VSYS CHGOUT CSNS_S CSNS_B FGSNSR BATREG BATSNSN VDDFLASH FLED FEN MEN nVBUSPOK MRSTB NOBAT
MUIC_D_P MUIC_D_N USB_AP_D_P USB_AP_D_N AP_UART_RXD AP_UART_TXD
IF_PMIC_SDA_1P8 IF_PMIC_SCL_1P8 CC_SDA_1P8 CC_SCL_1P8 TP6017 TP6018 TP6019 TP6020
L7M_VDDQ_1P8 IF_PMIC_INTB TYPE_C_INT NVBUSOK C6046
VBAT @VPH_PWR R6004 R6005 R6003 R6007 R6008 R6009 JTAG_USB_ID ZD6001 C6053 C6114
C6025 C6112 L6021 R6002 C6110 C6107 C6028 ZD6000 @PGND @VBAT_SENSING
VBATTERY_SR C6041 @CSP VBATTERY_CON VBAT_SENSE VBATGND_SENSE
L6007 C6113 C6166 LED6000 ZD6006 OPEN VOL
R6021 TP6010 TP6011 CAM_FLASH_EN CAM_TORCH_EN SW_ON
D6001 PM_SW_ON
REV01 0912
L7M_VDDQ_1P8 R6000 CHG_THM TH6000

OVP

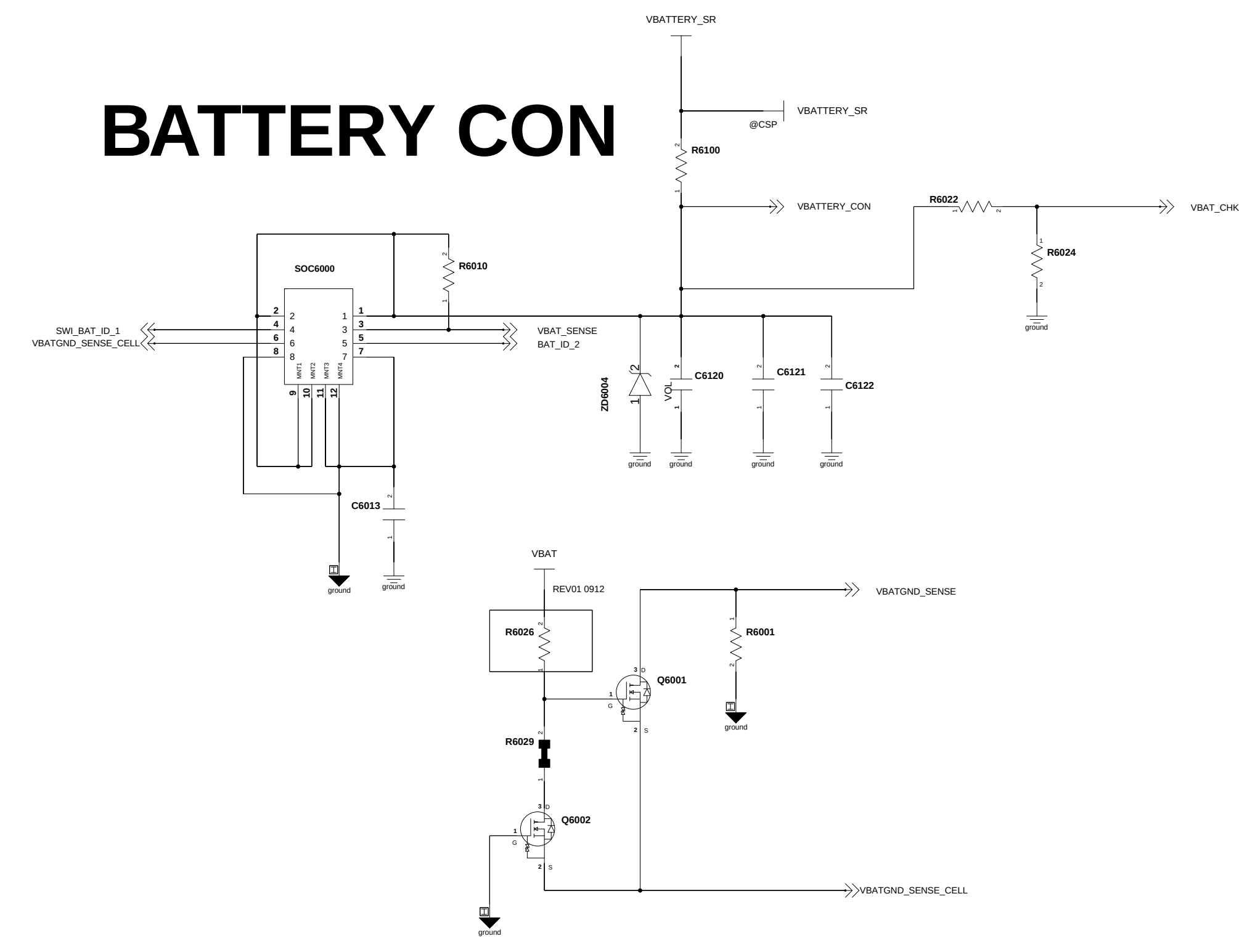

VBUS_OVP @VBUS_OVP_JTAG VBUS_IF
R6011 D6000 L6014
Q6000
VBUS_DISCHARGING
R6013 VOL
C6086
REV02 0918
U6006
IN OUT VDET OVLO ENB GND
C6162 C6087
REV01 0912

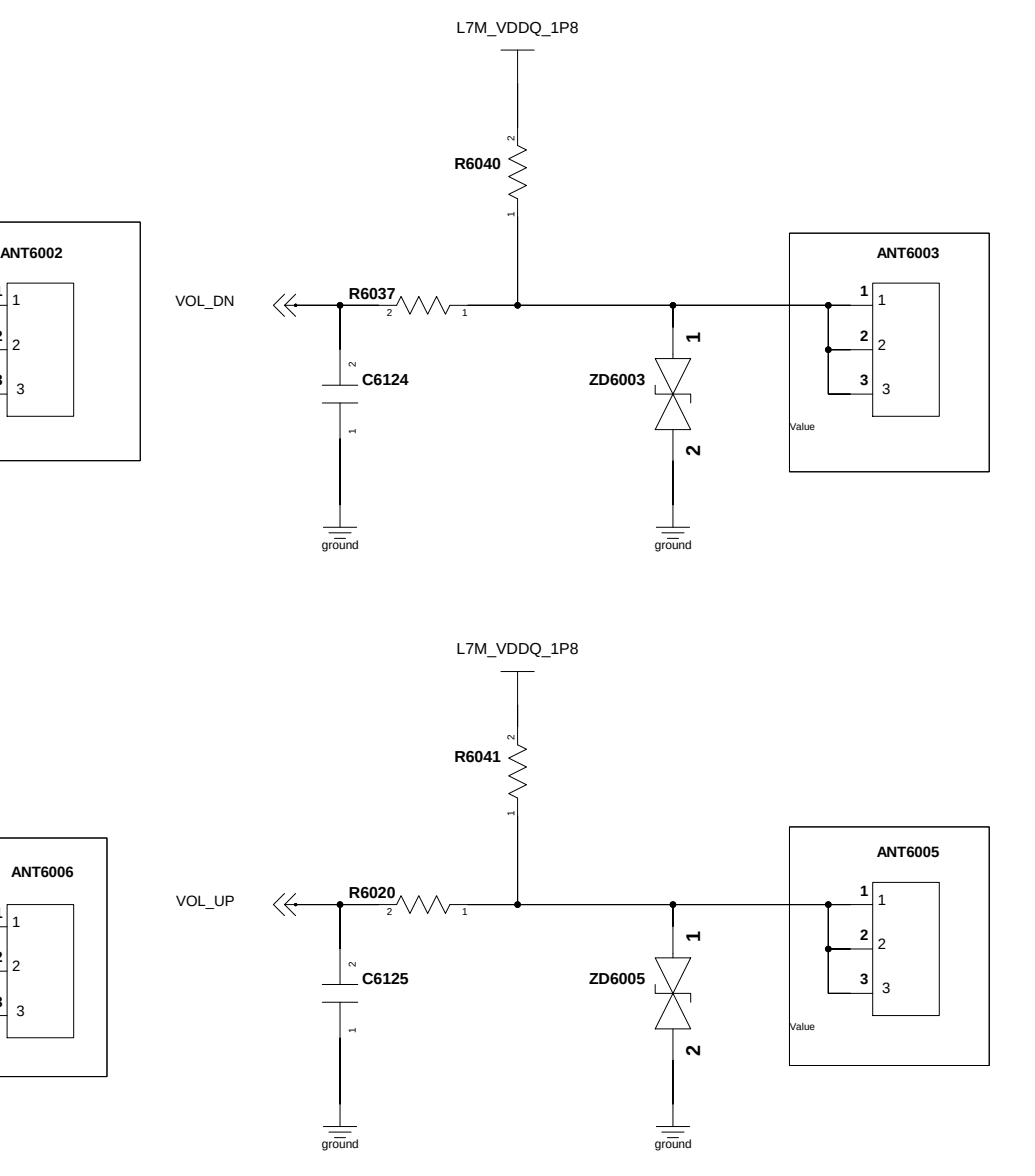
NA ONLY

DIRECT CHARGER


VBUS_IF
C6014
U6005
VBUS VAC SDA SCL |INT REGN NC GND
PMID BST1 BST2 CFH1 CFH2 VOUT CFL1 CFL2 SRP SRN BATP BATN NTC
IF_PMIC_SDA_1P8 IF_PMIC_SCL_1P8 TP6016
C6157 C6022 C6027 C6036 C6043 C6049 C6054 C6159
VBATTERY_SR C6091 C6092
VBAT_SENSE VBATGND_SENSE

BATTERY CON


VBATTERY_SR @CSP
R6100 VBATTERY_CON R6022 VBAT_CHK R6024
SOC6000 R6010
SWI_BAT_ID_1 VBATGND_SENSE_CELL VBAT_SENSE BAT_ID_2
ZD6004 VOL C6120 C6121 C6122 C6013
VBAT REV01 0912 R6026 Q6001 R6001 VBATGND_SENSE
R6029 Q6002 VBATGND_SENSE_CELL
ground


SW_ON R6036 C6123 ZD6002 ANT6002
L7M_VDDQ_1P8 R6040 VOL_DN R6037 C6124 ZD6003 ANT6003
L7M_VDDQ_1P8 R6041 VOL_UP R6020 C6125 ZD6005 ANT6005
ANT6006
Value
ground

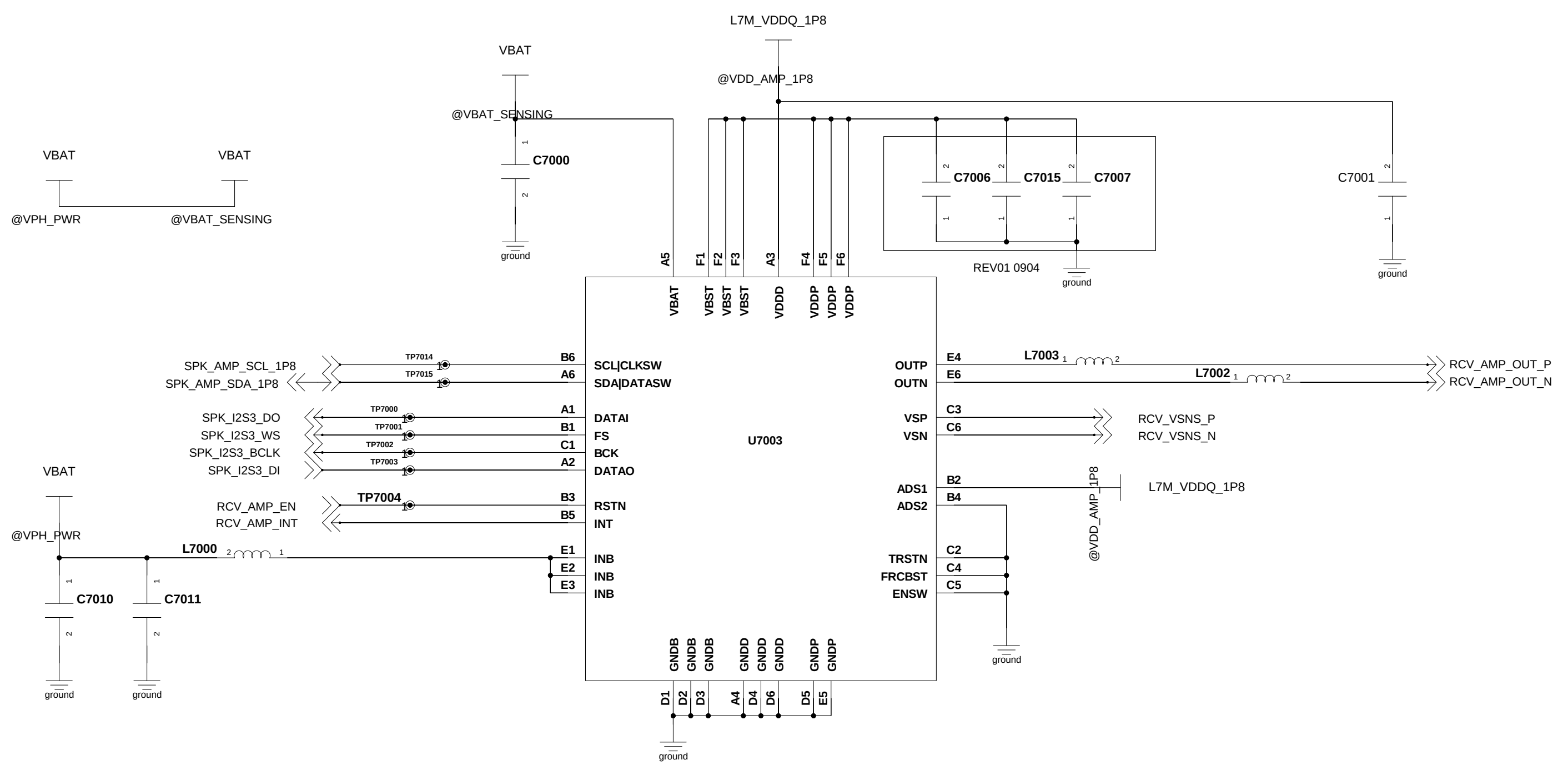
POWER & VOL KEY

Engineer:	A35
Drawn by:	
R&D CHK:	TITLE: POWER/MOT/KEY
DOC CTRL CHK:	Size:
MFG ENGR CHK:	
QA CHK:	REV: Drawing Number:

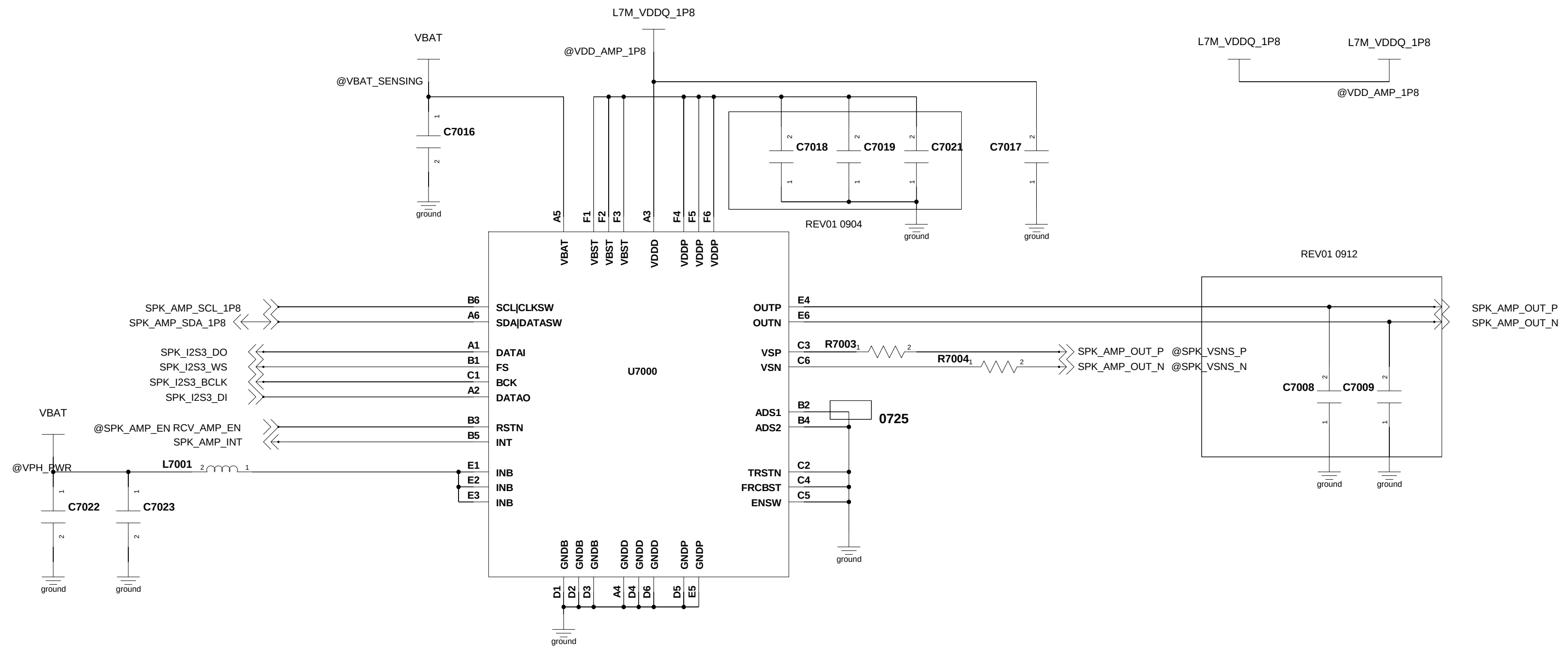
Changed by: Date Changed: Time Changed:

SAMSUNG CONFIDENTIAL
THIS DOCUMENT CONTAINS CONFIDENTIAL
PROPRIETARY INFORMATION THAT IS
SAMSUNG ELECTRONICS CO'S PROPERTY.
DO NOT DISCLOSE TO OR DUPLICATE FOR OTHERS
EXCEPT AS AUTHORIZED BY SAMSUNG.

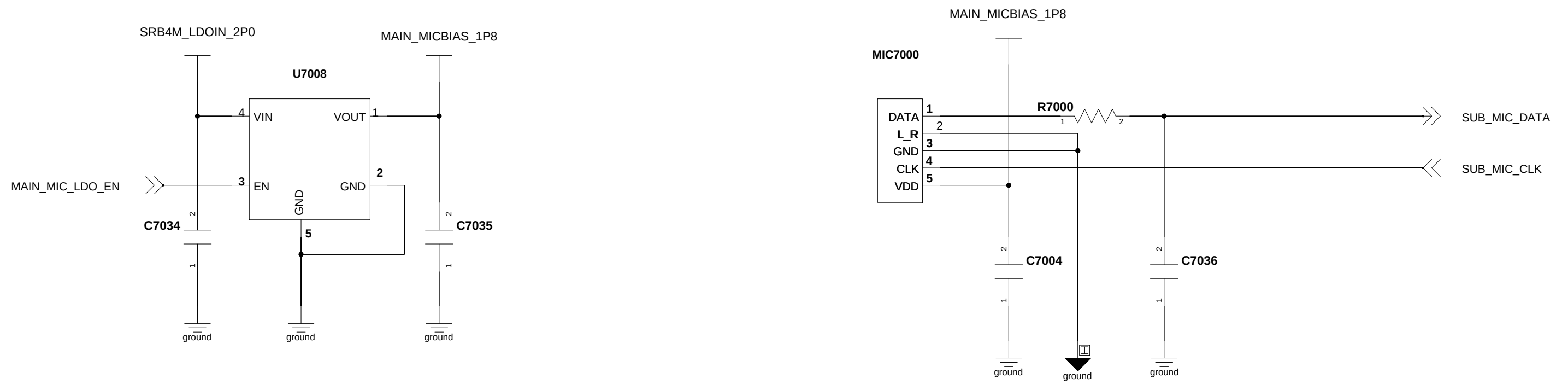
AUDIO & SENSOR PART


VBAT
VBAT
@VPH_PWR
@VBAT_SENSING
VBAT
@VBAT_SENSING
C7000
ground
L7M_VDDQ_1P8
@VDD_AMP_1P8
C7006
C7015
C7007
REV01 0904
ground
C7001
ground
A5
F1
F2
F3
A3
F4
F5
F6
VBAT
VBST
VBST
VBST
VDDD
VDDP
VDDP
VDDP
SPK_AMP_SCL_1P8
SPK_AMP_SDA_1P8
TP7014
TP7015
B6
A6
SCL|CLKSW
SDA|DATASW
SPK_I2S3_DO
SPK_I2S3_WS
SPK_I2S3_BCLK
SPK_I2S3_DI
TP7000
TP7001
TP7002
TP7003
A1
B1
C1
A2
DATAI
FS
BCK
DATAO
VBAT
RCV_AMP_EN
RCV_AMP_INT
TP7004
B3
B5
RSTN
INT
@VPH_PWR
L7000
E1
E2
E3
INB
INB
INB
C7010
C7011
ground
ground
U7003
OUTP
OUTN
E4
E6
L7003
L7002
RCV_AMP_OUT_P
RCV_AMP_OUT_N
VSP
VSN
C3
C6
RCV_VSNS_P
RCV_VSNS_N
ADS1
ADS2
B2
B4
@VDD_AMP_1P8
L7M_VDDQ_1P8
TRSTN
FRCBST
ENSW
C2
C4
C5
ground
GNDB
GNDB
GNDB
GNDD
GNDD
GNDD
GNDP
GNDP
D1
D2
D3
A4
D4
D6
D5
E5
ground

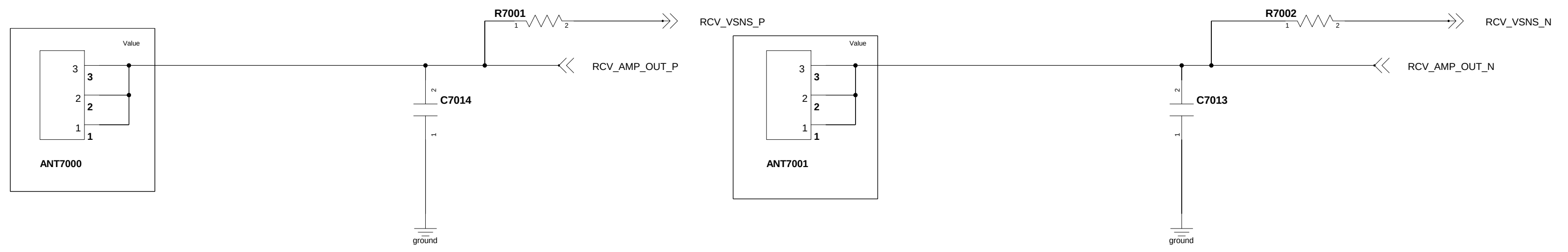
RCV AMP


L7M_VDDQ_1P8
VBAT
@VDD_AMP_1P8
@VBAT_SENSING
C7016
ground
C7018
C7019
C7021
C7017
REV01 0904
ground
ground
L7M_VDDQ_1P8
L7M_VDDQ_1P8
@VDD_AMP_1P8
A5
F1
F2
F3
A3
F4
F5
F6
VBAT
VBST
VBST
VBST
VDDD
VDDP
VDDP
VDDP
SPK_AMP_SCL_1P8
SPK_AMP_SDA_1P8
B6
A6
SCL|CLKSW
SDA|DATASW
SPK_I2S3_DO
SPK_I2S3_WS
SPK_I2S3_BCLK
SPK_I2S3_DI
A1
B1
C1
A2
DATAI
FS
BCK
DATAO
U7000
VBAT
@SPK_AMP_EN RCV_AMP_EN
SPK_AMP_INT
B3
B5
RSTN
INT
@VPH_PWR
L7001
E1
E2
E3
INB
INB
INB
C7022
C7023
ground
ground
OUTP
OUTN
E4
E6
REV01 0912
SPK_AMP_OUT_P
SPK_AMP_OUT_N
VSP
VSN
C3
C6
R7003
R7004
SPK_AMP_OUT_P @SPK_VSNS_P
SPK_AMP_OUT_N @SPK_VSNS_N
C7008
C7009
ground
ground
ADS1
ADS2
B2
B4
0725
TRSTN
FRCBST
ENSW
C2
C4
C5
ground
GNDB
GNDB
GNDB
GNDD
GNDD
GNDD
GNDP
GNDP
D1
D2
D3
A4
D4
D6
D5
E5
ground

SPK AMP

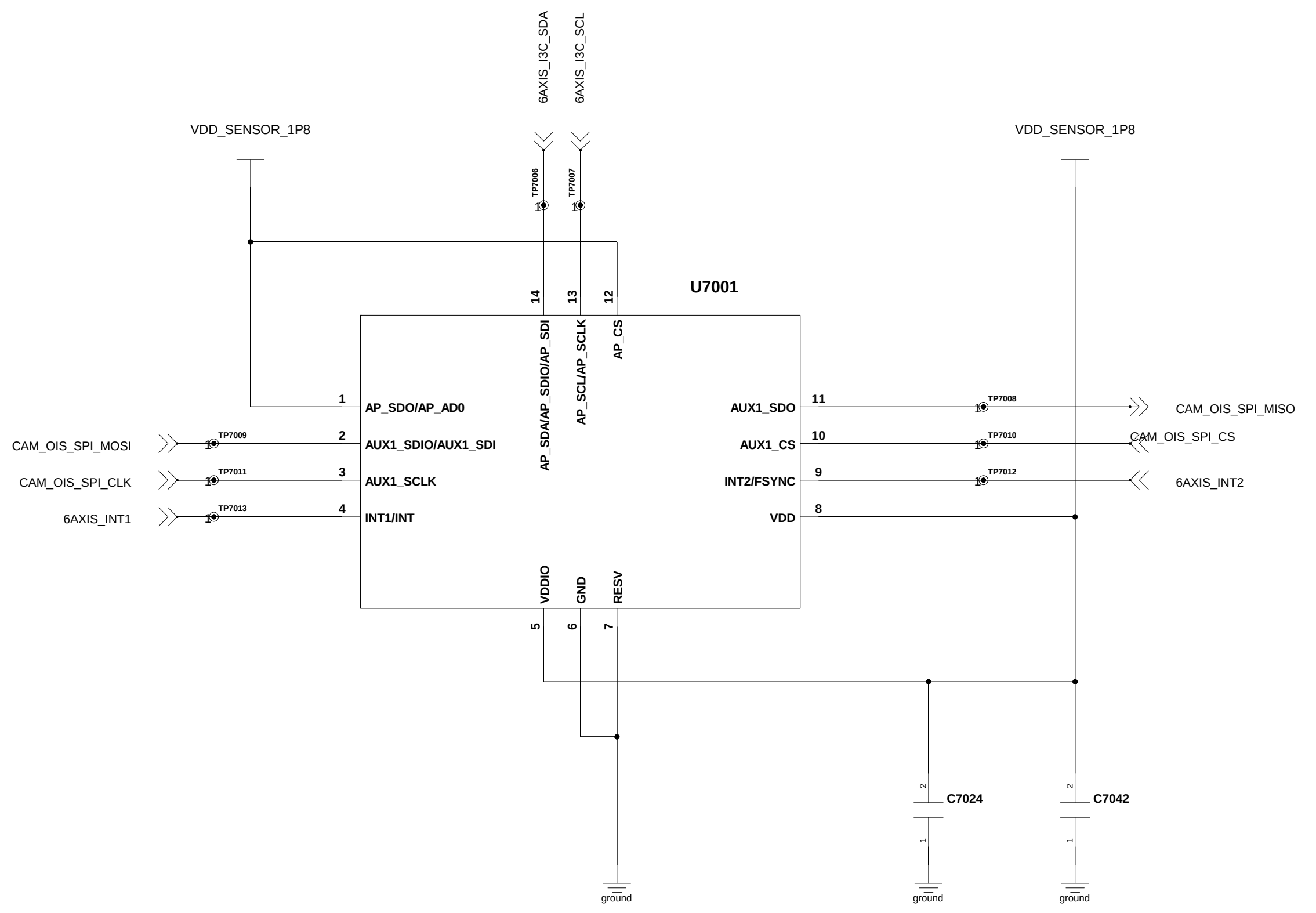

SRB4M_LDOIN_2P0
MAIN_MICBIAS_1P8
U7008
4
VIN
VOUT
1
MAIN_MIC_LDO_EN
3
EN
GND
GND
2
5
C7034
C7035
ground
ground
ground
MAIN_MICBIAS_1P8
MIC7000
DATA
L_R
GND
CLK
VDD
1
2
3
4
5
R7000
SUB_MIC_DATA
SUB_MIC_CLK
C7004
C7036
ground
ground
ground

2nd MIC

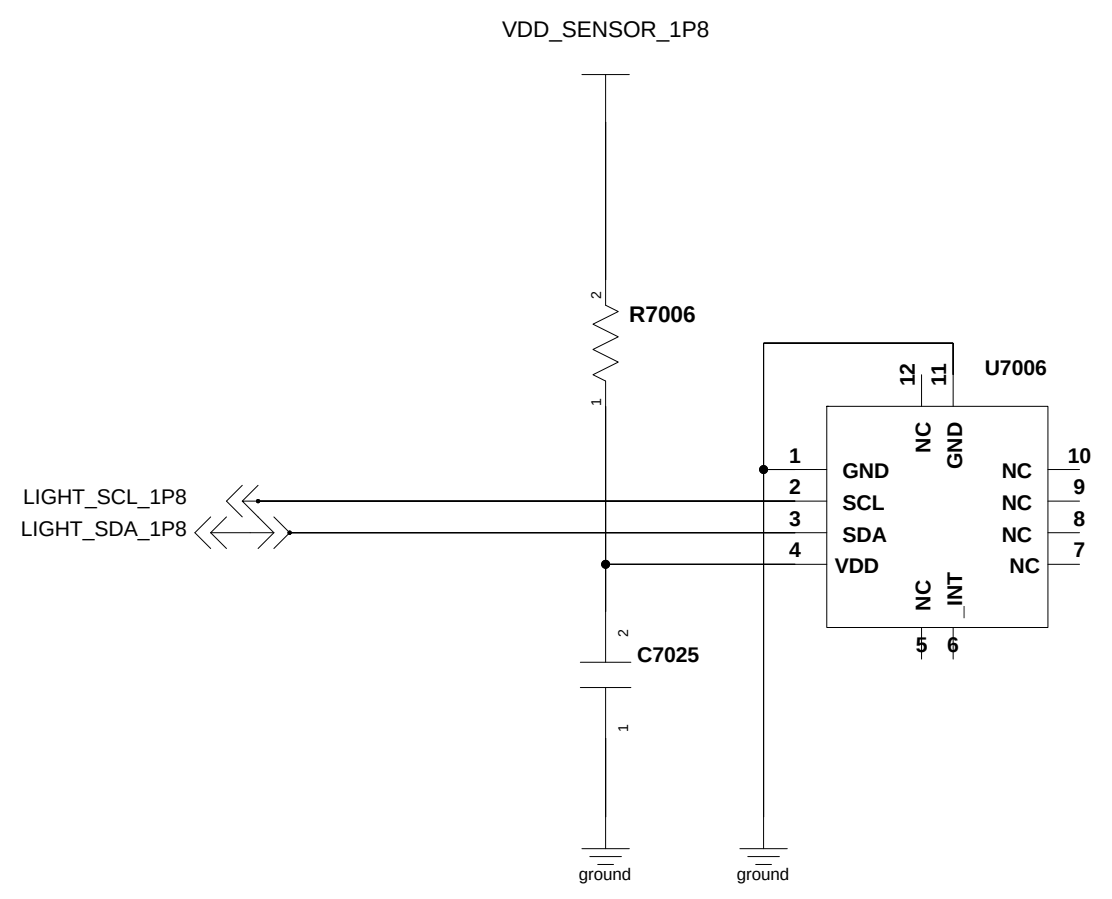

Value
3
2
1
ANT7000
R7001
RCV_VSNS_P
RCV_AMP_OUT_P
C7014
ground
Value
3
2
1
ANT7001
R7002
RCV_VSNS_N
RCV_AMP_OUT_N
C7013
ground

RCV CONTACT

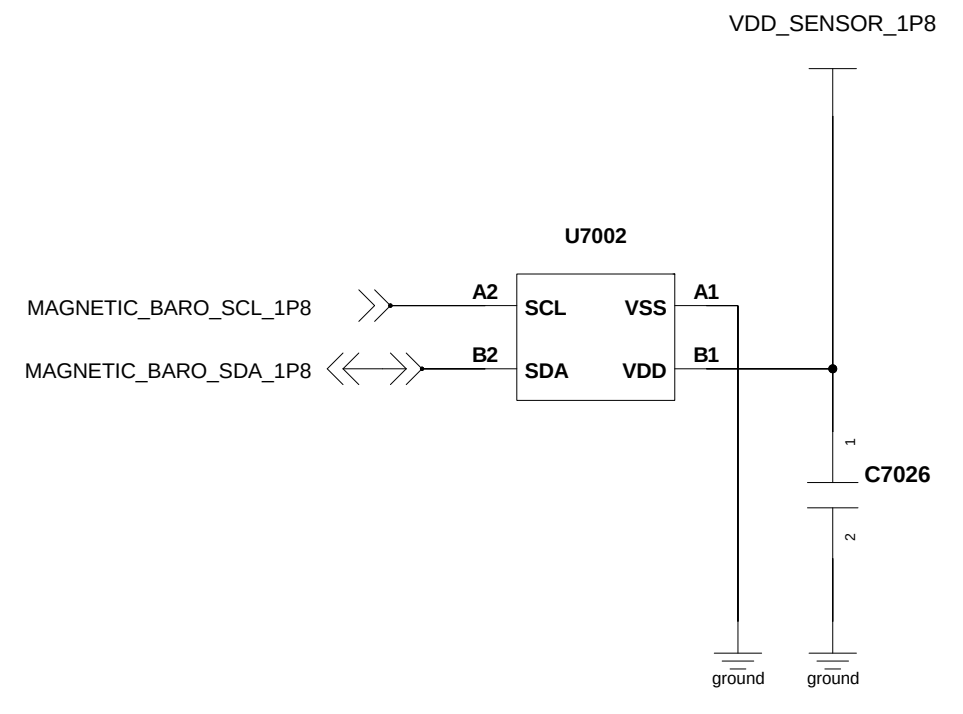
SENSOR


6AXIS_I3C_SDA
6AXIS_I3C_SCL
VDD_SENSOR_1P8
VDD_SENSOR_1P8
TP7006
TP7007
14
13
12
U7001
AP_SDA/AP_SDIO/AP_SDI
AP_SCL/AP_SCLK
AP_CS
1
AP_SDO/AP_AD0
CAM_OIS_SPI_MOSI
TP7009
2
AUX1_SDIO/AUX1_SDI
CAM_OIS_SPI_CLK
TP7011
3
AUX1_SCLK
6AXIS_INT1
TP7013
4
INT1/INT
AUX1_SDO
11
TP7008
CAM_OIS_SPI_MISO
AUX1_CS
10
TP7010
CAM_OIS_SPI_CS
INT2/FSYNC
9
TP7012
6AXIS_INT2
VDD
8
VDDIO
GND
RESV
5
6
7
C7024
C7042
ground
ground
ground

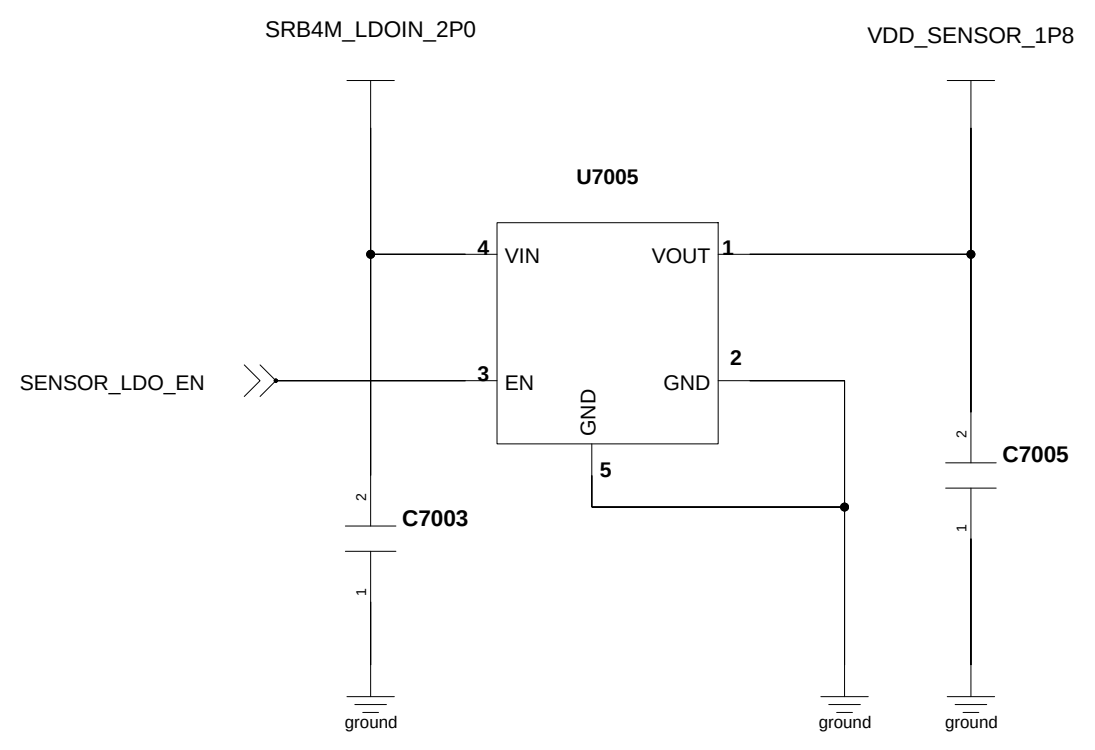
ACC&GYRO SENSOR

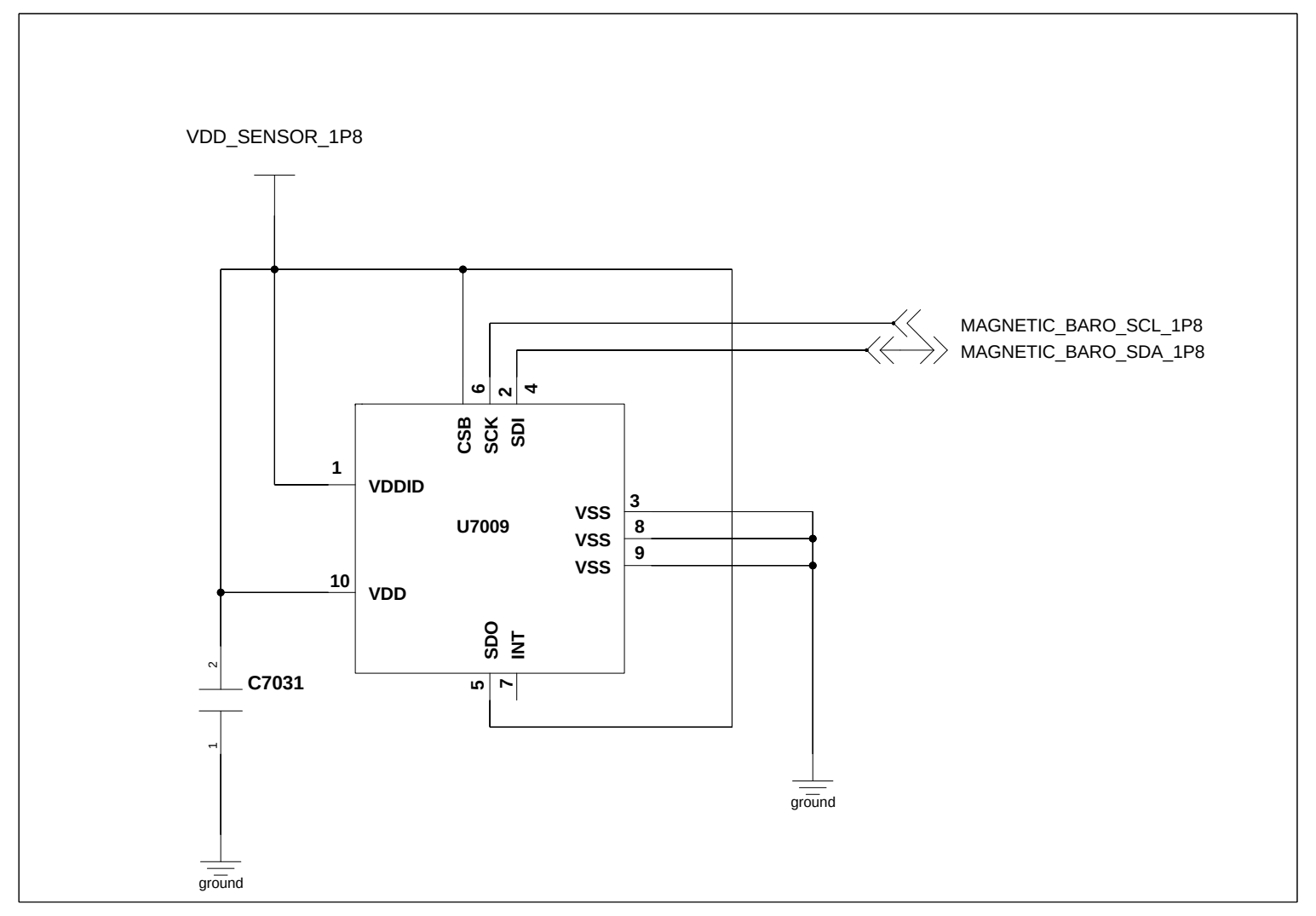

VDD_SENSOR_1P8
R7006
U7006
12
11
NC
GND
LIGHT_SCL_1P8
LIGHT_SDA_1P8
1
2
3
4
GND
SCL
SDA
VDD
NC
NC
NC
NC
10
9
8
7
NC
INT
5
6
C7025
ground
ground

AMBIENT SENSOR


VDD_SENSOR_1P8
U7002
MAGNETIC_BARO_SCL_1P8
MAGNETIC_BARO_SDA_1P8
A2
B2
SCL
SDA
VSS
VDD
A1
B1
C7026
ground
ground

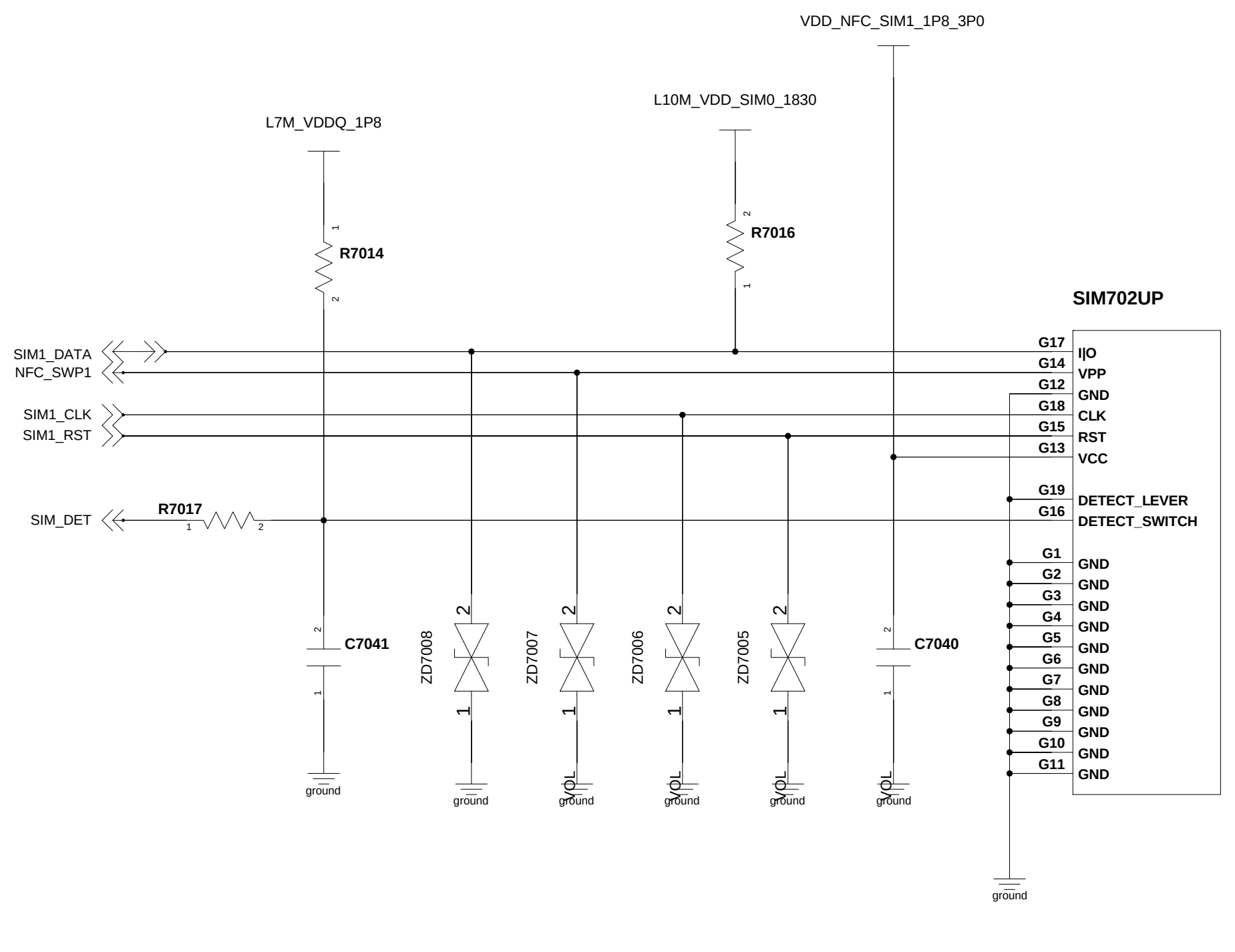
MAGNETIC SENSOR

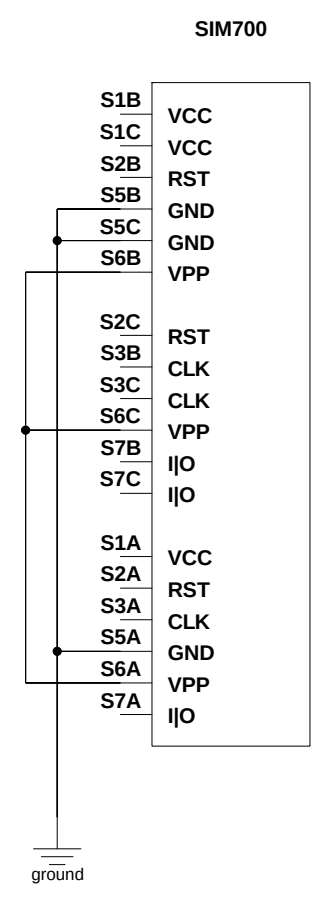

SRB4M_LDOIN_2P0
VDD_SENSOR_1P8
U7005
4
VIN
VOUT
1
SENSOR_LDO_EN
3
EN
GND
GND
2
5
C7003
C7005
ground
ground
ground

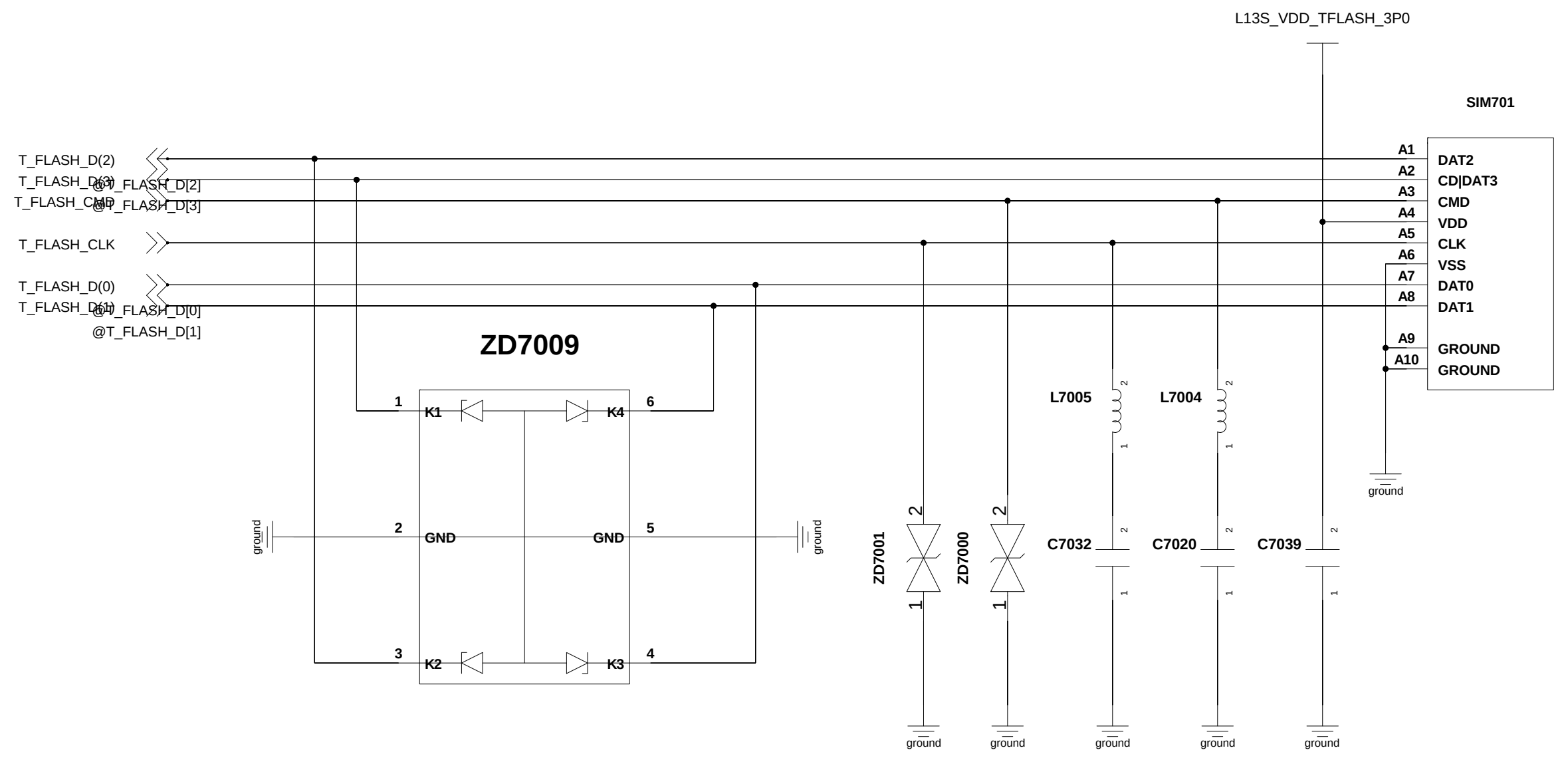

VDD_SENSOR_1P8
MAGNETIC_BARO_SCL_1P8
MAGNETIC_BARO_SDA_1P8
6
2
4
CSB
SCK
SDI
1
VDDID
U7009
VSS
VSS
VSS
3
8
9
10
VDD
SDO
INT
5
7
C7031
ground
ground

NA ONLY

DUAL SIM / T-FLASH (3in2)


VDD_NFC_SIM1_1P8_3P0
L10M_VDD_SIM0_1830
L7M_VDDQ_1P8
R7014
R7016
SIM702UP
SIM1_DATA
NFC_SWP1
SIM1_CLK
SIM1_RST
SIM_DET
R7017
G17
G14
G12
G18
G15
G13
I|O
VPP
GND
CLK
RST
VCC
G19
G16
DETECT_LEVER
DETECT_SWITCH
G1
G2
G3
G4
G5
G6
G7
G8
G9
G10
G11
GND
C7041
ZD7008
ZD7007
ZD7006
ZD7005
C7040
VOL
ground


SIM700
S1B
S1C
S2B
S5B
S5C
S6B
VCC
VCC
RST
GND
GND
VPP
S2C
S3B
S3C
S6C
S7B
S7C
RST
CLK
CLK
VPP
I|O
I|O
S1A
S2A
S3A
S5A
S6A
S7A
VCC
RST
CLK
GND
VPP
I|O
ground

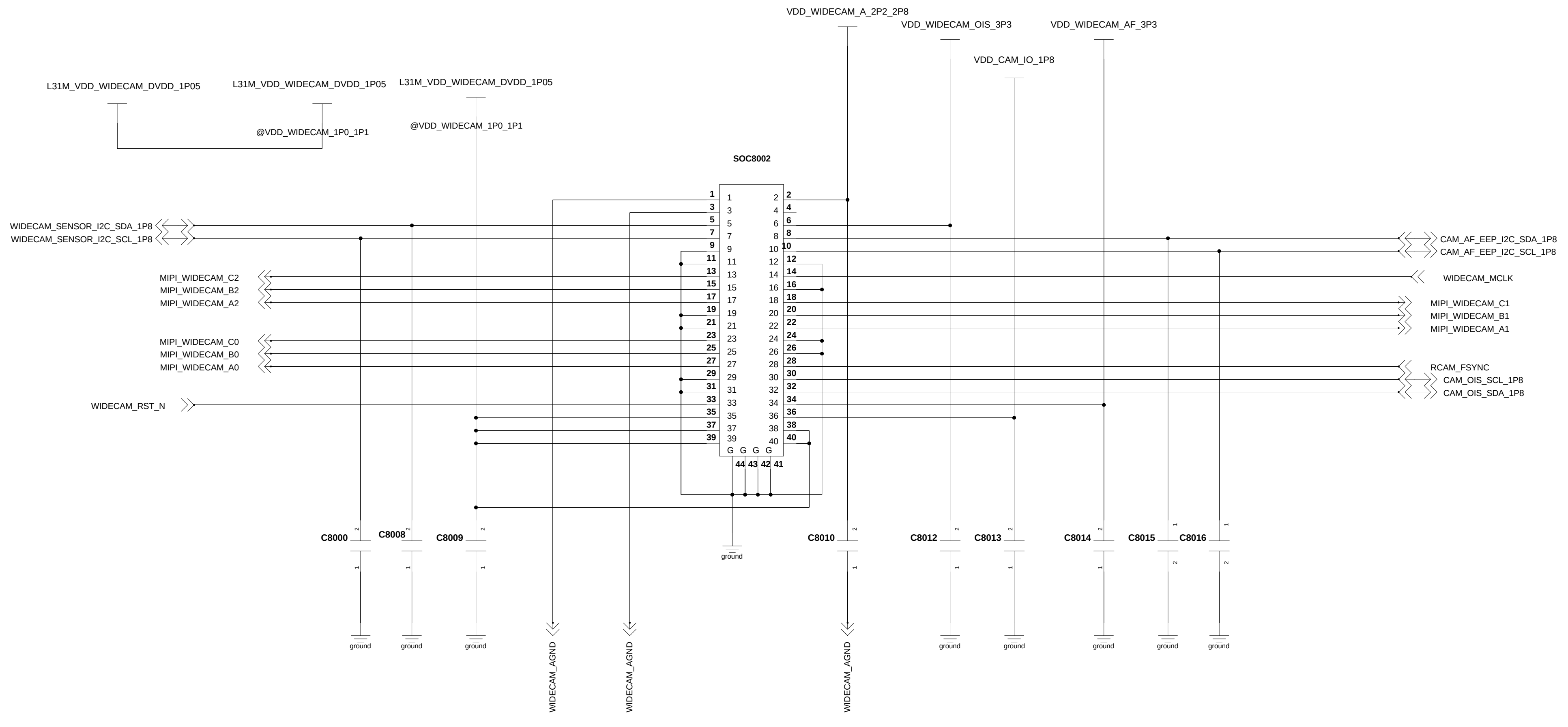

L13S_VDD_TFLASH_3P0
SIM701
T_FLASH_D(2)
T_FLASH_D(3)
T_FLASH_CMD
@T_FLASH_D[2]
@T_FLASH_D[3]
T_FLASH_CLK
T_FLASH_D(0)
T_FLASH_D(1)
@T_FLASH_D[0]
@T_FLASH_D[1]
A1
A2
A3
A4
A5
A6
A7
A8
DAT2
CD|DAT3
CMD
VDD
CLK
VSS
DAT0
DAT1
A9
A10
GROUND
GROUND
ZD7009
1
K1
K4
6
2
GND
GND
5
3
K2
K3
4
ground
ground
L7005
L7004
ZD7001
ZD7000
C7032
C7020
C7039
ground

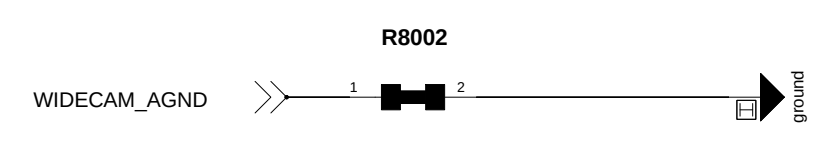
Stacked Dual SIM SOCKET (Reverse)

Engineer:
Drawn by:
A35
R&D CHK:
TITLE:
Size:
DOC CTRL CHK:
MFG ENGR CHK:
Changed by:
Date Changed:
Time Changed:
QA CHK:
REV:
Drawing Number:
Page:

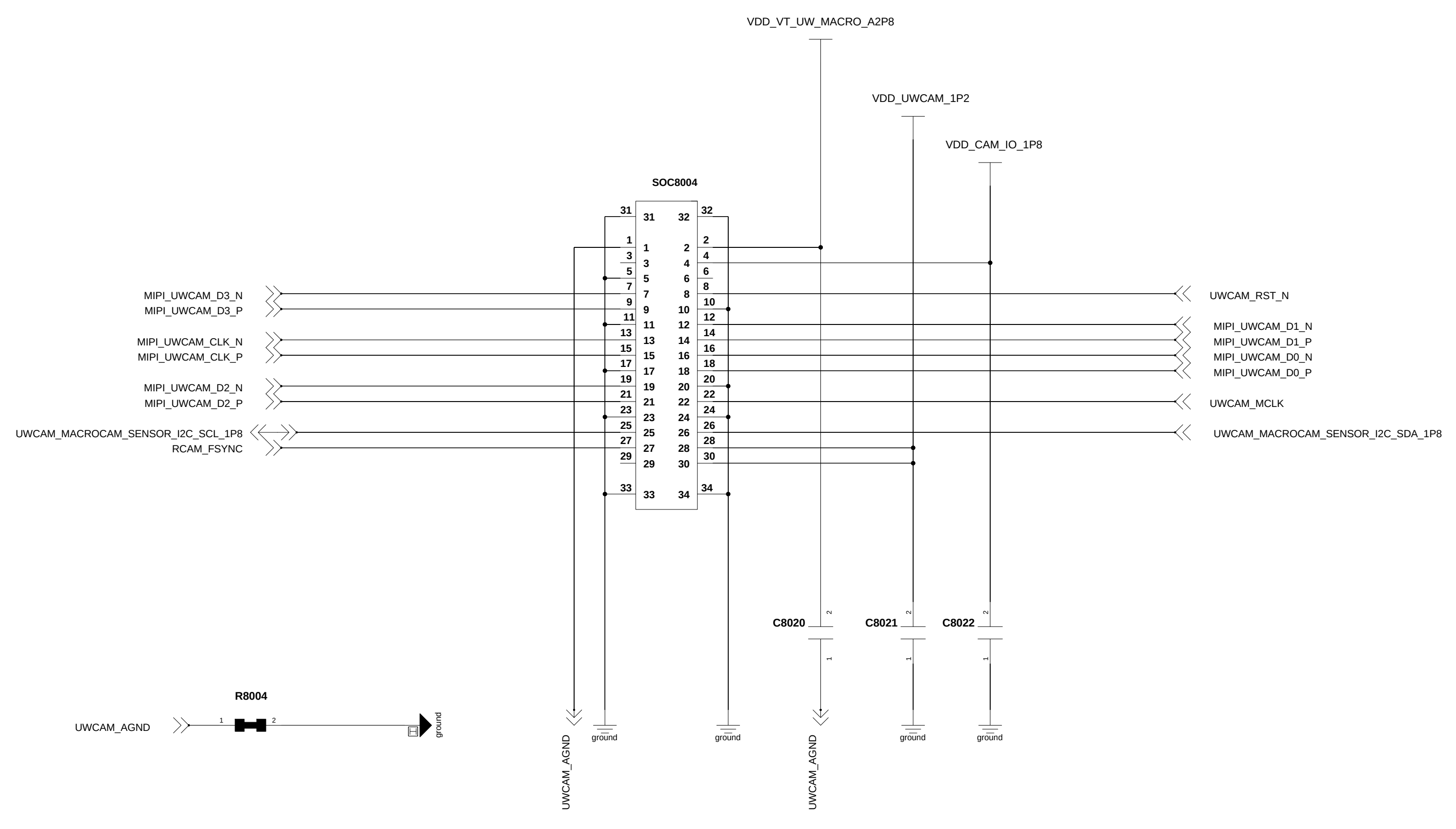
SAMSUNG CONFIDENTIAL
THIS DOCUMENT CONTAINS CONFIDENTIAL
PROPRIETARY INFORMATION THAT IS
SAMSUNG ELECTRONICS CO'S PROPERTY.
DO NOT DISCLOSE TO OR DUPLICATE FOR OTHERS
EXCEPT AS AUTHORIZED BY SAMSUNG.

CAMERA

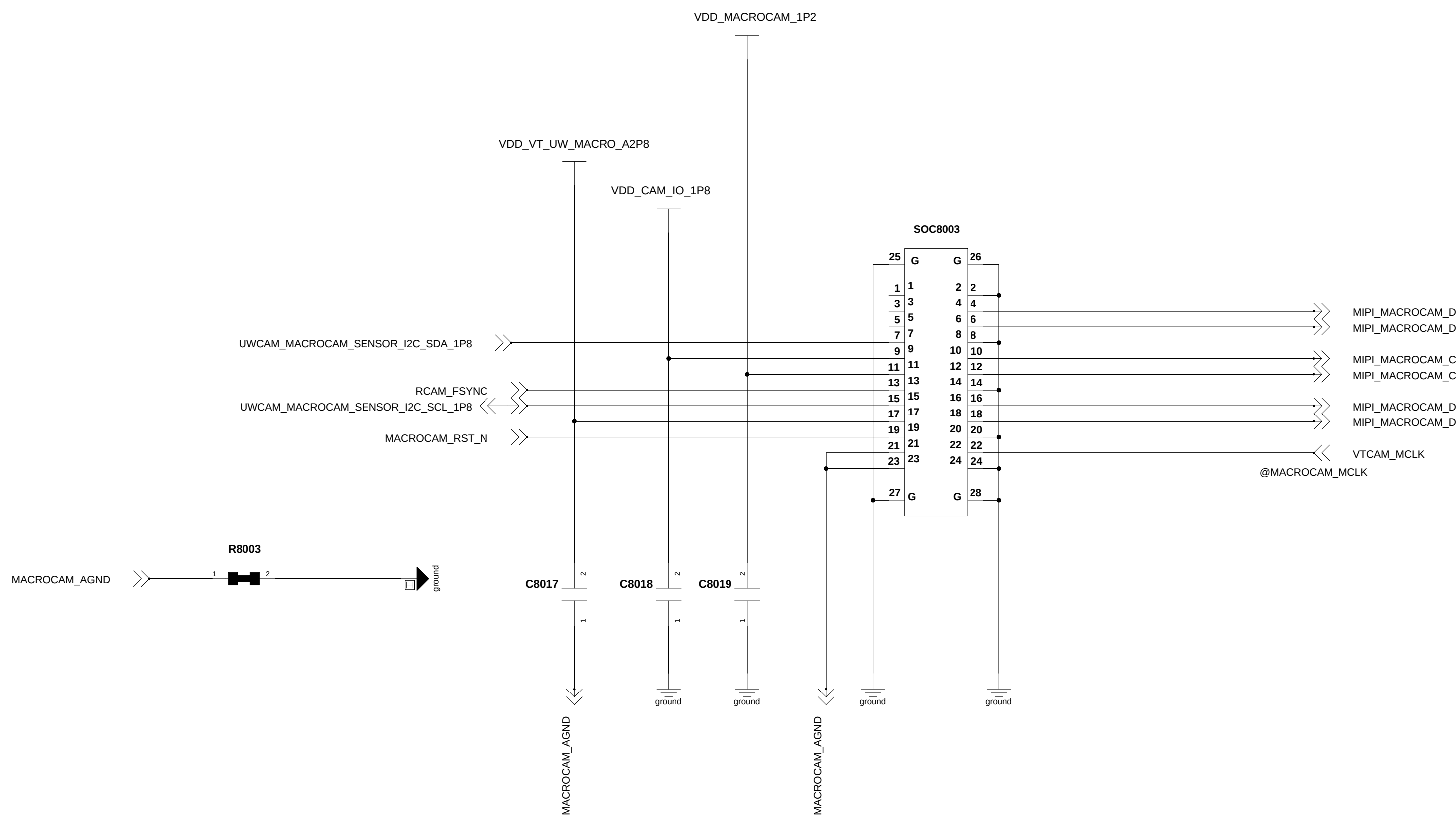

L31M_VDD_WIDECAM_DVDD_1P05
L31M_VDD_WIDECAM_DVDD_1P05
L31M_VDD_WIDECAM_DVDD_1P05
@VDD_WIDECAM_1P0_1P1
@VDD_WIDECAM_1P0_1P1
VDD_WIDECAM_A_2P2_2P8
VDD_WIDECAM_OIS_3P3
VDD_CAM_IO_1P8
VDD_WIDECAM_AF_3P3
SOC8002
WIDECAM_SENSOR_I2C_SDA_1P8
WIDECAM_SENSOR_I2C_SCL_1P8
MIPI_WIDECAM_C2
MIPI_WIDECAM_B2
MIPI_WIDECAM_A2
MIPI_WIDECAM_C0
MIPI_WIDECAM_B0
MIPI_WIDECAM_A0
WIDECAM_RST_N
CAM_AF_EEP_I2C_SDA_1P8
CAM_AF_EEP_I2C_SCL_1P8
WIDECAM_MCLK
MIPI_WIDECAM_C1
MIPI_WIDECAM_B1
MIPI_WIDECAM_A1
RCAM_FSYNC
CAM_OIS_SCL_1P8
CAM_OIS_SDA_1P8
G G G G
44 43 42 41
C8000
C8008
C8009
C8010
C8012
C8013
C8014
C8015
C8016
ground
WIDECAM_AGND
WIDECAM_AGND
WIDECAM_AGND


R8002
WIDECAM_AGND
1
2
ground

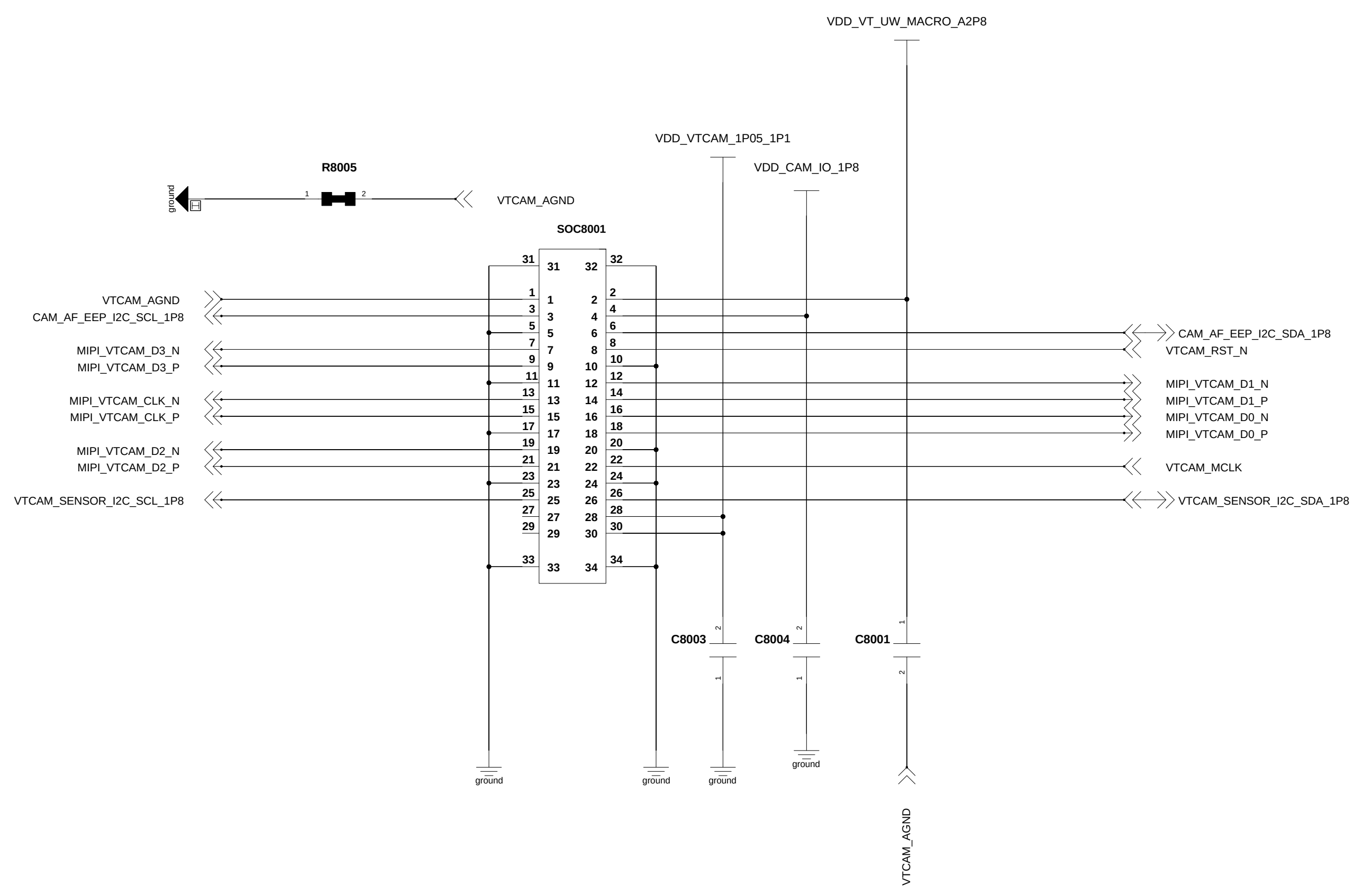
WIDECAM_50M OIS(40PIN)


VDD_VT_UW_MACRO_A2P8
VDD_UWCAM_1P2
VDD_CAM_IO_1P8
SOC8004
MIPI_UWCAM_D3_N
MIPI_UWCAM_D3_P
MIPI_UWCAM_CLK_N
MIPI_UWCAM_CLK_P
MIPI_UWCAM_D2_N
MIPI_UWCAM_D2_P
UWCAM_MACROCAM_SENSOR_I2C_SCL_1P8
RCAM_FSYNC
UWCAM_RST_N
MIPI_UWCAM_D1_N
MIPI_UWCAM_D1_P
MIPI_UWCAM_D0_N
MIPI_UWCAM_D0_P
UWCAM_MCLK
UWCAM_MACROCAM_SENSOR_I2C_SDA_1P8
C8020
C8021
C8022
R8004
UWCAM_AGND
ground
UWCAM_AGND
UWCAM_AGND

ULTRAWIDE_8M CAMERA(30pin)

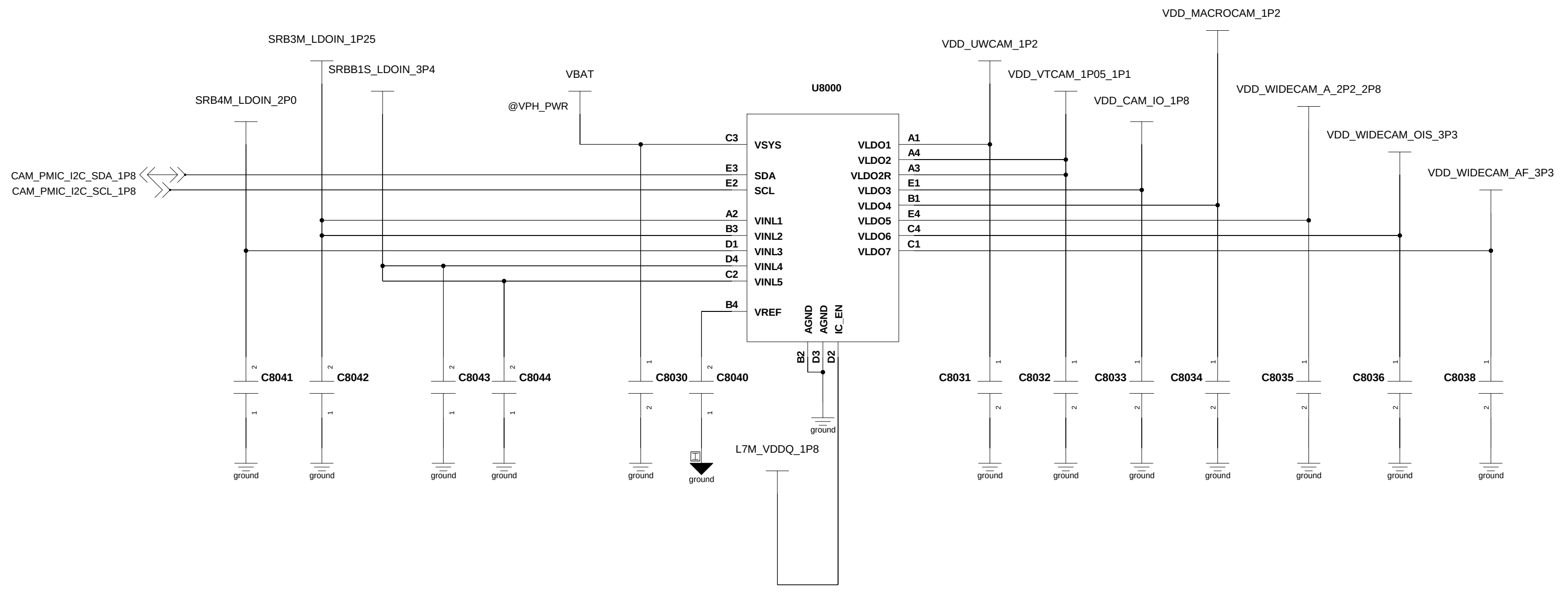

VDD_MACROCAM_1P2
VDD_VT_UW_MACRO_A2P8
VDD_CAM_IO_1P8
SOC8003
UWCAM_MACROCAM_SENSOR_I2C_SDA_1P8
RCAM_FSYNC
UWCAM_MACROCAM_SENSOR_I2C_SCL_1P8
MACROCAM_RST_N
MIPI_MACROCAM_D1_P
MIPI_MACROCAM_D1_N
MIPI_MACROCAM_CLK_P
MIPI_MACROCAM_CLK_N
MIPI_MACROCAM_D0_P
MIPI_MACROCAM_D0_N
VTCAM_MCLK
@MACROCAM_MCLK
R8003
MACROCAM_AGND
C8017
C8018
C8019
ground
MACROCAM_AGND
MACROCAM_AGND

MACRO_5M_CAMERA(24PIN)

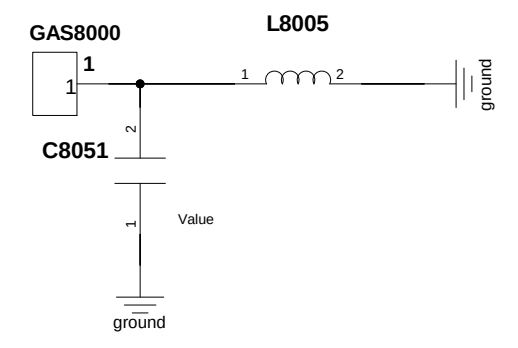

VDD_VT_UW_MACRO_A2P8
VDD_VTCAM_1P05_1P1
VDD_CAM_IO_1P8
R8005
ground
VTCAM_AGND
SOC8001
VTCAM_AGND
CAM_AF_EEP_I2C_SCL_1P8
MIPI_VTCAM_D3_N
MIPI_VTCAM_D3_P
MIPI_VTCAM_CLK_N
MIPI_VTCAM_CLK_P
MIPI_VTCAM_D2_N
MIPI_VTCAM_D2_P
VTCAM_SENSOR_I2C_SCL_1P8
CAM_AF_EEP_I2C_SDA_1P8
VTCAM_RST_N
MIPI_VTCAM_D1_N
MIPI_VTCAM_D1_P
MIPI_VTCAM_D0_N
MIPI_VTCAM_D0_P
VTCAM_MCLK
VTCAM_SENSOR_I2C_SDA_1P8
C8003
C8004
C8001
VTCAM_AGND

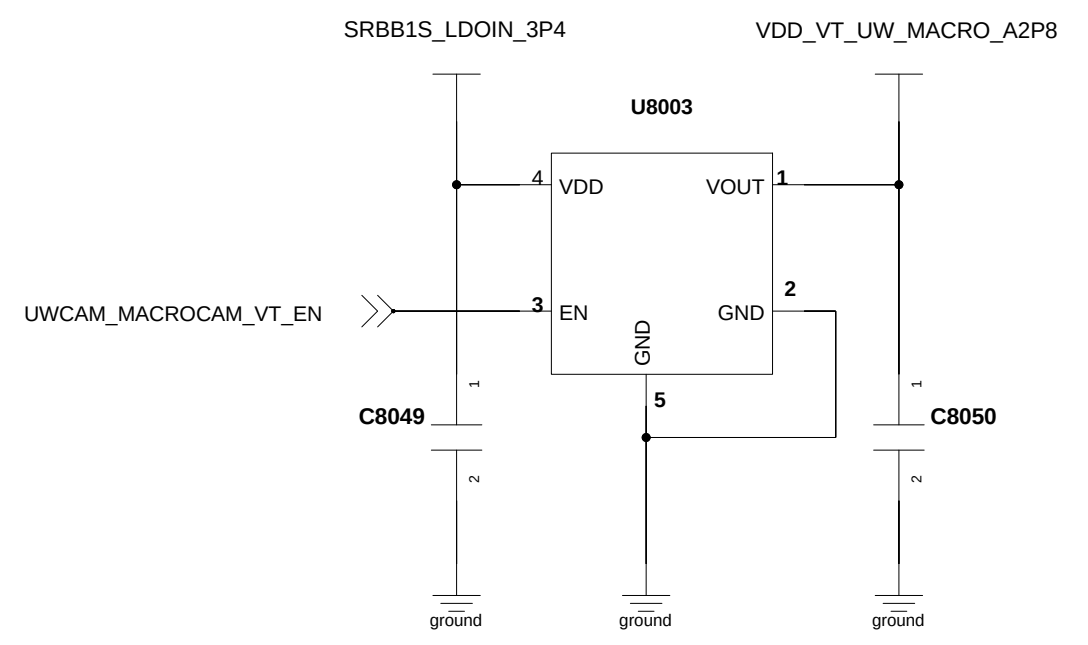
VTCAM_32M CAMERA (30PIN)

CAM PMIC


SRB3M_LDOIN_1P25
SRBB1S_LDOIN_3P4
SRB4M_LDOIN_2P0
VBAT
@VPH_PWR
U8000
CAM_PMIC_I2C_SDA_1P8
CAM_PMIC_I2C_SCL_1P8
C3 VSYS
E3 SDA
E2 SCL
A2 VINL1
B3 VINL2
D1 VINL3
D4 VINL4
C2 VINL5
B4 VREF
AGND
AGND
IC_EN
B2
D3
D2
VLDO1 A1
VLDO2 A4
VLDO2R A3
VLDO3 E1
VLDO4 B1
VLDO5 E4
VLDO6 C4
VLDO7 C1
VDD_UWCAM_1P2
VDD_VTCAM_1P05_1P1
VDD_CAM_IO_1P8
VDD_MACROCAM_1P2
VDD_WIDECAM_A_2P2_2P8
VDD_WIDECAM_OIS_3P3
VDD_WIDECAM_AF_3P3
C8041
C8042
C8043
C8044
C8030
C8040
C8031
C8032
C8033
C8034
C8035
C8036
C8038
L7M_VDDQ_1P8
ground

DECO CONTACT


GAS8000
1
L8005
C8051
Value
ground


SRBB1S_LDOIN_3P4
VDD_VT_UW_MACRO_A2P8
U8003
4 VDD
VOUT 1
UWCAM_MACROCAM_VT_EN
3 EN
GND 2
GND 5
C8049
C8050
ground

VTCAM_UWCAM_MACROCAM

Engineer:
Drawn by:
R&D CHK:
DOC CTRL CHK:
MFG ENGR CHK:
QA CHK:

A35

TITLE: CAMERA

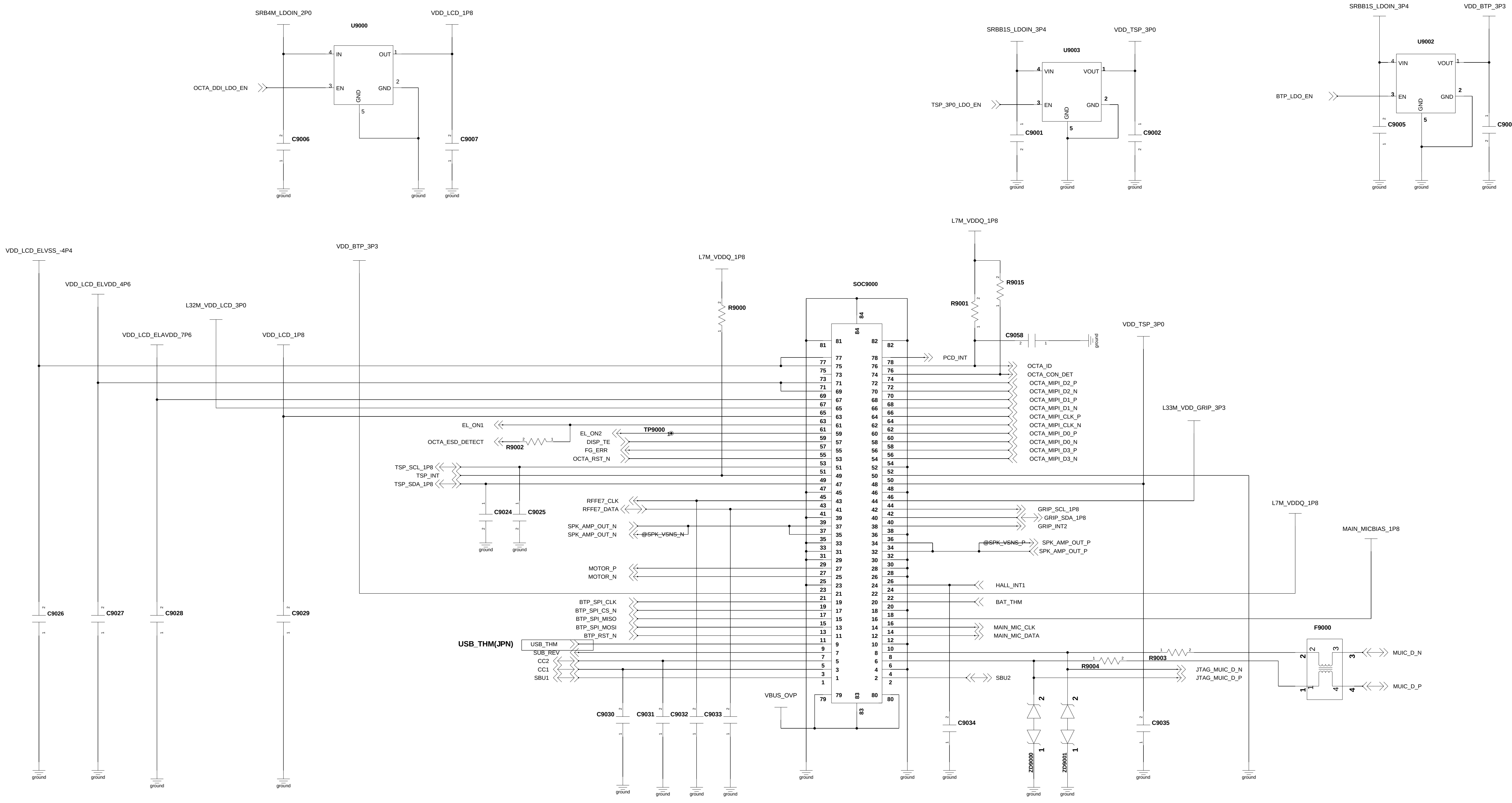
Size:

Changed by: Date Changed: Time Changed:

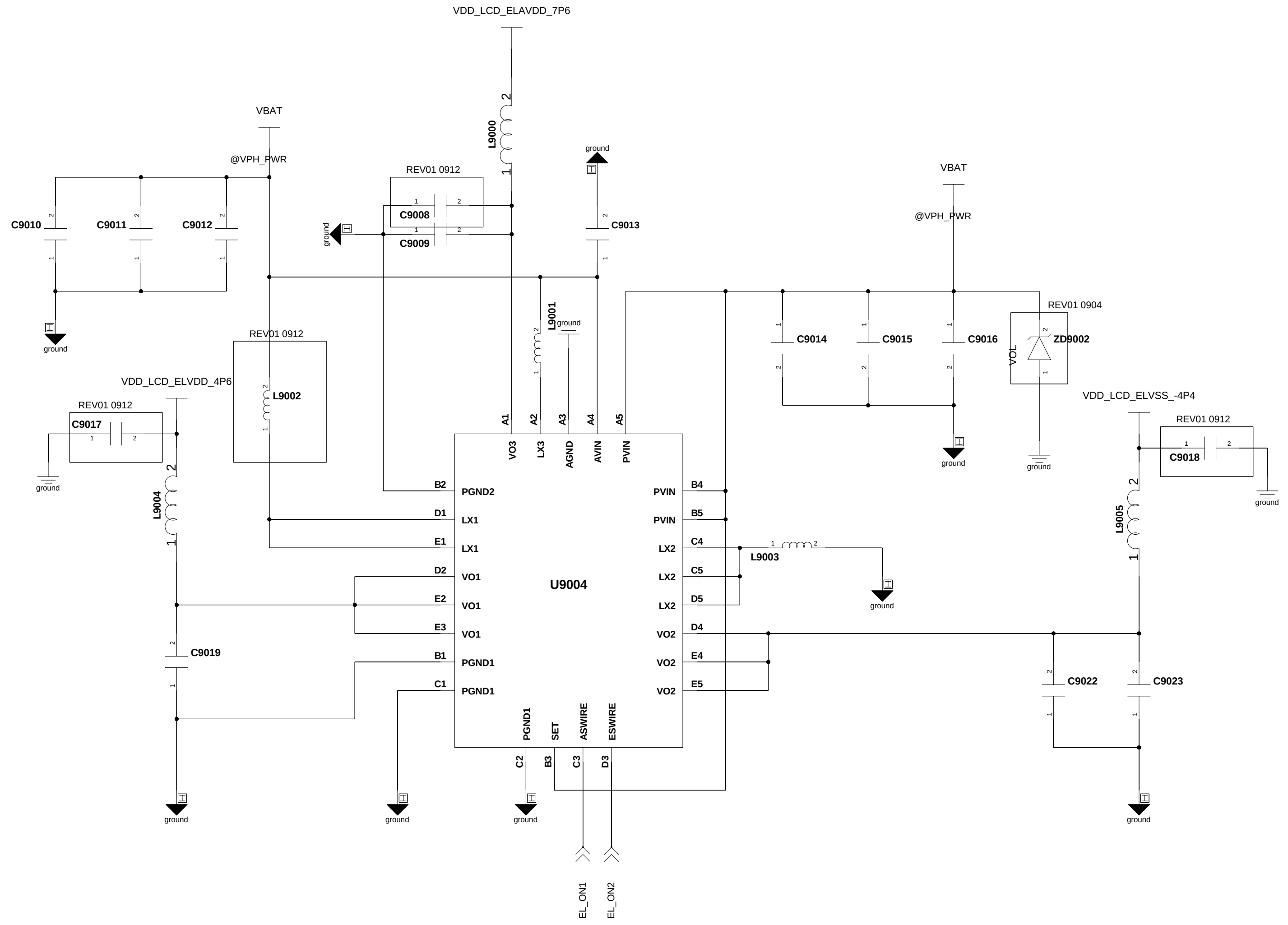
REV: Drawing Number: Page:

SAMSUNG CONFIDENTIAL
THIS DOCUMENT CONTAINS CONFIDENTIAL
PROPRIETARY INFORMATION THAT IS
SAMSUNG ELECTRONICS CO'S PROPERTY.
DO NOT DISCLOSE TO OR DUPLICATE FOR OTHERS
EXCEPT AS AUTHORIZED BY SAMSUNG.

C2C (Conn to Conn FPCB)

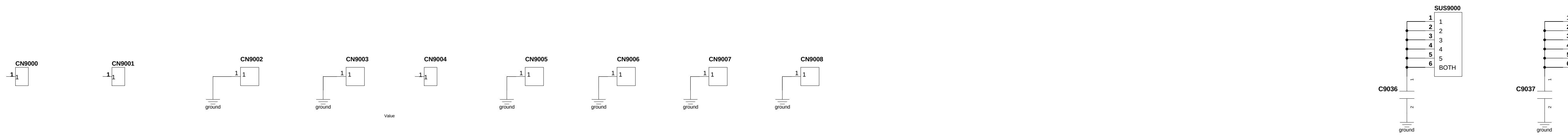

SRB4M_LDOIN_2P0
VDD_LCD_1P8
U9000
IN
OUT
OCTA_DDI_LDO_EN
EN
GND
C9006
C9007
ground
SRBB1S_LDOIN_3P4
VDD_TSP_3P0
U9003
VIN
VOUT
TSP_3P0_LDO_EN
C9001
C9002
SRBB1S_LDOIN_3P4
VDD_BTP_3P3
U9002
BTP_LDO_EN
C9005
C9003
VDD_LCD_ELVSS_-4P4
VDD_LCD_ELVDD_4P6
L32M_VDD_LCD_3P0
VDD_LCD_ELAVDD_7P6
VDD_LCD_1P8
VDD_BTP_3P3
L7M_VDDQ_1P8
R9000
SOC9000
L7M_VDDQ_1P8
R9001
R9015
C9058
PCD_INT
OCTA_ID
OCTA_CON_DET
OCTA_MIPI_D2_P
OCTA_MIPI_D2_N
OCTA_MIPI_D1_P
OCTA_MIPI_D1_N
OCTA_MIPI_CLK_P
OCTA_MIPI_CLK_N
OCTA_MIPI_D0_P
OCTA_MIPI_D0_N
OCTA_MIPI_D3_P
OCTA_MIPI_D3_N
VDD_TSP_3P0
L33M_VDD_GRIP_3P3
EL_ON1
EL_ON2
TP9000
OCTA_ESD_DETECT
R9002
DISP_TE
FG_ERR
OCTA_RST_N
TSP_SCL_1P8
TSP_INT
TSP_SDA_1P8
C9024
C9025
RFFE7_CLK
RFFE7_DATA
SPK_AMP_OUT_N
SPK_AMP_OUT_N
@SPK_VSNS_N
GRIP_SCL_1P8
GRIP_SDA_1P8
GRIP_INT2
@SPK_VSNS_P
SPK_AMP_OUT_P
SPK_AMP_OUT_P
MOTOR_P
MOTOR_N
HALL_INT1
BAT_THM
L7M_VDDQ_1P8
MAIN_MICBIAS_1P8
BTP_SPI_CLK
BTP_SPI_CS_N
BTP_SPI_MISO
BTP_SPI_MOSI
BTP_RST_N
MAIN_MIC_CLK
MAIN_MIC_DATA
USB_THM(JPN)
USB_THM
SUB_REV
CC2
CC1
SBU1
SBU2
F9000
MUIC_D_N
MUIC_D_P
R9003
R9004
JTAG_MUIC_D_N
JTAG_MUIC_D_P
VBUS_OVP
C9026
C9027
C9028
C9029
C9030
C9031
C9032
C9033
C9034
C9035
ZD9000
ZD9001

DISPLAY

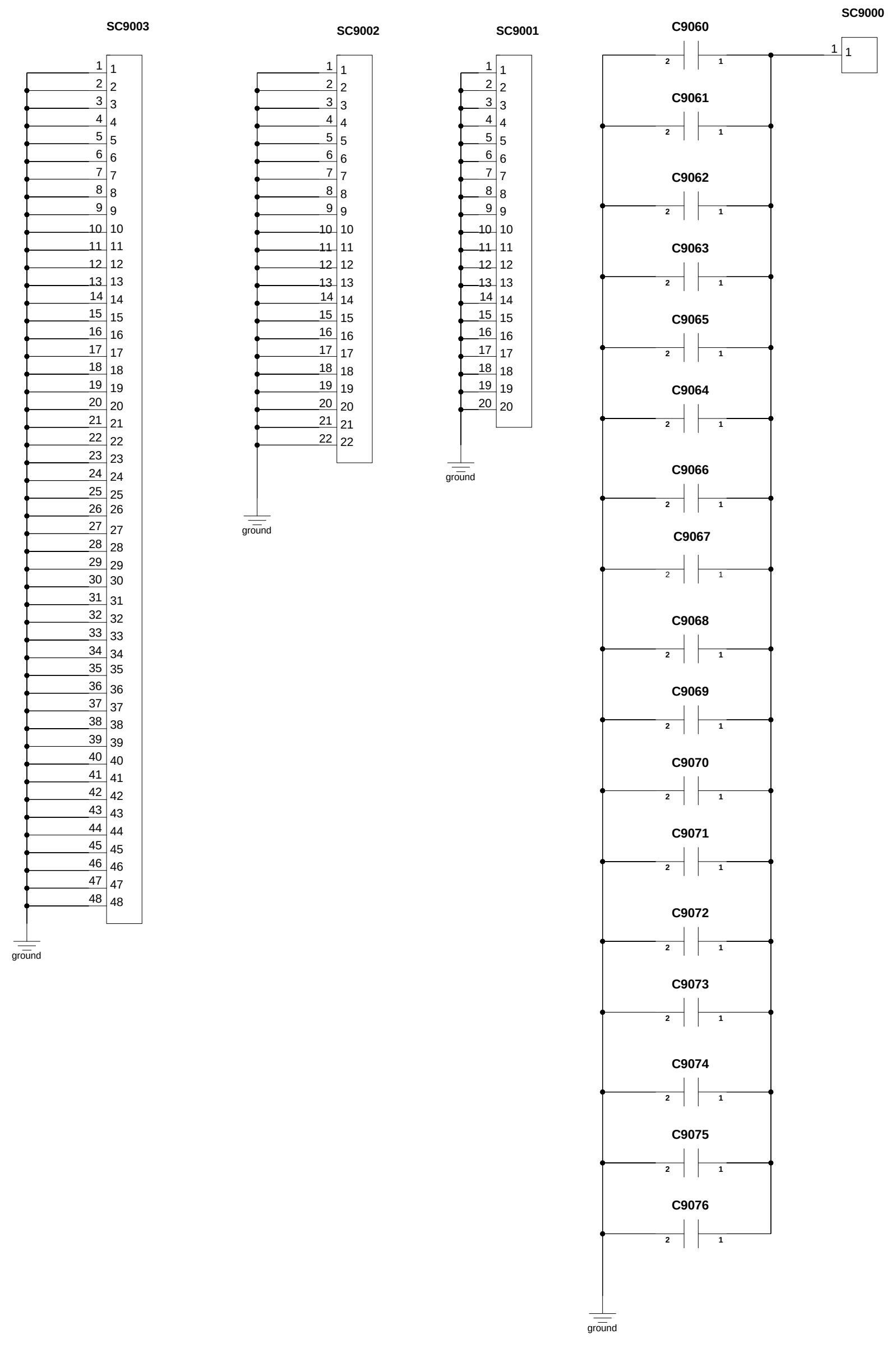

VDD_LCD_ELAVDD_7P6
L9000
VBAT
@VPH_PWR
REV01 0912
C9008
C9009
C9010
C9011
C9012
C9013
ground
L9001
VBAT
@VPH_PWR
C9014
C9015
C9016
REV01 0904
ZD9002
VOL
REV01 0912
L9002
VDD_LCD_ELVDD_4P6
REV01 0912
C9017
L9004
C9019
VDD_LCD_ELVSS_-4P4
REV01 0912
C9018
L9005
U9004
A1 VO3
A2 LX3
A3 AGND
A4 AVIN
A5 PVIN
B2 PGND2
D1 LX1
E1 LX1
D2 VO1
E2 VO1
E3 VO1
B1 PGND1
C1 PGND1
B4 PVIN
B5 PVIN
C4 LX2
C5 LX2
D5 LX2
D4 VO2
E4 VO2
E5 VO2
C2 PGND1
B3 SET
C3 ASWIRE
D3 ESWIRE
L9003
C9022
C9023
EL_ON1
EL_ON2

DISPLAY PMIC

BOTH HOLE


CN9000
CN9001
CN9002
CN9003
CN9004
CN9005
CN9006
CN9007
CN9008
Value
ground
SUS9000
SUS9001
BOTH
C9036
C9037

SHIELD CAN

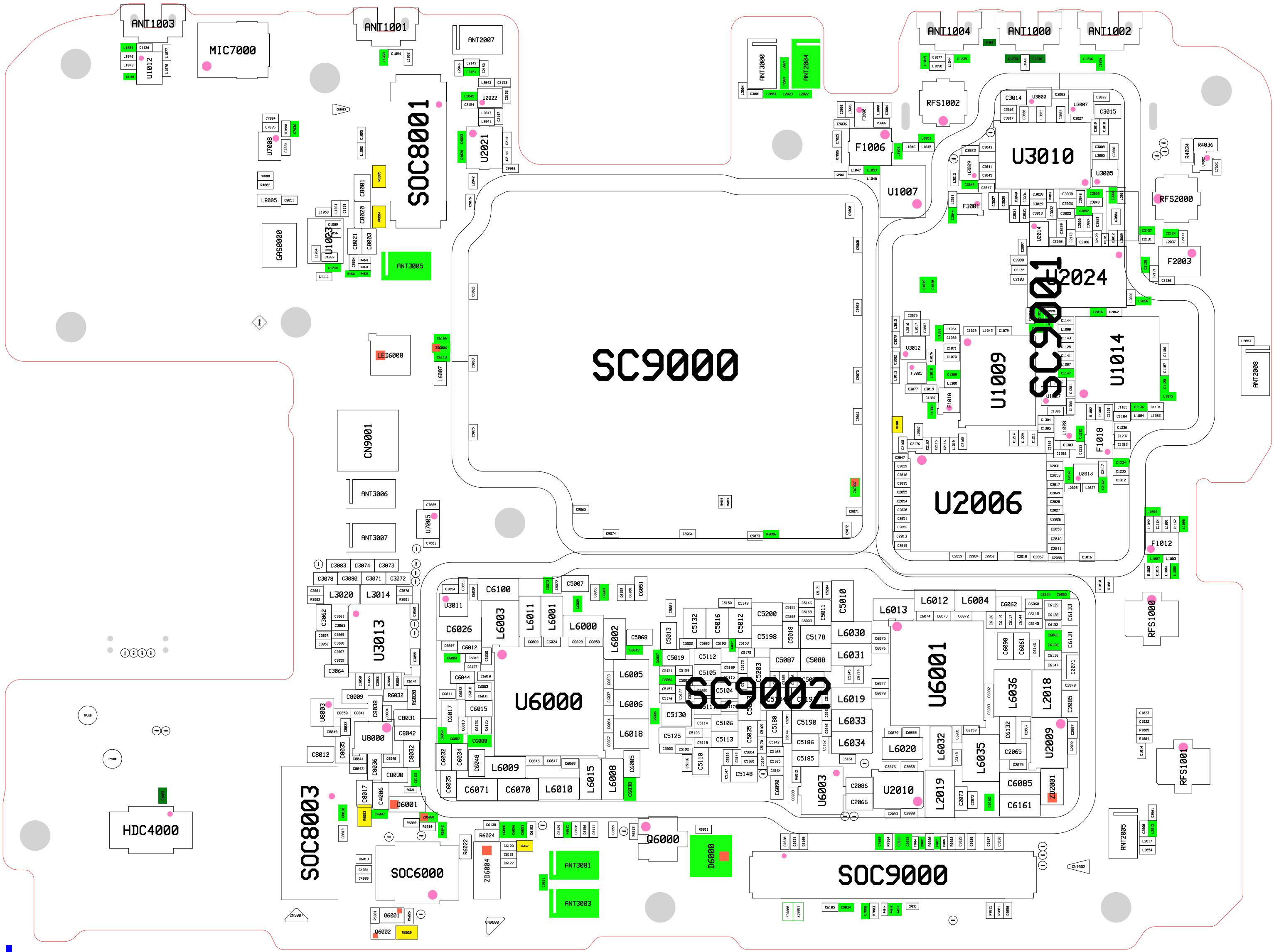

SC9003
SC9002
SC9001
SC9000
C9060
C9061
C9062
C9063
C9065
C9064
C9066
C9067
C9068
C9069
C9070
C9071
C9072
C9073
C9074
C9075
C9076
ground

REGION		MEMORY		
R9006	U_W_S356V	R9012	6/128	
		R9013	8/128	
		R9014	8/256	

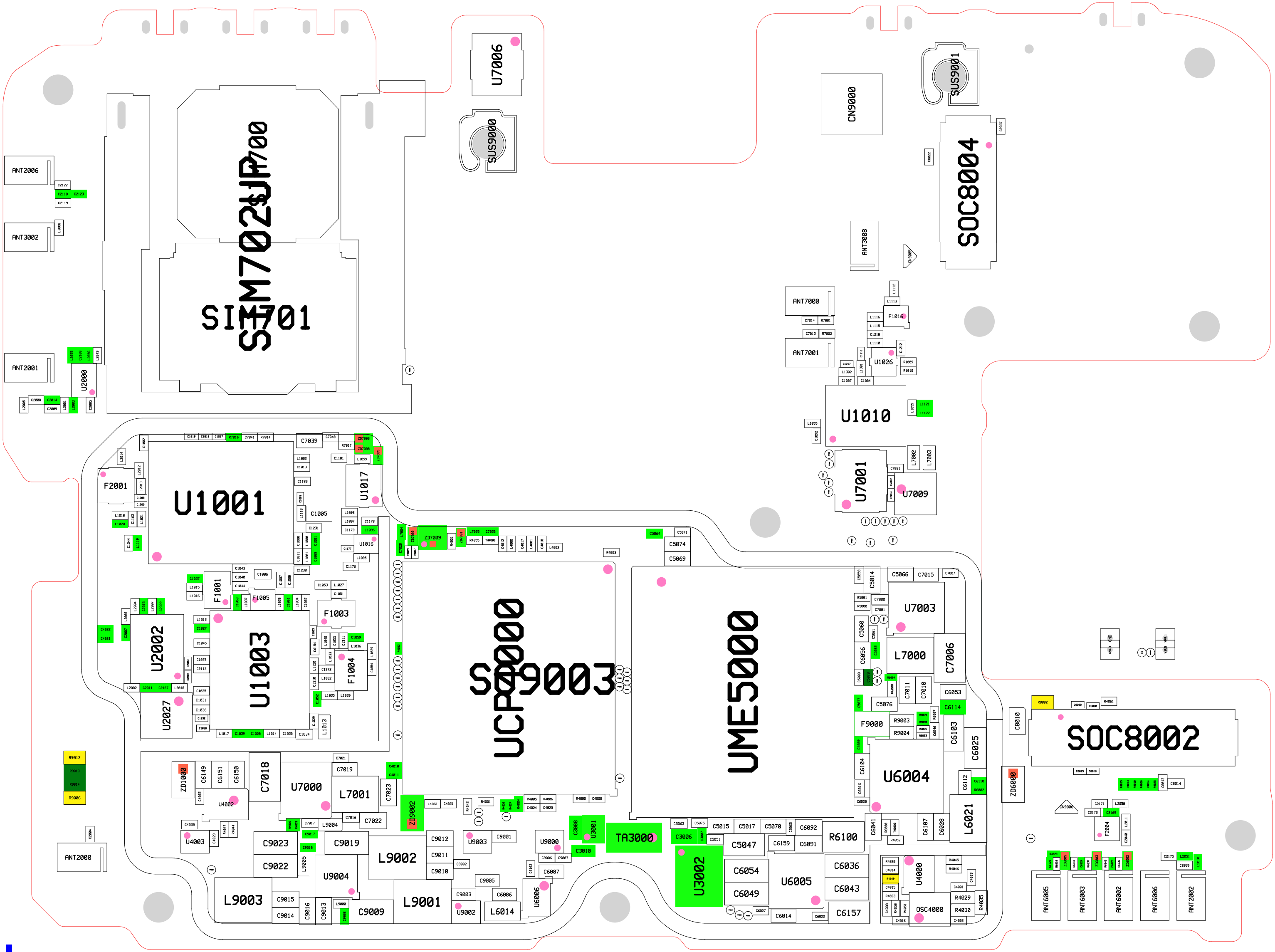
REV01 0904

			Engineer:	A35		
			Drawn by:			
			R&D CHK:	TITLE:		Size:
			DOC CTRL CHK:			
			MFG ENGR CHK:			
Changed by:	Date Changed:	Time Changed:	QA CHK:	REV:	Drawing Number:	Page:

A356U


ANT1003
MIC7000
ANT1001
ANT2007
ANT3000
ANT2004
ANT1004
ANT1000
ANT1002
L1081
C1126
L1076
L1073
C1118
U1012
L1077
L1078
L1060
C1094
L1062
L2046
C2149
C2151
C2150
L2043
C2153
L2045
U2022
C2156
C2154
L2047
L2041
C2147
C4023
C4026
U2021
C2141
C2144
C9066
L2042
C9076
SOC8001
CN9003
C7004
C7035
R7000
C7036
U7008
C7034
C1095
L1082
R8005
C8001
R8004
C8020
TH4001
R4002
L8005
C8051
L1058
L1061
C1131
C1089
L1056
U1023
L1064
C1097
C1245
L1111
C8021
C8003
C8004
R4042
R4041
R4031
R4032
ANT3005
GAS8000
L3004
C3001
L3003
L2023
L2022
L2024
C2081
R1000
L1049
C1077
L1050
L1044
C1239
C1238
C1086
C1240
C1246
C1096
RFS1002
C3002
L3006
F3000
L3008
C3004
R3007
C9036
C7025
R7006
F1006
L1051
L1052
L1046
L1045
L1047
L1053
L1048
C9067
U1007
C3014
U3000
C3003
U3007
C3033
C3015
C3016
C3017
C3008
L3002
C3025
C3027
C3019
C3018
U3010
C3023
C3043
U3009
C3041
C3045
L3012
C3042
C3047
C3009
L3005
C3008
U3005
L3011
F3001
C3044
C3037
C3039
C3040
C3034
C3028
C3051
C3029
C3038
C3036
C3048
C3050
C3049
C3086
L3018
C3031
C3035
C3013
C3032
C3022
C3052
U3004
U2014
C2099
C3020
C3024
C3021
C2100
C2173
C2108
C2129
C2132
C3012
L3009
C2157
C2131
C2124
L2037
L2039
R4034
R4036
U7002
C7026
RFS2000
F2003
C2120
C2121
C2126
C2097
C2098
C2172
C2103
U2024
L2026
L2028
L2016
C2062
C4019
C4020
C3075
L3015
L3016
L3017
C3087
C3079
U3012
C3076
C3082
F3002
L3018
L3013
C3077
L3019
C1307
C1081
L1054
C1082
C1071
C1070
C1078
L1043
C1079
C1309
L1300
C1308
F1010
U1009
SC9001
C1144
L1088
C1143
C1125
C1141
L1087
C1147
U1027
C1301
C1300
C1306
U1014
C1106
C1107
C1120
L1072
R1002
TH1000
C1101
C1105
C1136
C1134
C1104
L1084
L1083
C1304
C1305
U1028
C1232
C1303
C1233
C1302
F1018
C1236
C1237
C1313
C1161
C1214
C1229
C1211
R1000
L2057
C2160
C2176
C2163
C2115
C2116
L1019
C2165
C2047
C2029
C2016
C2035
C2055
C2054
C2030
C2051
C2052
C2013
C2019
U2006
C2059
C2034
C2056
C2018
C2057
C2058
C1016
C2031
C2053
C2017
C2049
C2028
C2027
C2026
C2050
C2046
C2041
C2161
U2013
C2117
L2025
L2027
C2162
C1234
C1235
C1312
L1093
L1092
C1164
L1091
C1162
L1090
F1012
L1007
L1003
R1003
C1015
L1004
L1005
C1010
R1001
L2053
ANT2008
SC9000
C9060
C9068
C9069
C9070
C9061
C9062
C9063
C9075
C9065
R4018
R4019
ZD7007
C9071
C9072
C9074
C9064
C9073
R3006
LED6000
C6166
ZD6006
C6113
L6007
CN9004
CN9001
ANT3006
ANT3007
C7005
U7005
C7003
C3083
C3074
C3073
C3078
C3080
C3071
C3072
C3081
R3002
L3020
L3014
C3070
R3001
C3062
C3061
C3063
C3057
C3069
C3056
C3068
C3067
C3059
U3013
C3060
C3055
C3064
C3058
C3065
C3066
R3005
R3004
C6141
C8009
R6032
R6028
U8003
C8050
C8041
C8038
C8034
C8031
C8049
C8033
U8000
C8042
C8012
C8035
C8044
C8043
C8036
C8040
C8032
C8030
C6163
R6021
C8017
C4006
D6001
C8018
R8003
C4007
ZD6001
R6009
R6010
R6043
C8019
SOC8003
TP_UB
TP4005
C4005
HDC4000
C6013
C4004
C4009
SOC6000
CN9007
R6001
Q6001
R6026
Q6002
R6029
C3054
C3053
C6039
C6100
C5073
C5072
C5007
C6055
C6081
C6109
C6108
C6051
U3011
L6003
L6011
L6001
C6009
L6000
C6026
C6069
C6024
C6029
C6058
L6002
C5068
C6042
C6097
C6012
C6004
C6040
C6050
C6137
C6044
C6010
C6003
C6031
C6011
C6023
C6018
C6017
C6015
U6000
C6033
L6005
C6037
L6006
C6019
C6136
C6135
C6059
C6052
C6000
C6084
L6018
C6067
C6032
C6034
C6048
L6009
C6045
C6047
C6068
L6015
L6008
C6005
C6035
C6071
C6070
L6010
C6038
C6006
C5001
C5013
C5132
C5016
C5012
C5150
C5149
C5200
C5155
C5158
C5202
C5003
C5146
C5011
C5171
C5204
C5010
C5000
C5005
C5193
R4017
C5153
C5198
C5018
C5178
C6057
C5019
C5112
C5175
C5173
C5203
C5087
C5088
C5151
C5159
C5105
C5109
C5115
C6007
C5157
C5177
C5176
C5104
SC9002
C5145
C5172
C5130
C5114
C5106
C5035
C5169
C5180
C5201
C5190
C5125
C5126
C5118
C5113
C5170
C5142
C5144
C5186
C5046
C5162
C5053
C5192
C5004
C5160
C5168
C5165
C5164
C5185
C5161
C5116
C5110
C5152
C5143
C5167
C5147
C5148
C6090
R6012
C6089
U6003
L6030
L6031
L6019
L6033
L6034
L6013
L6012
L6004
C6062
C6134
C6065
C6074
C6073
C6072
C6126
C6119
C6117
C6144
C6060
C6115
C6145
C6129
C6128
C6152
C6133
C6075
C6076
U6001
C6098
C6061
C6146
C6063
C6130
C6131
C6116
C6147
C2071
C6077
C6078
L6036
L2018
C6082
C6083
C2070
C2085
C6079
C6080
L6020
L6032
C6081
C6153
C6148
L6035
C6132
C2067
C2065
C2075
U2009
C2087
C2092
C2076
C2068
C2086
C2066
U2010
L2019
C2073
C2072
C6165
C6085
C6161
ZD2001
C2093
C2088
RFS1000
C1023
C1022
R1005
R1004
C1014
RFS1001
ANT2005
C2061
C2060
L2015
L2017
L2054
C6138
R6024
C6088
C6096
C6093
C6102
C6139
R6015
C6030
C6106
C6111
C6099
R6013
Q6000
R6011
D6000
R6022
ZD6004
C6120
C6167
C6121
C6122
ANT3001
C3011
ANT3003
CN9008
C9030
C9031
C6160
C7009
R7004
C9033
C9032
C9024
R4022
R9000
R4037
C9025
R9002
C9029
C9028
C9027
C9026
SOC9000
CN9002
C6105
C9034
ZD9000
ZD9001
C7008
R7003
R4014
R4012
R4011
C9035
R9015
R9001
C9058

A356U


U7006
SUS9000
CN9000
SUS9001
SOC8004
C9037
C8022
ANT3008
CN9005
ANT2006
C2122
C2118
C2123
C2119
ANT3002
L3008
SIM7002
SIM701
ANT2001
L2055
C2168
L2056
L2049
U2000
L2005
C2008
C2014
C2009
L2001
L2003
C2005
ANT7000
C7014
R7001
C7013
R7002
ANT7001
L1112
L1113
F1016
L1116
L1115
C1218
L1110
C1212
C1216
U1026
R1009
R1010
C1217
L1302
L1301
C1087
C1084
U1010
L1059
L1121
L1122
L1055
C1092
U7001
L7002
L7003
C7031
C7042
C7024
U7009
C1019
C1018
C1017
R7016
C7041
R7014
C7039
C7040
R7017
ZD7006
ZD7008
ZD7005
C1002
L2014
L2012
L2013
C1208
C1209
F2001
U1001
L1002
C1013
C1181
L1099
C1180
U1017
C1083
L1118
C1005
L1098
L1097
C1179
C1178
L1096
C1231
U1016
C1177
L1095
C1176
L1018
L1020
C1163
L1021
C1244
L1119
C1000
L1008
C1001
C1011
L1001
C1009
C1230
C1043
C1040
C1044
C1006
C1007
C1008
C1037
L1015
L1016
F1001
C1060
L1037
F1005
L1038
C1061
L1034
C1057
C1053
L1027
C1051
F1003
L2004
C2015
L2007
C2022
L2000
C2007
C4022
C4021
U2002
C2003
C2002
L1012
C1027
C1045
C1075
C2113
U1003
C1033
C6154
L1040
C1055
C1311
C1059
L1036
L1029
L1033
F1004
L1120
C1242
L1032
C1054
C1310
L2002
C2011
C2167
L2048
C1035
C1031
C1036
C1032
C1030
U2027
C1052
L1035
L1039
C1029
L1013
L1017
C1039
C1028
L1014
C1038
C1034
R9012
R9013
R9014
R9006
C2004
ANT2000
ZD1000
C6149
C6151
C6150
C7018
C4003
U4002
U7000
C7021
C7019
L7001
C4010
C4011
C7023
ZD9002
C4030
R4047
R4044
U4003
C4029
R4016
R4033
C7017
L9004
C7016
C7022
C9017
C9023
C9018
C9019
C9022
L9005
U9004
L9002
C9012
C9011
C9010
C9002
L9003
C9015
C9014
C9016
C9013
L9000
C9008
C9009
L9001
L7004
ZD7000
C7020
R4009
R4007
ZD7009
R4021
ZD7001
L7005
C7032
R4055
TH4000
C4012
L4000
C4017
L4001
C4018
L4002
R4003
C5064
C5071
C5074
C5069
R4053
SUS9003
UCP9000
UME5000
L4003
C4031
R4043
R4001
R4026
R4027
R4004
R4005
R4006
C4024
C4025
R4000
C4000
C3008
U3001
TA3000
U9003
C9001
U9000
C9006
C9007
C3010
C9005
C6162
C6087
C9003
C6086
U9002
L6014
U6006
C5063
C5075
C3006
C3007
C5051
C5015
C5017
C5070
C5065
C6092
R6100
C5047
C6159
C6091
U3002
C6054
C6049
U6005
C6036
C6043
C6027
C6014
C6022
C6157
C5050
C5014
C5066
C7015
C7007
R5001
R5000
C7000
C7001
U7003
C5060
C5061
C6056
C5062
L7000
C7006
C5008
C5078
R6004
R6008
C7011
C7010
C6053
C5077
C5076
C6114
F9000
R9003
R9004
R4039
R4040
R6005
R6003
R6007
C6046
C6103
C6025
C5009
C6104
U6004
C6016
C6020
C6112
C6110
R6002
L6021
C6041
R6000
TH6000
R4052
C6107
C6028
R4020
C4014
R4049
C4015
R4023
C4008
R4050
R4051
C4016
U4000
OSC4000
R4045
R4046
C4013
C4001
R4029
R4030
R4035
C4002
GND
GND_1
VBUS
VBUS_1
ID
R8002
C8008
C8009
R4061
C8010
SOC8002
ZD6000
C8015
C8016
R4015
R4013
R4010
R4008
R4024
R4025
C8013
C8014
CN9006
C2171
L2050
C2170
C2169
F2004
L2011
C2040
R4028
C6125
R6028
ZD6005
R6041
C6124
R6037
ZD6003
R6040
C6123
R6036
ZD6002
C2175
L2051
C2039
L2010
ANT6005
ANT6003
ANT6002
ANT6006
ANT2002